

CHAPTER 3

Further Preliminaries, with Applications to the Analysis of Deformation

after Steigmann and Shirani,
Principles of Continuum Mechanics

Every result of the chapter is derived from the algebra of Chapters 1 and 2, with no step omitted. Section and equation numbers match the book. The four worked deformations of §3.5 are carried out in full, the seven figures of the chapter are redrawn, and all nine problems of §3.8 are solved.

Orientation. Chapter 2 produced the deformation gradient $\underline{\underline{F}}$ and, from it, the Cauchy–Green tensors $\underline{\underline{C}} = \underline{\underline{F}}^t \underline{\underline{F}}$ and $\underline{\underline{B}} = \underline{\underline{F}} \underline{\underline{F}}^t$. It stopped there, because the next question cannot be answered with the algebra of Chapter 1 alone. The question is this. A deformation both *stretches* material and *rotates* it. The stretch is physical: it costs energy, it is what a material resists. The rotation is not: carrying a rigid body across the room changes $\underline{\underline{F}}$ but does nothing to the material. If the two are mixed together inside a single $\underline{\underline{F}}$, no constitutive law can be written. So we need a theorem that separates them, cleanly and uniquely.

That theorem is the polar decomposition $\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}} = \underline{\underline{V}} \underline{\underline{R}}$ of §3.4. Getting to it requires three pieces of linear algebra, and the chapter assembles them in order:

- §3.1, the spectral theorem: a symmetric tensor is a set of three stretches along three orthogonal axes, and nothing more.
- §3.2, positive definiteness: the criterion that says the three numbers are positive, so that square roots exist.
- §3.3, the square root theorem: $\underline{\underline{U}} = \sqrt{\underline{\underline{C}}}$ exists and is unique, which is what makes $\underline{\underline{R}} = \underline{\underline{F}} \underline{\underline{U}}^{-1}$ well defined.

§3.5 then computes everything explicitly for four deformations, §3.6 isolates the class in which $\underline{\underline{F}}$ is uniform, and §3.7 shows that rigid motions are exactly the case $\underline{\underline{U}} = \hat{\underline{\underline{I}}}$. Two ideas that the chapter uses constantly but states only in passing, the meaning of a *material* vector and the difference between a principal direction and a material direction, are given their own treatment in Remarks 3.26 and 3.27.

3.1 Principal values and vectors: spectral representation of a symmetric tensor

Why we want this before anything else

A tensor is a machine that eats a vector and returns a vector, and in general it does two things at once: it changes the length of what it eats and it turns it. Most vectors come out pointing somewhere new. But it can happen that for some special input direction the output points the same way, and only the length has changed. Such a direction is a direction the tensor does not rotate, and along it the tensor is nothing but multiplication by a number. If we could find enough such directions to build a basis, the tensor would be completely described by that handful of numbers: it would be three independent stretches along three fixed axes, with no turning at all.

That is what the spectral theorem delivers, and it delivers it whenever the tensor is symmetric. The relevance to Chapter 2 is immediate: $\underline{C}$ and $\underline{B}$ were proved symmetric there, at (2.85), so the theorem applies to them, and the numbers it produces will turn out to be the squares of the stretches of (2.83).

DEFINITION 3.1 (Principal value, principal vector). A unit vector $\underline{u}$ is a *principal vector* of a tensor $\underline{A}$ if there is a scalar λ , a *principal value*, such that

$$\underline{A}\underline{u} = \lambda\underline{u}, \quad \text{or} \quad (\underline{A} - \lambda\underline{I})\underline{u} = \underline{0}. \quad (3.1)$$

For definiteness vectors are taken in $\mathbb{E}^3$ and tensors as linear maps of $\mathbb{E}^3$ to itself. Everything below holds verbatim in $\hat{\mathbb{E}}^3$ with hats attached,

and this is used without comment when $\underline{C}$ and $\underline{U}$ are treated in §3.4.

REMARK 3.2 (why the second form is the useful one). The two forms in (3.1) say the same thing, but the second is the one that can be attacked. Read as a statement about the tensor $\underline{A} - \lambda \underline{I}$, it says that this tensor kills a non-zero vector, hence is not invertible, hence has zero determinant by Theorem 1.52 of Chapter 1. The unknown λ has thereby been isolated in a scalar equation, and the unknown $\underline{u}$ has been eliminated. This is the whole trick.

Because $\underline{u}$ is a unit vector it is not $\underline{0}$, so $\underline{A} - \lambda \underline{I}$ is singular and, by Problem 23 of Chapter 1,

$$0 = \det(\underline{A} - \lambda \underline{I}) = -\lambda^3 + I_1(\underline{A})\lambda^2 - I_2(\underline{A})\lambda + I_3(\underline{A}), \quad (3.2)$$

where, by Problem 22 of Chapter 1,

$$I_1(\underline{A}) = \text{tr } \underline{A}, \quad I_2(\underline{A}) = \frac{1}{2}[(\text{tr } \underline{A})^2 - \text{tr}(\underline{A}^2)], \quad I_3(\underline{A}) = \det \underline{A} \quad (3.3)$$

are the *principal invariants* of $\underline{A}$. Equation (3.2) is the *characteristic equation*.

REMARK 3.3 (invariant means basis free, and why that matters here). The three numbers (3.3) were constructed in Chapter 1 out of box products alone, with no basis anywhere in the definition, so they do not change when the basis changes. Consequently the polynomial (3.2) does not change, and therefore neither do its roots. This is not a technicality: it is the statement that the principal values are properties of the *tensor*, not of the arithmetic we use to compute them. If they depended on the basis they could not be used in a constitutive law, and the whole chapter would collapse.

Equation (3.2) is a real cubic. Its complex roots occur in conjugate pairs, so it has either three real roots or exactly one; in particular *every tensor has at least one real principal value*. For symmetric tensors the situation is better still.

THEOREM 3.4 (Reality). *If $\underline{A} = \underline{A}^t$ then every root of (3.2) is real.*

Proof. Suppose a root λ and the associated $\underline{u}$ are complex,

$$\lambda = a + ib, \quad \underline{u} = u_j \underline{e}_j \quad \text{with} \quad u_j = a_j + ib_j, \quad (3.4)$$

where $i^2 = -1$ and a, b, a_j, b_j are real. Here the basis $\{\underline{e}_j\}$ is still real, and the inner product is extended to complex components *bilinearly*, $\underline{u} \cdot \underline{v} = u_j v_j$, without conjugation; this is the only reading under which the algebra of Chapter 1 continues to hold unchanged, and it is what makes the computation below work.

The components A_{ij} are real, so conjugating (3.1) term by term, and using that the conjugate of a product is the product of the conjugates,

$$\underline{A}\bar{\underline{u}} = \bar{\lambda}\bar{\underline{u}}, \quad (3.5)$$

where

$$\bar{\lambda} = a - ib, \quad \bar{\underline{u}} = \bar{u}_j \underline{e}_j \quad \text{with} \quad \bar{u}_j = a_j - ib_j. \quad (3.6)$$

Dot (3.1) with $\bar{\underline{u}}$ and (3.5) with $\underline{u}$:

$$\bar{\underline{u}} \cdot \underline{A}\underline{u} = \lambda \bar{\underline{u}} \cdot \underline{u}, \quad \underline{u} \cdot \underline{A}\bar{\underline{u}} = \bar{\lambda} \underline{u} \cdot \bar{\underline{u}}. \quad (3.7)$$

Now use the symmetry of $\underline{A}$ on the left-hand side of the second:

$$\underline{u} \cdot \underline{A}\bar{\underline{u}} = \underline{A}^t \underline{u} \cdot \bar{\underline{u}} = \underline{A}\underline{u} \cdot \bar{\underline{u}} = \bar{\underline{u}} \cdot \underline{A}\underline{u},$$

so the two left-hand sides in (3.7) are equal. Subtracting,

$$(\lambda - \bar{\lambda}) \bar{\underline{u}} \cdot \underline{u} = 0, \quad (3.8)$$

in which, by the bilinear reading of the dot product,

$$\bar{\underline{u}} \cdot \underline{u} = \bar{u}_j u_j = (a_j - ib_j)(a_j + ib_j) = a_j a_j + b_j b_j,$$

the cross terms $ia_j b_j$ and $-ib_j a_j$ cancelling. This is a sum of squares of real numbers, hence strictly positive unless every a_j and every b_j vanishes, that is unless $\underline{u} = \underline{0}$; and $\underline{u} = \underline{0}$ is excluded by hypothesis. Therefore $\lambda - \bar{\lambda} = 0$, i.e. $2ib = 0$, i.e. $b = 0$ and λ is real. $\square$

THEOREM 3.5 (Orthogonality). *If $\underline{A} = \underline{A}^t$ and $\underline{A}\underline{u}_1 = \lambda_1\underline{u}_1$, $\underline{A}\underline{u}_2 = \lambda_2\underline{u}_2$ with $\lambda_1 \neq \lambda_2$, then $\underline{u}_1 \cdot \underline{u}_2 = 0$.*

Proof. Move $\underline{A}$ across the dot product twice:

$$\lambda_1\underline{u}_1 \cdot \underline{u}_2 = \underline{A}\underline{u}_1 \cdot \underline{u}_2 = \underline{u}_1 \cdot \underline{A}^t\underline{u}_2 = \underline{u}_1 \cdot \underline{A}\underline{u}_2 = \lambda_2\underline{u}_1 \cdot \underline{u}_2, \quad (3.9-3.10)$$

the third equality being the symmetry of $\underline{A}$. Hence $(\lambda_1 - \lambda_2)\underline{u}_1 \cdot \underline{u}_2 = 0$, and since $\lambda_1 \neq \lambda_2$ the second factor must vanish. $\square$

So if the three roots of (3.2) are distinct, the three principal vectors are mutually orthogonal and, being unit vectors, form an orthonormal basis. The book records that this remains true when roots repeat and refers the argument to its Appendix D. Since the whole chapter rests on it, here is a complete proof that needs nothing beyond Chapter 1 and does not care about multiplicities at all.

LEMMA 3.6 (The orthogonal complement of a principal direction is invariant). *Let $\underline{A} = \underline{A}^t$ and $\underline{A}\underline{u} = \lambda\underline{u}$ with $|\underline{u}| = 1$. Put $P = \{\underline{u}\}^\perp$, the plane through the origin perpendicular to $\underline{u}$. Then $\underline{A}\underline{w} \in P$ for every $\underline{w} \in P$.*

Proof. Let $\underline{w} \cdot \underline{u} = 0$. Then

$$\underline{u} \cdot \underline{A}\underline{w} = \underline{A}^t\underline{u} \cdot \underline{w} = \underline{A}\underline{u} \cdot \underline{w} = \lambda\underline{u} \cdot \underline{w} = \lambda \cdot 0 = 0,$$

so $\underline{A}\underline{w}$ is perpendicular to $\underline{u}$, that is $\underline{A}\underline{w} \in P$. $\square$

THEOREM 3.7 (Spectral theorem). *Every symmetric $\underline{A}$ possesses an orthonormal basis $\{\underline{u}_1, \underline{u}_2, \underline{u}_3\}$ of principal vectors, with real principal values $\lambda_1, \lambda_2, \lambda_3$.*

Proof. The cubic (3.2) has at least one real root λ_1 , and Theorem 3.4 says every root is real in any case. Since $\det(\underline{A} - \lambda_1\underline{I}) = 0$, the tensor $\underline{A} - \lambda_1\underline{I}$ is not invertible, so by Theorem 1.52 of Chapter 1 it is not injective, and there is a non-zero $\underline{p}$ with $(\underline{A} - \lambda_1\underline{I})\underline{p} = \underline{0}$. Normalise, $\underline{u}_1 = \underline{p}/|\underline{p}|$.

By Lemma 3.6 the plane $P_1 = \{\underline{u}_1\}^\perp$ is carried into itself by $\underline{A}$, so the restriction $\underline{A}|_{P_1}$ is a linear map of the two-dimensional space P_1 to itself, and it is symmetric there because $\underline{w} \cdot \underline{A}\underline{s} = \underline{A}\underline{w} \cdot \underline{s}$ holds for all vectors, hence in particular for those lying in P_1 . Choose any orthonormal pair $\{\underline{s}_1, \underline{s}_2\}$ spanning P_1 ; the matrix of $\underline{A}|_{P_1}$ is $\begin{pmatrix} \alpha & \beta \\ \beta & \delta \end{pmatrix}$ with real entries, whose characteristic equation $\mu^2 - (\alpha + \delta)\mu + (\alpha\delta - \beta^2) = 0$ has discriminant

$$(\alpha + \delta)^2 - 4(\alpha\delta - \beta^2) = (\alpha - \delta)^2 + 4\beta^2 \geq 0,$$

a sum of squares. So there is a real root λ_2 and a unit $\underline{u}_2 \in P_1$ with $\underline{A}\underline{u}_2 = \lambda_2\underline{u}_2$, and by construction $\underline{u}_1 \cdot \underline{u}_2 = 0$.

Finally let $\underline{u}_3 = \underline{u}_1 \times \underline{u}_2$, a unit vector orthogonal to both. It lies in P_1 , and it is orthogonal to $\underline{u}_2$ inside P_1 ; applying Lemma 3.6 once more, this time inside P_1 with $\underline{u}_2$ in the role of $\underline{u}$, gives $\underline{A}\underline{u}_3 \perp \underline{u}_2$, while $\underline{A}\underline{u}_3 \in P_1$ means $\underline{A}\underline{u}_3 \perp \underline{u}_1$. A vector of $\mathbb{E}^3$ orthogonal to both $\underline{u}_1$ and $\underline{u}_2$ is a multiple of $\underline{u}_3$, so $\underline{A}\underline{u}_3 = \lambda_3\underline{u}_3$ for some real λ_3 . The triad is orthonormal by construction. $\square$

REMARK 3.8 (what the proof buys that the determinant argument does not). Nothing above counted multiplicities. If $\lambda_1 = \lambda_2$, the argument still produces two orthogonal principal vectors, because the second was obtained by restricting to a plane rather than by looking for a second root. This is exactly the case that a naive treatment mishandles, and it is not exotic: it occurs in §3.5.2 below whenever $f' = f/R$, and in every isotropic dilatation, where all three principal values coincide and *every* direction is principal.

Having an orthonormal basis of principal vectors, we can write the tensor down. First,

$$\underline{I} = \underline{u}_i \otimes \underline{u}_i, \tag{3.11}$$

which is checked by feeding it an arbitrary $\underline{v}$: expanding $\underline{v} = (\underline{v} \cdot \underline{u}_i)\underline{u}_i$ on the orthonormal basis and using $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$,

$$(\underline{u}_i \otimes \underline{u}_i)\underline{v} = (\underline{u}_i \cdot \underline{v})\underline{u}_i = \underline{v} \quad \text{for every } \underline{v}, \quad \text{so} \quad \underline{u}_i \otimes \underline{u}_i = \underline{I}.$$

Then, using Problem 17(b) of Chapter 1, $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$,

$$\underline{A} = \underline{A}\underline{I} = \underline{A}(\underline{u}_i \otimes \underline{u}_i) = (\underline{A}\underline{u}_i) \otimes \underline{u}_i = \sum_{i=1}^3 \lambda_i \underline{u}_i \otimes \underline{u}_i, \quad (3.12)$$

the *spectral representation*. The explicit summation sign is written because the repeated index i now also sits on λ_i , and the summation convention would be ambiguous.

The matrix of $\underline{A}$ in the basis $\{\underline{u}_i \otimes \underline{u}_j\}$ is $(A_{ij}) = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, since $A_{ij} = \underline{u}_i \cdot \underline{A}\underline{u}_j = \lambda_j \underline{u}_i \cdot \underline{u}_j = \lambda_j \delta_{ij}$ (no sum). A symmetric tensor can therefore be diagonalised by resolving it on its own principal vectors.

REMARK 3.9 (reading (3.12) as a machine). Equation (3.12) says that $\underline{A}$ works like this: take the input $\underline{v}$, measure its three components $\underline{u}_i \cdot \underline{v}$ along the principal axes, multiply the i th by λ_i , and reassemble. Explicitly,

$$\underline{A}\underline{v} = \sum_i \lambda_i (\underline{u}_i \cdot \underline{v}) \underline{u}_i.$$

There is no rotation anywhere in that description. This is the precise content of the sentence “a symmetric tensor is pure stretch”, which is the sentence §3.4 will need. A dyad $\underline{u}_i \otimes \underline{u}_i$ is the projection $\mathbb{P}_{(\underline{u}_i)}$ of Problem 10 of Chapter 1 onto the i th axis, so (3.12) reads: *decompose, scale each piece, add up*.

REMARK 3.10 (powers and functions of a symmetric tensor). From (3.12) and the dyad product rule $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c}) \underline{a} \otimes \underline{d}$,

$$\begin{aligned} \underline{A}^2 &= \left(\sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i \right) \left(\sum_j \lambda_j \underline{u}_j \otimes \underline{u}_j \right) \\ &= \sum_{i,j} \lambda_i \lambda_j \delta_{ij} \underline{u}_i \otimes \underline{u}_j = \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i, \end{aligned}$$

the Kronecker delta arising from $\underline{u}_i \cdot \underline{u}_j = \delta_{ij}$ and collapsing the double sum to a single one. By induction $\underline{A}^n = \sum_i \lambda_i^n \underline{u}_i \otimes \underline{u}_i$, and if all $\lambda_i \neq 0$ the same formula with $n = -1$ gives the inverse, as one verifies at once by multiplying. *Every* function of $\underline{A}$ is obtained by applying that function to the three numbers and leaving the axes alone. This single observation is the entire content of §3.3.

REMARK 3.11 (what the three invariants *are*: lengths, areas, volume). Chapter 1 produced I_1, I_2, I_3 by a formal recipe, three ways of inserting $\underline{A}$ into a box product, and the pattern looked arbitrary. For a symmetric tensor the spectral representation makes the pattern obvious, and it is worth seeing, because it is the reason these three and no others appear in every constitutive law.

Let $\underline{A}$ be symmetric with principal values $\lambda_i > 0$ and principal axes $\underline{u}_i$, and apply it to the unit cube built on those axes. By (3.12) the cube goes to a rectangular box with edges λ_i along the same axes. Then, by Problem 7 of §3.8,

$$\begin{aligned} I_1 &= \lambda_1 + \lambda_2 + \lambda_3 && \text{sum of the three } \textit{edge} \text{ lengths,} \\ I_2 &= \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1 && \text{sum of the three } \textit{face} \text{ areas,} \\ I_3 &= \lambda_1\lambda_2\lambda_3 && \text{the } \textit{volume}. \end{aligned}$$

So the three invariants measure how $\underline{A}$ scales length, area and volume, in that order. Nothing else of that kind exists in three dimensions, because a box has only edges, faces and an interior.

This also explains Problem 22(a) of Chapter 1, $I_2(\underline{A}) = \text{tr } \underline{A}^*$, which there looked like an identity with no content. The cofactor is the tensor that carries *areas*, Remark 1.55; its trace collects the three face areas; and that is exactly I_2 . The same three levels appear again in Chapter 2 as the three deformation measures: $\underline{F}$ carries line elements, (2.34); $\underline{F}^*$ carries area elements, (2.54); and $J = \det \underline{F}$ carries volume elements, (2.50).

For the deformation tensors themselves the reading is immediate. With $\underline{C} = \underline{U}^2$ and λ_i the principal stretches,

$$\begin{aligned} I_1(\underline{C}) &= \lambda_1^2 + \lambda_2^2 + \lambda_3^2, & I_2(\underline{C}) &= \sum_{i < j} \lambda_i^2 \lambda_j^2, \\ I_3(\underline{C}) &= (\lambda_1 \lambda_2 \lambda_3)^2 = J^2, \end{aligned}$$

so $I_1(\underline{C})$ is the sum of squared *length* stretches, $I_2(\underline{C})$ the sum of squared *area* stretches, and $I_3(\underline{C})$ the squared *volume* stretch. An isotropic elastic material stores energy as a function of these three numbers and nothing else, which is why they are worth naming.

3.2 Positive definiteness and the extremal property of principal values

The square root of §3.3 will need the λ_i to be positive. The criterion is the following, and Chapter 2 has already verified it for $\underline{C}$ at (2.84).

DEFINITION 3.12 (Positive definite). A tensor $\underline{A}$ is *positive definite* if

$$\underline{a} \cdot \underline{A}\underline{a} > 0 \quad \text{for all } \underline{a} \neq \underline{0}. \quad (3.13)$$

Two observations before proceeding, both from Chapter 1. By Problem 14 any $\underline{A}$ splits as $\underline{A} = \underline{S} + \underline{W}$ with $\underline{S} = \text{sym } \underline{A}$ symmetric and $\underline{W} = \text{skw } \underline{A}$ skew, and by Problem 6(a) $\underline{a} \cdot \underline{W}\underline{a} = 0$ for skew $\underline{W}$; hence

$$\underline{a} \cdot \underline{A}\underline{a} = \underline{a} \cdot \underline{S}\underline{a} + \underline{a} \cdot \underline{W}\underline{a} = \underline{a} \cdot \underline{S}\underline{a}.$$

So (3.13) sees only the symmetric part, and a skew tensor can never be positive definite. We therefore assume $\underline{A}$ symmetric with no loss.

THEOREM 3.13 (Positive definite $\iff$ positive principal values). *For symmetric $\underline{A}$ with principal values λ_i ,*

$$\underline{A} \text{ positive definite} \iff \lambda_i > 0, \quad i = 1, 2, 3.$$

Moreover $\det \underline{A} = \lambda_1 \lambda_2 \lambda_3 > 0$.

Proof. ($\Rightarrow$) Put $\underline{a} = \underline{u}$, a principal vector with value λ . Then $\underline{u} \neq \underline{0}$ and

$$0 < \underline{u} \cdot \underline{A}\underline{u} = \underline{u} \cdot \lambda \underline{u} = \lambda \underline{u} \cdot \underline{u} = \lambda, \quad (3.14)$$

using $|\underline{u}| = 1$. Applying this to each principal vector,

$$\lambda_i > 0; \quad i = 1, 2, 3. \quad (3.15)$$

($\Leftarrow$) Conversely, if every $\lambda_i > 0$ then, by the spectral representation and $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$,

$$\underline{a} \cdot \underline{A}\underline{a} = \underline{a} \cdot \sum_i \lambda_i (\underline{u}_i \cdot \underline{a}) \underline{u}_i = \sum_i \lambda_i (\underline{a} \cdot \underline{u}_i)^2 > 0 \quad \text{for all } \underline{a} \neq \underline{0}, \quad (3.16)$$

the strict inequality holding because the λ_i are positive and, $\{u_i\}$ being a basis, at least one component $\underline{a} \cdot \underline{u}_i$ is non-zero when $\underline{a} \neq \underline{0}$; were all three zero, $\underline{a}$ would be orthogonal to a basis and hence $\underline{0}$.

For the determinant, use the box-product definition with the principal triad itself as the test triad, which is legitimate because $\{u_i\}$ is a basis and therefore linearly independent, so $[\underline{u}_1, \underline{u}_2, \underline{u}_3] \neq 0$. Feed the triad to $\underline{A}$, one member at a time:

$$\underline{A}\underline{u}_1 = \lambda_1\underline{u}_1, \quad \underline{A}\underline{u}_2 = \lambda_2\underline{u}_2, \quad \underline{A}\underline{u}_3 = \lambda_3\underline{u}_3,$$

by the very definition of a principal vector. Now pull the three scalars out of the box product one slot at a time, using that it is linear in each slot *separately*: first the λ_1 from the first slot, leaving the other two untouched, then the λ_2 from the second, then the λ_3 from the third,

$$\begin{aligned} \det \underline{A} &= \frac{[\underline{A}\underline{u}_1, \underline{A}\underline{u}_2, \underline{A}\underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} = \frac{[\lambda_1\underline{u}_1, \lambda_2\underline{u}_2, \lambda_3\underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} \\ &= \frac{\lambda_1[\underline{u}_1, \lambda_2\underline{u}_2, \lambda_3\underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} = \frac{\lambda_1\lambda_2\lambda_3[\underline{u}_1, \underline{u}_2, \underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} \\ &= \lambda_1\lambda_2\lambda_3 > 0. \end{aligned} \tag{3.17}$$

The last step is a cancellation of one and the same non-zero number, $[\underline{u}_1, \underline{u}_2, \underline{u}_3]$, which appears identically above and below. For an orthonormal triad that number is $+1$ or -1 according to handedness, but its value never enters: what matters is only that it is the *same* in numerator and denominator, so the answer cannot depend on which of the two orientations the triad happens to have. $\square$

The extremal property

The remaining result of the section says that the principal values bracket the diagonal entries in *any* orthonormal basis. It is proved by optimisation, and the optimisation is worth following because the same argument, constrained stationarity, recurs throughout mechanics.

Fix a symmetric positive definite $\underline{A}$ and define

$$f(\underline{u}) = \underline{u} \cdot \underline{A}\underline{u} \quad \text{with} \quad |\underline{u}| = 1. \tag{3.18}$$

f is continuous and positive on the unit sphere $S = \{\underline{v} : |\underline{v}| = 1\}$, which is closed and bounded, hence compact; a continuous function on a compact set attains its extrema. So there exist $\underline{u}_1, \underline{u}_2 \in S$ with

$$\begin{aligned} 0 < \lambda_1 &= \min_{\underline{u} \in S} f(\underline{u}), & 0 < \lambda_2 &= \max_{\underline{u} \in S} f(\underline{u}), \\ \lambda_1 &= f(\underline{u}_1), & \lambda_2 &= f(\underline{u}_2). \end{aligned} \quad (3.19)$$

REMARK 3.14 (a warning about the labels). In this subsection λ_1 and λ_2 denote the smallest and largest principal values, whereas everywhere else in the book the subscript is a plain enumeration index. In three dimensions the extremal pair is the smallest and the largest, so the middle principal value is not named here at all. Keeping the two conventions apart is the only difficulty in (3.26) below.

At an extremum f is stationary. Differentiating (3.18) with the product rule and using symmetry,

$$0 = df(\underline{u}) = d\underline{u} \cdot \underline{A}\underline{u} + \underline{u} \cdot \underline{A}(d\underline{u}) = d\underline{u} \cdot \underline{A}\underline{u} + \underline{A}^t \underline{u} \cdot d\underline{u} = 2 \underline{A}\underline{u} \cdot d\underline{u}, \quad (3.20)$$

so we seek $\underline{u}$ such that

$$\underline{A}\underline{u} \cdot d\underline{u} = 0. \quad (3.21)$$

The increments $d\underline{u}$ are not arbitrary: differentiating the constraint $\underline{u} \cdot \underline{u} = 1$ gives $2\underline{u} \cdot d\underline{u} = 0$, so $d\underline{u}$ ranges over the plane $\{\underline{u}\}^\perp$ and nothing more. Thus (3.21) says that $\underline{A}\underline{u}$ is orthogonal to every vector of $\{\underline{u}\}^\perp$. By Problem 10 of Chapter 1 the orthogonal complement of $\{\underline{u}\}^\perp$ is the line spanned by $\underline{u}$, so $\underline{A}\underline{u}$ must be a multiple of $\underline{u}$:

$$\underline{A}\underline{u} = \mu \underline{u} \quad (3.22)$$

for some scalar μ . Dotting with $\underline{u}$ identifies the multiplier as $\mu = \underline{u} \cdot \underline{A}\underline{u} = f(\underline{u})$. Applying this at the minimiser and at the maximiser,

$$\underline{A}\underline{u}_1 = \lambda_1 \underline{u}_1, \quad (3.23)$$

$$\underline{A}\underline{u}_2 = \lambda_2 \underline{u}_2, \quad (3.24)$$

so λ_1 and λ_2 are principal values of $\underline{A}$: the extremes of the quadratic form on the unit sphere are attained *at* principal directions, and *equal* the corresponding principal values. By construction,

$$\lambda_1 = f(\underline{u}_1) \leq f(\underline{u}) \leq f(\underline{u}_2) = \lambda_2 \quad \text{for every } \underline{u} \in S. \quad (3.25)$$

Finally take $\underline{u} = \underline{e}_1$, an element of an arbitrary orthonormal basis. Then $f(\underline{e}_1) = \underline{e}_1 \cdot \underline{A}\underline{e}_1 = A_{11}$, and similarly for $\underline{e}_2, \underline{e}_3$, so (3.25) gives the *extremal property*

$$\lambda_1 \leq \min\{A_{11}, A_{22}, A_{33}\} \leq \max\{A_{11}, A_{22}, A_{33}\} \leq \lambda_2. \quad (3.26)$$

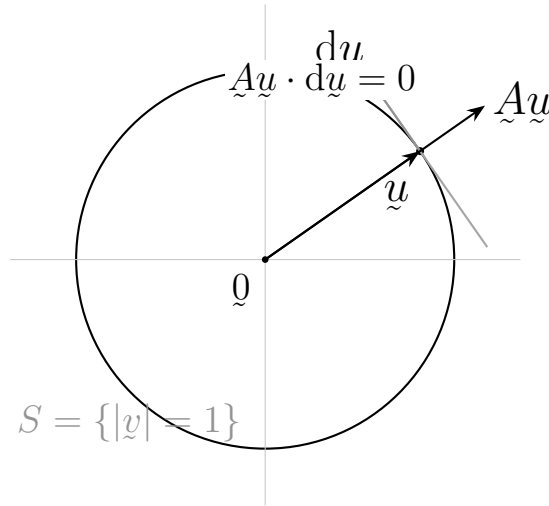


Figure 3.1: The unit sphere S . The constraint $|\underline{u}| = 1$ confines the admissible increments $d\underline{u}$ to the tangent plane at $\underline{u}$, so stationarity of $f(\underline{u}) = \underline{u} \cdot \underline{A}\underline{u}$ requires $\underline{A}\underline{u}$ to be orthogonal to that whole plane, hence parallel to $\underline{u}$ itself. This is (3.22): an extremum of the quadratic form is a principal direction.

REMARK 3.15 (what (3.26) is good for). Two things. First, it is a free bound: having computed the matrix of $\underline{C}$ in any convenient basis, one reads off three diagonal entries and knows immediately that the largest principal stretch squared is at least the largest of them and the smallest is at most the smallest. In the simple shear of §3.5.1, (C_{AB}) has diagonal $1, 1 + \gamma^2, 1$, so $\lambda_{\max}^2 \geq 1 + \gamma^2$ and $\lambda_{\min}^2 \leq 1$ without solving anything. Second, it explains why the principal values are called principal: they

are the extreme values of the stretch over all directions, so the material's worst-stretched fibre lies along a principal axis. That is the fact a failure criterion needs.

3.3 Square root theorem

Remark 3.10 already contains the answer: a symmetric positive definite tensor is three positive numbers attached to three orthogonal axes, and the square root is obtained by taking the square roots of the three numbers. What has to be proved is that this is the *only* possible answer, because it is uniqueness, not existence, that makes $\underline{\underline{R}} = \underline{\underline{F}}\underline{\underline{U}}^{-1}$ a definition rather than a choice.

THEOREM 3.16 (Square root). *Let $\underline{\underline{A}}$ be symmetric and positive definite. Then there is exactly one symmetric positive definite $\underline{\underline{U}}$ with $\underline{\underline{A}} = \underline{\underline{U}}^2$. We write $\underline{\underline{U}} = \sqrt{\underline{\underline{A}}}$.*

Proof. Existence. Write $\underline{\underline{A}}$ in spectral form, which Theorem 3.7 permits and Theorem 3.13 makes positive:

$$\underline{\underline{A}} = \sum_i \lambda_i \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i, \quad \lambda_i > 0, \quad (3.27)$$

and define

$$\underline{\underline{U}} = \sum_i \sqrt{\lambda_i} \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i. \quad (3.28)$$

This is symmetric, because each dyad $\underline{\underline{u}}_i \otimes \underline{\underline{u}}_i$ is symmetric by Problem 17(a) of Chapter 1, $(\underline{\underline{a}} \otimes \underline{\underline{b}})^t = \underline{\underline{b}} \otimes \underline{\underline{a}}$, and a sum of symmetric tensors is symmetric. It is positive definite, because its principal values are the positive numbers $\sqrt{\lambda_i}$ and Theorem 3.13 applies. And squaring it,

$$\begin{aligned} \underline{\underline{U}}^2 &= \underline{\underline{U}} \left(\sum_i \sqrt{\lambda_i} \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i \right) = \sum_i \sqrt{\lambda_i} (\underline{\underline{U}} \underline{\underline{u}}_i) \otimes \underline{\underline{u}}_i \\ &= \sum_i \sqrt{\lambda_i} (\sqrt{\lambda_i} \underline{\underline{u}}_i) \otimes \underline{\underline{u}}_i = \sum_i \lambda_i \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i = \underline{\underline{A}}, \end{aligned} \quad (3.29)$$

where the second equality is Problem 17(b), $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$, applied term by term, and the third uses $\underline{U}\underline{u}_i = \sqrt{\lambda_i}\underline{u}_i$, which is read off (3.28) by feeding it $\underline{u}_i$ and using orthonormality. So this $\underline{U}$ has the stated properties.

A preparatory fact. Suppose $\underline{U}$ is symmetric positive definite with $\underline{U}^2\underline{u} = \lambda\underline{u}$ for some unit $\underline{u}$ and $\lambda > 0$. Then, since $\underline{U}$ commutes with $\underline{I}$,

$$\underline{0} = (\underline{U}^2 - \lambda\underline{I})\underline{u} = (\underline{U} + \sqrt{\lambda}\underline{I})(\underline{U} - \sqrt{\lambda}\underline{I})\underline{u} = (\underline{U} + \sqrt{\lambda}\underline{I})\underline{v}, \quad (3.30)$$

where $\underline{v} := (\underline{U} - \sqrt{\lambda}\underline{I})\underline{u}$,

the factorisation being legitimate because $(\underline{U} + \sqrt{\lambda}\underline{I})(\underline{U} - \sqrt{\lambda}\underline{I}) = \underline{U}^2 - \sqrt{\lambda}\underline{U} + \sqrt{\lambda}\underline{U} - \lambda\underline{I} = \underline{U}^2 - \lambda\underline{I}$. Hence

$$\underline{U}\underline{v} = -\sqrt{\lambda}\underline{v}. \quad (3.31)$$

If $\underline{v} \neq \underline{0}$ this exhibits $-\sqrt{\lambda} < 0$ as a principal value of $\underline{U}$, contradicting Theorem 3.13. Therefore $\underline{v} = \underline{0}$, that is

$$\underline{U}\underline{u} = \sqrt{\lambda}\underline{u}.$$

In words: *a symmetric positive definite square root of $\underline{A}$ is forced to act on each principal vector of $\underline{A}$ by the positive square root of the corresponding principal value.* It has no freedom at all.

Uniqueness. Let $\tilde{\underline{U}}$ be symmetric positive definite with $\tilde{\underline{U}}^2 = \underline{A}$. Applying the preparatory fact to $\tilde{\underline{U}}$ with $\underline{u} = \underline{u}_i$ and $\lambda = \lambda_i$ gives $\tilde{\underline{U}}\underline{u}_i = \sqrt{\lambda_i}\underline{u}_i = \underline{U}\underline{u}_i$ for each i . Two linear maps agreeing on a basis agree everywhere: for arbitrary $\underline{w} = w_i\underline{u}_i$,

$$\tilde{\underline{U}}\underline{w} = w_i\tilde{\underline{U}}\underline{u}_i = w_i\underline{U}\underline{u}_i = \underline{U}\underline{w},$$

so $\tilde{\underline{U}} = \underline{U}$. □

REMARK 3.17 (why positive definiteness is not decoration). Drop it and uniqueness fails immediately. The tensor $\underline{I}$ has, among symmetric square roots, not only $\underline{I}$ but also $\underline{I} - 2\underline{e}_1 \otimes \underline{e}_1 = \text{diag}(-1, 1, 1)$, whose square is $\text{diag}(1, 1, 1) = \underline{I}$. There are eight such sign patterns. Positive definiteness is precisely the condition that picks one. In the polar decomposition

this matters physically: a fibre must stretch by a positive factor, and the sign ambiguity is exactly the ambiguity between a stretch and a stretch followed by a reflection.

REMARK 3.18 (what §§3.1–3.3 really buy: every strain measure at once). Chapter 2 remarked, just after (2.99), that “there are infinitely many equivalent ways to define strain tensors” and sent the reader to Ogden. It did not say what makes them definable, and the answer is the two sections just completed. Once $\underline{U}$ exists and is diagonal on a fixed orthonormal triad, *any* function of a single real variable can be applied to a deformation:

$$g(\underline{U}) := \sum_i g(\lambda_i) \underline{u}_i \otimes \underline{u}_i.$$

This is well defined precisely because Theorem 3.7 supplies the triad and Theorem 3.16 makes $\underline{U}$ unique, so there is nothing to choose.

A strain measure is such a g subject to two demands: it must vanish when nothing happens, $g(1) = 0$, so that rigid motions are strain free by (3.122); and it must be increasing near $\lambda = 1$ with $g'(1) = 1$, so that it reduces to the elongation for small stretches. The standard family satisfying both is the Seth–Hill family

$$\underline{E}^{(m)} = \frac{1}{m}(\underline{U}^m - \hat{\underline{I}}) \quad (m \neq 0), \quad \underline{E}^{(0)} = \ln \underline{U} = \sum_i \ln \lambda_i \underline{u}_i \otimes \underline{u}_i,$$

which contains everything the literature uses:

$$\begin{array}{ll} m = 2 : & \frac{1}{2}(\underline{U}^2 - \hat{\underline{I}}) = \frac{1}{2}(\underline{C} - \hat{\underline{I}}) = \underline{E} \quad \text{Lagrange, (2.97),} \\ m = 1 : & \underline{U} - \hat{\underline{I}} \quad \text{Biot,} \\ m = -2 : & \frac{1}{2}(\hat{\underline{I}} - \underline{C}^{-1}) \quad \text{referential Euler,} \\ m = 0 : & \ln \underline{U} \quad \text{Hencky.} \end{array}$$

They all agree to first order, and that is why linear elasticity cannot tell them apart. Put $\underline{U} = \hat{\underline{I}} + \underline{H}$ with $\underline{H}$ small. Then $\underline{U}^m = \hat{\underline{I}} + m\underline{H} + O(|\underline{H}|^2)$ by the binomial expansion applied to each principal value, so

$$\underline{E}^{(m)} = \frac{1}{m}(m\underline{H} + O(|\underline{H}|^2)) = \underline{H} + O(|\underline{H}|^2) \quad \text{for every } m,$$

and $\ln(1 + h) = h + O(h^2)$ gives the same for $m = 0$. Every member of the family is the same tensor $\underline{\underline{H}}$ to first order. It is only in finite deformation that the choice matters, and then it matters a great deal.

Why $m = 0$ is the distinguished member. Two properties single out the logarithmic strain, and neither holds for any other m .

First, its trace is *exactly* the logarithmic volume change, with no approximation:

$$\text{tr}(\ln \underline{\underline{U}}) = \sum_i \ln \lambda_i = \ln(\lambda_1 \lambda_2 \lambda_3) = \ln \det \underline{\underline{U}} = \ln J,$$

using (3.17) and $\det \underline{\underline{U}} = \det \underline{\underline{F}}$ from (3.35) with $\det \underline{\underline{R}} = 1$. For comparison $\text{tr} \underline{\underline{E}} = \frac{1}{2}(I_1(\underline{\underline{C}}) - 3) = \frac{1}{2} \sum (\lambda_i^2 - 1)$, which is not any simple function of J . So Hencky strain, and only Hencky strain, splits cleanly into a volumetric part $\frac{1}{3}(\ln J)\hat{\underline{\underline{I}}}$ and a traceless distortional part.

Second, coaxial stretches *add*. If $\underline{\underline{U}}_2$ and $\underline{\underline{U}}_1$ share principal axes, then $\ln(\underline{\underline{U}}_2 \underline{\underline{U}}_1) = \ln \underline{\underline{U}}_1 + \ln \underline{\underline{U}}_2$, because on each axis the principal values multiply and their logarithms add. Stretching by 2 and then again by 2 gives Hencky strain $2 \ln 2$, exactly twice the strain of a single doubling, which is the property one expects of a strain and which $\underline{\underline{E}}$ conspicuously lacks: there $\frac{1}{2}(4 - 1) = 1.5$ against $\frac{1}{2}(16 - 1) = 7.5$, five times as much rather than twice. The additivity fails as soon as the axes rotate, which is why the logarithmic strain is easy in one dimension and delicate in three.

3.4 Polar decomposition theorem

THEOREM 3.19 (Polar decomposition). *Let $\underline{\underline{F}}$ have $\det \underline{\underline{F}} > 0$, regarded as a two-point tensor from $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$. Then there exist unique symmetric positive definite $\underline{\underline{U}}$ and $\underline{\underline{V}}$, and a unique rotation $\underline{\underline{R}}$, such that*

$$\underline{\underline{F}} = \underline{\underline{R}}\underline{\underline{U}} = \underline{\underline{V}}\underline{\underline{R}}. \quad (3.32)$$

In components,

$$F_{iA} = R_{iB}U_{BA} = V_{ij}R_{jA}. \quad (3.33)$$

The construction runs as follows, and each ingredient has already been prepared.

Step 1: build $\underline{U}$. By (2.85) and (2.84), $\underline{C} = \underline{F}^t \underline{F}$ is symmetric and positive definite, and by (2.86) it maps $\hat{\mathbb{E}}^3$ to itself, so it is referential. Theorem 3.16, applied in $\hat{\mathbb{E}}^3$, produces a unique symmetric positive definite $\underline{U}$ with

$$\underline{U}^2 = \underline{C}.$$

It is invertible, its principal values being positive, and by §3.3 it too is referential. Hence $\underline{R} := \underline{F} \underline{U}^{-1}$ maps $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$, a two-point tensor, and $\underline{V}$, defined below, will map $\mathbb{E}^3$ to itself.

Step 2: the determinant bookkeeping. For a linearly independent $\{\underline{a}, \underline{b}, \underline{c}\}$ in $\hat{\mathbb{E}}^3$, using linearity of the box product and $\underline{U}\underline{a}, \underline{U}\underline{b}, \underline{U}\underline{c} \in \hat{\mathbb{E}}^3$,

$$\begin{aligned} (\det \underline{F})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{R}(\underline{U}\underline{a}), \underline{R}(\underline{U}\underline{b}), \underline{R}(\underline{U}\underline{c})] \\ &= (\det \underline{R})[\underline{U}\underline{a}, \underline{U}\underline{b}, \underline{U}\underline{c}] \\ &= (\det \underline{R})(\det \underline{U})[\underline{a}, \underline{b}, \underline{c}], \end{aligned} \tag{3.34}$$

so that, cancelling the non-zero box product,

$$\det \underline{F} = (\det \underline{R})(\det \underline{U}). \tag{3.35}$$

Since $\det \underline{U} = \lambda_1 \lambda_2 \lambda_3 > 0$ by (3.17) and $\det \underline{F} > 0$ by hypothesis,

$$\det \underline{R} > 0. \tag{3.36}$$

The book disposes of the other factorisation with “in the same way we easily establish”. Written out, it is the same three lines with the factors interchanged. Using (3.32)₂, and noting that $\underline{R}\underline{a}, \underline{R}\underline{b}, \underline{R}\underline{c}$ all lie in $\mathbb{E}^3$, which is what makes the middle box product legitimate,

$$\begin{aligned} (\det \underline{F})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{V}(\underline{R}\underline{a}), \underline{V}(\underline{R}\underline{b}), \underline{V}(\underline{R}\underline{c})] \\ &= (\det \underline{V})[\underline{R}\underline{a}, \underline{R}\underline{b}, \underline{R}\underline{c}] \\ &= (\det \underline{V})(\det \underline{R})[\underline{a}, \underline{b}, \underline{c}], \end{aligned}$$

so $\det \underline{F} = (\det \underline{V})(\det \underline{R})$. Comparing with (3.35) and cancelling the non-zero $\det \underline{R}$ gives the by-product $\det \underline{V} = \det \underline{U}$, which is not obvious

from the definitions and will be confirmed at (3.42), where $\underline{U}$ and $\underline{V}$ are found to have the same principal values. Since $\det \underline{F} > 0$ and $\det \underline{R} > 0$, also $\det \underline{V} > 0$.

REMARK 3.20 (an extension smuggled in). Problem 21(b) of Chapter 1 proved $\det(\underline{A}\underline{B}) = \det \underline{A} \det \underline{B}$ for tensors of a single space. Equation (3.34) silently extends it to the case where the first factor is two-point and the second referential. The extension is legitimate for exactly the reason given after (2.47): at each step the three vectors inside a box product belong to a single space, so every box product written is defined. Nothing else is needed.

Step 3: $\underline{R}$ is orthogonal. Using $(\underline{A}\underline{B})^t = \underline{B}^t \underline{A}^t$ from Problem 20, $\underline{U}^{-t} = \underline{U}^{-1}$ from the symmetry of $\underline{U}$ (its inverse is symmetric because its spectral form has the same axes, see Remark 3.10), and $\underline{U}\underline{U} = \underline{C}$,

$$\underline{R}^t \underline{R} = \underline{U}^{-t} \underline{F}^t \underline{F} \underline{U}^{-1} = \underline{U}^{-1} \underline{C} \underline{U}^{-1} = \underline{U}^{-1} \underline{U} \underline{U} \underline{U}^{-1} = \underline{\hat{I}}. \quad (3.37)$$

So $\underline{R}$ is orthogonal, with $\underline{R}^{-1} = \underline{R}^t : \mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$.

Step 4: rotate the principal frame. Write the spectral forms of $\underline{U}$ and, by squaring, of $\underline{C}$:

$$\underline{U} = \sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i, \quad \underline{C} = \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i, \quad (3.38)$$

with $\lambda_i > 0$ the *principal stretches* and $\underline{u}_i \in \hat{\mathbb{E}}^3$ orthonormal. Set

$$\underline{v}_i = \underline{R} \underline{u}_i \in \mathbb{E}^3. \quad (3.39)$$

These are orthonormal:

$$\underline{v}_i \cdot \underline{v}_j = \underline{R} \underline{u}_i \cdot \underline{R} \underline{u}_j = \underline{u}_i \cdot \underline{R}^t \underline{R} \underline{u}_j = \underline{u}_i \cdot \underline{\hat{I}} \underline{u}_j = \underline{u}_i \cdot \underline{u}_j = \delta_{ij}. \quad (3.40)$$

Consequently, by (3.11) applied in $\mathbb{E}^3$ and then Problems 17(b),(c) of Chapter 1 in the combined form $(\underline{R}\underline{a}) \otimes (\underline{R}\underline{b}) = \underline{R}(\underline{a} \otimes \underline{b})\underline{R}^t$,

$$\underline{I} = \underline{v}_i \otimes \underline{v}_i = (\underline{R} \underline{u}_i) \otimes (\underline{R} \underline{u}_i) = \underline{R}(\underline{u}_i \otimes \underline{u}_i) \underline{R}^t = \underline{R} \underline{\hat{I}} \underline{R}^t = \underline{R} \underline{R}^t. \quad (3.41)$$

The intermediate identity deserves its own line, since it is used again in §3.5. For any $\underline{w} \in \mathbb{E}^3$,

$$\begin{aligned} [(\underline{Ra}) \otimes (\underline{Rb})]\underline{w} &= (\underline{Rb} \cdot \underline{w}) \underline{Ra} = (\underline{b} \cdot \underline{R}^t \underline{w}) \underline{Ra} \\ &= \underline{R}[(\underline{a} \otimes \underline{b})\underline{R}^t \underline{w}] = [\underline{R}(\underline{a} \otimes \underline{b})\underline{R}^t]\underline{w}. \end{aligned}$$

Step 5: build $\underline{V}$. Take (3.32)₂ as the definition of $\underline{V}$, i.e. $\underline{V} := \underline{F}\underline{R}^t$; then $\underline{V} = \underline{V}\underline{I} = \underline{V}\underline{R}\underline{R}^t = \underline{F}\underline{R}^t$ is consistent, and

$$\begin{aligned} \underline{V} &= \underline{R}\underline{U}\underline{R}^t = \underline{R}\left(\sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i\right)\underline{R}^t \\ &= \sum_i \lambda_i (\underline{R}\underline{u}_i) \otimes (\underline{R}\underline{u}_i) = \sum_i \lambda_i \underline{v}_i \otimes \underline{v}_i. \end{aligned} \tag{3.42}$$

This is manifestly symmetric and positive definite, and it exhibits the spectral decomposition $\underline{B} = \underline{V}^2 = \sum_i \lambda_i^2 \underline{v}_i \otimes \underline{v}_i$ of the left Cauchy–Green tensor. Note that $\underline{U}$ and $\underline{V}$ have the *same* principal values and differently placed axes. Also

$$\underline{F} = \underline{R}\underline{U} = \underline{R}\left(\sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i\right) = \sum_i \lambda_i (\underline{R}\underline{u}_i) \otimes \underline{u}_i = \sum_i \lambda_i \underline{v}_i \otimes \underline{u}_i. \tag{3.43}$$

Step 6: $\det \underline{R} = 1$. Take the principal triad $\{\underline{u}_i\}$ of $\underline{U}$ as the test triad in the two-point determinant (2.47), which is allowed because it is a basis of $\hat{\mathbb{E}}^3$ and hence independent. By the definition (3.39) of $\underline{v}_i$,

$$(\det \underline{R})[\underline{u}_1, \underline{u}_2, \underline{u}_3] = [\underline{R}\underline{u}_1, \underline{R}\underline{u}_2, \underline{R}\underline{u}_3] = [\underline{v}_1, \underline{v}_2, \underline{v}_3].$$

Both triads are orthonormal, $\{\underline{u}_i\}$ by Theorem 3.7 and $\{\underline{v}_i\}$ by (3.40), so by Lemma 3.28

$$[\underline{u}_1, \underline{u}_2, \underline{u}_3] = \pm 1, \quad [\underline{v}_1, \underline{v}_2, \underline{v}_3] = \pm 1,$$

each sign being + or – according to the handedness of its triad. Substituting, $\det \underline{R} = (\pm 1)/(\pm 1)$, so $\det \underline{R} = \pm 1$ and the value is +1 exactly when the two triads have the *same* handedness. But (3.36) has already established $\det \underline{R} > 0$, which rules out –1. Therefore

$$\det \underline{R} = +1,$$

and as a by-product the two triads must have the same handedness: $\underline{R}$ cannot turn a right-handed frame into a left-handed one. Thus $\underline{R}$ is a two-point rotation and Theorem 3.19 is proved apart from uniqueness.

Step 7: uniqueness. Suppose $\underline{F} = \underline{R}\underline{U} = \underline{R}'\underline{U}'$ with $\underline{U}, \underline{U}'$ symmetric positive definite and $\underline{R}, \underline{R}'$ rotations. Then

$$\underline{C} = \underline{F}^t \underline{F} = (\underline{R}'\underline{U}')^t (\underline{R}'\underline{U}') = \underline{U}'^t \underline{R}'^t \underline{R}' \underline{U}' = \underline{U}'^t \hat{I} \underline{U}' = \underline{U}'^2,$$

so $\underline{U}'$ is a symmetric positive definite square root of $\underline{C}$; by Theorem 3.16 it equals $\underline{U}$. Then $\underline{R}' = \underline{F}\underline{U}'^{-1} = \underline{F}\underline{U}^{-1} = \underline{R}$. Finally $\underline{V}' = \underline{F}\underline{R}'^t = \underline{F}\underline{R}^t = \underline{V}$. The three factors are therefore uniquely determined by $\underline{F}$, and only the positive definiteness of $\underline{U}$ was used to get there. $\square$

REMARK 3.21 (what the theorem says in words). $\underline{F} = \underline{R}\underline{U}$ says: *first* stretch by λ_i along the three referential axes $\underline{u}_i$, *then* rotate the result rigidly. $\underline{F} = \underline{V}\underline{R}$ says: *first* rotate, *then* stretch by the same three factors along the rotated axes $\underline{v}_i = \underline{R}\underline{u}_i$. The order can be exchanged provided the axes are carried along, and the stretches themselves are the same numbers in both readings. The physically important half is $\underline{U}$, or equivalently $\underline{C} = \underline{U}^2$: it is the part that a material feels. The rotation $\underline{R}$ costs nothing, which is why constitutive laws are written in terms of $\underline{C}$ and never in terms of $\underline{F}$ alone.

REMARK 3.22 (why it is called *polar*). The name is never explained in the text and is not decoration. Every non-zero complex number has exactly one representation

$$z = r e^{i\vartheta}, \quad r > 0,$$

as a positive magnitude times a unit-modulus factor: the polar form. Theorem 3.19 is the same statement for tensors, item by item:

$$\begin{array}{llll} \text{the object:} & z \neq 0 & \longleftrightarrow & \underline{F}, \det \underline{F} > 0 \\ \text{the magnitude:} & r > 0 & \longleftrightarrow & \underline{U} \text{ sym. pos. def.} \\ \text{the phase:} & |e^{i\vartheta}| = 1 & \longleftrightarrow & \underline{R}^t \underline{R} = \hat{I} \\ \text{how it is got:} & r = \sqrt{\bar{z}z} & \longleftrightarrow & \underline{U} = \sqrt{\underline{F}^t \underline{F}} \\ \text{rigidity:} & |z| = 1 & \longleftrightarrow & \underline{U} = \hat{I}, \quad (3.122) \end{array}$$

Conjugation $z \mapsto \bar{z}$ corresponds to transposition $\underline{F} \mapsto \underline{F}^t$, and $\bar{z}z = |z|^2$ to $\underline{F}^t \underline{F} = \underline{C} = \underline{U}^2$. Even the proof runs parallel: one forms $\bar{z}z$ to eliminate the phase, takes the positive square root, and divides it out. That is exactly Steps 1 and 3 of §3.4.

The analogy also shows where tensors are harder, and the difference is worth stating because it is the source of most of the difficulty in finite elasticity. Complex multiplication commutes, so $re^{i\vartheta} = e^{i\vartheta}r$ and there is only *one* polar form. Tensor multiplication does not, so $\underline{R}\underline{U} \neq \underline{U}\underline{R}$ in general and there are *two*, $\underline{F} = \underline{R}\underline{U}$ and $\underline{F} = \underline{V}\underline{R}$, with $\underline{V} = \underline{R}\underline{U}\underline{R}^t \neq \underline{U}$ unless $\underline{R}$ happens to fix the principal axes. Nothing in the complex case prepares one for the fact that “stretch then rotate” and “rotate then stretch” require different stretch tensors. Note also that $\underline{U}$ and $\underline{V}$ are not independent: by (3.42) they have the same principal values, only differently placed, so there is exactly one set of three stretches and two ways to say where they act.

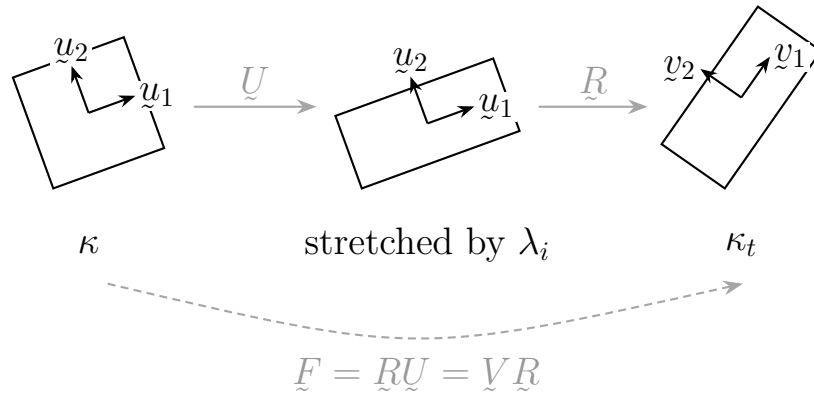


Figure A: The polar decomposition. Reading $\underline{F} = \underline{R}\underline{U}$, the referential principal axes $\underline{u}_i$ are stretched by λ_i and the stretched configuration is then rotated rigidly onto κ_t . Reading $\underline{F} = \underline{V}\underline{R}$ reverses the order and stretches along the rotated axes $\underline{v}_i = \underline{R}\underline{u}_i$ instead, by the same factors. The two routes reach the same κ_t , and by Step 7 the factors are unique.

Principal stretches are stretches: the link back to (2.67)

Nothing so far has said why the numbers λ_i deserve the name *stretch*. The following computation supplies the reason and, at the same time, forces us to be careful about what a material vector is.

Fix an instant t_* and a material point with referential position $\underline{X}$. The principal vectors $\underline{u}_i$ of $\underline{U}(\underline{X}, t_*)$ are three directions in $\hat{\mathbb{E}}^3$. Since $\hat{\mathbb{E}}^3$ is where material directions also live, we may select a material curve through $\underline{X}$ whose unit tangent $\underline{M} = \underline{X}'(S)$ happens to equal a given $\underline{u} \in \{\underline{u}_i\}$ at that point and at that instant. For this curve, (2.67) reads $\mu \underline{m} = \underline{F} \underline{M}$, so using (3.32)₁, (3.38)₁ and (3.39),

$$\begin{aligned} \mu \underline{m} &= \underline{F} \underline{M} = \underline{F} \underline{u} = \underline{R}(\underline{U} \underline{u}) \\ &= \underline{R}(\lambda \underline{u}) \\ &= \lambda \underline{R} \underline{u} \\ &= \lambda \underline{v}, \end{aligned} \tag{3.44}$$

where λ is the principal value belonging to $\underline{u}$ and $\underline{v} \in \{\underline{v}_i\}$. Since $|\underline{v}| = 1$ and $\mu > 0$, comparing lengths gives $\mu = \lambda$ and comparing directions gives $\underline{m} = \underline{v}$. So: *a material fibre momentarily aligned with the i th principal direction of $\underline{U}$ stretches by exactly λ_i , and ends up aligned with $\underline{v}_i$* . That is the justification for the name.

The same conclusion from the other factorisation:

$$\mu \underline{m} = \underline{F} \underline{M} = \underline{F} \underline{u} = \underline{V}(\underline{R} \underline{u}) = \underline{V} \underline{v} = \lambda \underline{v}, \tag{3.45}$$

using (3.42) and $\underline{V} \underline{v}_i = \lambda_i \underline{v}_i$. Line (3.44) reads the deformation as *stretch then rotate*, line (3.45) as *rotate then stretch*.

Computing $\underline{U}$ and $\underline{R}$ without eigenvectors

Theorem 3.19 proves that $\underline{U}$ exists but constructs it through (3.28), which needs the principal triad. Finding that triad means solving a cubic and then three linear systems, and for a $\underline{U}$ that varies from point to point, as in §§3.5.2–3.5.4, that is unusable. Problem 9 of §3.8 gives a

closed form in two dimensions. Here is the three-dimensional one. It is not in the book, and it makes every example of §3.5 a finite computation.

THEOREM 3.23 (Closed form for the right stretch tensor). *Let $\underline{C} = \underline{F}^t \underline{F}$ have principal invariants I_1, I_2, I_3 , and let $i_1 = \text{tr } \underline{U}$, $i_2, i_3 = \det \underline{U}$ be those of $\underline{U} = \sqrt{\underline{C}}$. Then*

$$\underline{U} = \frac{1}{i_1 i_2 - i_3} \left[-\underline{C}^2 + (i_1^2 - i_2) \underline{C} + i_1 i_3 \hat{\underline{I}} \right], \quad (*_1)$$

where $i_3 = \sqrt{I_3}$, $i_2 = \frac{1}{2}(i_1^2 - I_1)$, and i_1 is the largest real root of

$$i_1^4 - 2I_1 i_1^2 - 8\sqrt{I_3} i_1 + (I_1^2 - 4I_2) = 0. \quad (*_2)$$

The rotation then follows from $\underline{R} = \underline{F} \underline{U}^{-1}$, with $\underline{U}^{-1}$ supplied by Cayley–Hamilton as $\underline{U}^{-1} = (\underline{C} - i_1 \underline{U} + i_2 \hat{\underline{I}})/i_3$.

Proof. The invariant relations. Since $\underline{U}$ and $\underline{C} = \underline{U}^2$ share principal axes with values λ_i and λ_i^2 , Problem 7 of §3.8 gives

$$I_1 = \sum_i \lambda_i^2, \quad I_2 = \sum_{i < j} \lambda_i^2 \lambda_j^2, \quad I_3 = (\lambda_1 \lambda_2 \lambda_3)^2,$$

while $i_1 = \sum \lambda_i$, $i_2 = \sum_{i < j} \lambda_i \lambda_j$, $i_3 = \lambda_1 \lambda_2 \lambda_3$. Squaring i_1 and subtracting twice i_2 ,

$$i_1^2 - 2i_2 = \left(\sum_i \lambda_i \right)^2 - 2 \sum_{i < j} \lambda_i \lambda_j = \sum_i \lambda_i^2 = I_1,$$

the cross terms cancelling exactly; squaring i_2 and subtracting $2i_1 i_3$ gives I_2 by the identical computation one level up; and $i_3^2 = I_3$ is immediate. Hence

$$i_3 = \sqrt{I_3} > 0, \quad i_2 = \frac{1}{2}(i_1^2 - I_1), \quad i_2^2 = I_2 + 2i_1 i_3,$$

the positive root being forced by $\lambda_i > 0$. Substituting the middle relation into the last,

$$\frac{1}{4}(i_1^2 - I_1)^2 = I_2 + 2i_1 \sqrt{I_3},$$

and clearing the fraction gives $(*_2)$. Of its real roots, i_1 is the sum of three positive numbers and is the largest.

The formula. Apply Cayley–Hamilton, Problem 7(b), to $\underline{U}$:

$$\underline{U}^3 - i_1 \underline{U}^2 + i_2 \underline{U} - i_3 \hat{\underline{I}} = \hat{\underline{O}}.$$

Multiply by $\underline{U}$ and use $\underline{U}^2 = \underline{C}$ repeatedly to eliminate all odd powers except one:

$$\underline{U}^4 - i_1 \underline{U}^3 + i_2 \underline{U}^2 - i_3 \underline{U} = \hat{\underline{O}}, \quad \text{i.e.} \quad \underline{C}^2 - i_1 \underline{U} \underline{C} + i_2 \underline{C} - i_3 \underline{U} = \hat{\underline{O}},$$

since $\underline{U}^4 = \underline{C}^2$ and $\underline{U}^3 = \underline{U} \underline{C}$. From the first display, $\underline{U} \underline{C} = \underline{U}^3 = i_1 \underline{C} - i_2 \underline{U} + i_3 \hat{\underline{I}}$; substituting,

$$\underline{C}^2 - i_1 (i_1 \underline{C} - i_2 \underline{U} + i_3 \hat{\underline{I}}) + i_2 \underline{C} - i_3 \underline{U} = \hat{\underline{O}},$$

that is

$$(i_1 i_2 - i_3) \underline{U} = -\underline{C}^2 + (i_1^2 - i_2) \underline{C} + i_1 i_3 \hat{\underline{I}},$$

which is $(*_1)$. The coefficient never vanishes: expanded, $i_1 i_2 - i_3 = (\lambda_1 + \lambda_2)(\lambda_2 + \lambda_3)(\lambda_3 + \lambda_1)$, a product of three positive numbers.

The inverse. Divide the Cayley–Hamilton identity by $i_3 \neq 0$ and rearrange, using $\underline{U}^2 = \underline{C}$ again: $\underline{U}^{-1} = (\underline{U}^2 - i_1 \underline{U} + i_2 \hat{\underline{I}})/i_3 = (\underline{C} - i_1 \underline{U} + i_2 \hat{\underline{I}})/i_3$. $\square$

REMARK 3.24 (the same statement one dimension down). Run the proof in two dimensions and it collapses to Problem 9. There Cayley–Hamilton reads $\underline{U}^2 - I \underline{U} + J \hat{\underline{I}} = \hat{\underline{O}}$ with $I = i_1$, $J = i_2$, and $\underline{U}^2 = \underline{C}$ turns it directly into $I \underline{U} = \underline{C} + J \hat{\underline{I}}$ with no need to multiply by $\underline{U}$ at all; the quartic $(*_2)$ degenerates to $I^2 = \text{tr } \underline{C} + 2J$ and $J = \sqrt{\det \underline{C}}$. The extra work in three dimensions is entirely the price of the third invariant.

Two checks on $(*_1)$. Put $\underline{C} = \hat{\underline{I}}$: then $I_1 = 3$, $I_2 = 3$, $I_3 = 1$, and $(*_2)$ becomes $i_1^4 - 6i_1^2 - 8i_1 + 3 = 0$, whose largest root is $i_1 = 3$; then $i_2 = \frac{1}{2}(9 - 3) = 3$, $i_3 = 1$, and $(*_1)$ gives $\underline{U} = \frac{1}{9-1}[-\hat{\underline{I}} + 6\hat{\underline{I}} + 3\hat{\underline{I}}] = \frac{8}{8}\hat{\underline{I}} = \hat{\underline{I}}$, as it must. Put $\underline{C} = \text{diag}(\lambda_1^2, \lambda_2^2, \lambda_3^2)$: every tensor in $(*_1)$ is diagonal, and the k th entry of the right side is $[-\lambda_k^4 + (i_1^2 - i_2)\lambda_k^2 + i_1 i_3]/(i_1 i_2 - i_3)$, which equals λ_k precisely because λ_k^4 can be reduced by the characteristic equation of $\underline{U}$, $\lambda_k^3 = i_1 \lambda_k^2 - i_2 \lambda_k + i_3$, multiplied by λ_k .

Material vectors

The word “material” has been used repeatedly since Chapter 2 and never defined sharply. It is the hinge of the argument just given, so it is worth pinning down completely.

DEFINITION 3.25 (Material vector). A vector field $\underline{M}$ attached to the body is *material* if, expressed as a function of the referential position and time, it does not depend on time:

$$\hat{\underline{M}} = \hat{\underline{M}}(\underline{X}), \quad \text{equivalently} \quad \dot{\underline{M}} = \frac{\partial \hat{\underline{M}}}{\partial t} \Big|_{\underline{X}} = \underline{0}.$$

REMARK 3.26 (what a material vector really is, with examples). The definition looks like a triviality and is not. Here is the content.

A material vector is a *label glued to matter*. It is decided once and for all in the reference configuration and then never touched again. It does not move, because κ does not move; it is $\underline{X}$ -dependent data, like a serial number painted on a fibre. What moves is its *image* $\underline{F}\underline{M}$, and that is a different object living in a different space.

The test is mechanical: write the field as a function of $(\underline{X}, t)$ and look for a t . If there is none, the field is material.

EXAMPLES OF MATERIAL VECTORS.

(i) *The tangent to a material curve.* Let $\mathcal{C} \subset \kappa$ be a curve consisting of a fixed set of material points, with arclength parametrisation $\underline{X}(S)$, and let $\underline{M}(S) = \underline{X}'(S)$ as in (2.66). Since $\mathcal{C}$ is a fixed subset of the fixed configuration κ , the function $\underline{X}(S)$ carries no t , so neither does $\underline{M}$. Material.

(ii) *The normal to a material surface.* If $d\underline{X}$ and $d\underline{Y}$ span the tangent plane of a material surface at $\underline{X}$, then $\underline{N} dA = d\underline{X} \times d\underline{Y}$ of (2.51) involves only referential data. Material.

(iii) *The referential basis $\{\hat{\underline{E}}_A\}$ and the position $\underline{X}$.* Both are fixed by hypothesis. Material, trivially.

(iv) *A fibre direction in a fibre-reinforced solid.* The direction in which the reinforcing fibres were laid up is decided when the material is

manufactured, that is, in κ . Material. This is the example that makes the concept indispensable: an anisotropic constitutive law must be written in terms of such a vector, and it must be a *referential* one, or the law would change when the body is merely carried across the room.

EXAMPLES THAT ARE *not* MATERIAL VECTORS.

(v) *The current tangent* $\underline{m}$. By (2.67), $\underline{m} = \underline{F}\underline{M}/\mu$ with $\underline{F} = \underline{F}(\underline{X}, t)$; the t is right there. Not material. It also lives in $\mathbb{E}^3$ rather than $\hat{\mathbb{E}}^3$, so it could not be material in the sense of Definition 3.25 even in principle.

(vi) *The current normal* $\underline{n}$. By Piola–Nanson (2.54), $\underline{n} \, da = \underline{F}^* \underline{N} \, dA$, again t -dependent. Not material.

(vii) *The principal directions* $\underline{u}_i$ of $\underline{U}$. These are referential, so they at least live in the right space, but they are properties of $\underline{U}(\underline{X}, t)$ and therefore inherit its time dependence. Not material. This is the case the book flags immediately after (3.44), and it is genuinely subtle, because $\underline{u}_i$ passes the “lives in $\hat{\mathbb{E}}^3$ ” test and fails only the “no t ” test. Remark 3.27 computes $\dot{\underline{u}}_1 \neq \underline{0}$ explicitly.

The algebraic consequence of being material is worth stating once. If $\underline{M}$ is material then, from (2.67),

$$\left. \frac{\partial}{\partial t} \right|_{\underline{X}} (\mu \underline{m}) = \left. \frac{\partial}{\partial t} \right|_{\underline{X}} (\underline{F} \underline{M}) = \dot{\underline{F}} \underline{M},$$

with no term $\underline{F} \dot{\underline{M}}$, because $\dot{\underline{M}} = \underline{0}$. Every rate formula for material fibres in Chapter 4 rests on the disappearance of that second term. If $\underline{M}$ were allowed to depend on time the formulas of Chapter 4 would all acquire an extra term and would describe nothing physical, since one would be comparing a different fibre at each instant.

REMARK 3.27 (the principal directions are not material: the algebra). Take the simple shear of §3.5.1 below and let the amount of shear grow with time, $\gamma = \gamma(t)$, which is what happens in any actual shearing experiment. By (3.62) and (3.64) the first principal direction of $\underline{U}$ is

$$\underline{u}_1 = \cos A \, \hat{\underline{E}}_1 + \sin A \, \hat{\underline{E}}_2, \quad \tan A = \frac{1}{2}(\gamma + \sqrt{4 + \gamma^2}).$$

Differentiate at fixed $\tilde{X}$, remembering that $\hat{\tilde{E}}_A$ is constant:

$$\dot{\underline{u}}_1 = \dot{A}(-\sin A \hat{\tilde{E}}_1 + \cos A \hat{\tilde{E}}_2) = \dot{A} \underline{u}_2, \quad \dot{A} = \frac{dA}{d\gamma} \dot{\gamma}.$$

From $\tan A = \frac{1}{2}(\gamma + \sqrt{4 + \gamma^2})$,

$$\sec^2 A \frac{dA}{d\gamma} = \frac{1}{2} \left(1 + \frac{\gamma}{\sqrt{4 + \gamma^2}} \right) = \frac{\sqrt{4 + \gamma^2} + \gamma}{2\sqrt{4 + \gamma^2}} = \frac{\tan A}{\sqrt{4 + \gamma^2}},$$

so $dA/d\gamma = \sin A \cos A / \sqrt{4 + \gamma^2} > 0$ for $A \in (0, \pi/2)$. Hence $\dot{\underline{u}}_1 = \dot{A} \underline{u}_2 \neq \underline{0}$ whenever $\dot{\gamma} \neq 0$: the principal axes rotate through the material as the shear proceeds. At $\gamma = 0$ the rate is $dA/d\gamma = (\frac{1}{\sqrt{2}})(\frac{1}{\sqrt{2}})/2 = \frac{1}{4}$, so $A \approx \pi/4 - \dots$ turns at a quarter of the rate of γ .

Contrast this with $\underline{M} = \hat{\tilde{E}}_1$, which is material. Its image under the deformation is, from (3.51),

$$\underline{F} \hat{\tilde{E}}_1 = \underline{\tilde{E}}_1 + \gamma \underline{e}_1 (\hat{\tilde{E}}_2 \cdot \hat{\tilde{E}}_1) = \underline{e}_1 + \gamma \underline{e}_1 \cdot \underline{0} = \underline{e}_1, \quad \mu = |\underline{e}_1| = 1.$$

So the X_1 fibres neither stretch nor turn, at any γ : they are the fibres lying in the shearing direction. Meanwhile $\underline{M} = \hat{\tilde{E}}_2$, also material, has

$$\underline{F} \hat{\tilde{E}}_2 = \underline{e}_2 + \gamma \underline{e}_1, \quad \mu = \sqrt{1 + \gamma^2}, \quad \underline{m} = \frac{\gamma \underline{e}_1 + \underline{e}_2}{\sqrt{1 + \gamma^2}},$$

so those fibres stretch and tilt. Two material fibres that were perpendicular are no longer so: by (2.106),

$$\begin{aligned} \cos \theta &= \frac{\hat{\tilde{E}}_1 \cdot \underline{C} \hat{\tilde{E}}_2}{\sqrt{(\hat{\tilde{E}}_1 \cdot \underline{C} \hat{\tilde{E}}_1)(\hat{\tilde{E}}_2 \cdot \underline{C} \hat{\tilde{E}}_2)}} = \frac{C_{12}}{\sqrt{C_{11}C_{22}}} \\ &= \frac{\gamma}{\sqrt{1 \cdot (1 + \gamma^2)}} = \frac{\gamma}{\sqrt{1 + \gamma^2}}, \end{aligned}$$

using the matrix (3.55); the same number follows directly from $\underline{e}_1 \cdot \underline{m}$ above, which is the point of (2.106).

The moral: *material* directions are the ones you may follow through the motion and still be talking about the same matter; *principal* directions are the ones along which, at one instant, the stretching is extremal. They coincide instant by instant, never as fields.

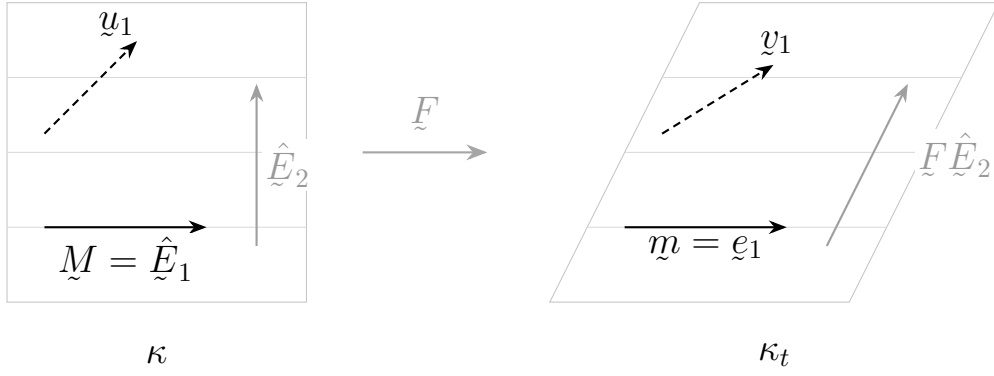


Figure B: Material versus principal directions in simple shear. The solid arrow is the material fibre $\hat{E}_1$: it is decided in κ and its image is $\underline{e}_1$, unstretched and unturned for every γ . The grey arrow is the material fibre $\hat{E}_2$, whose image tilts and lengthens by $\sqrt{1 + \gamma^2}$. The dashed arrow is the principal direction $\underline{u}_1$: it is *not* attached to any fixed matter, and as γ grows it sweeps across the fibres at the rate computed in Remark 3.27.

3.5 Examples

Four deformations are now carried through completely. The first is homogeneous and is the standard test case for every idea in the chapter; the other three are cylindrical and are the standard test cases for elasticity. In each, the recipe is the same and is worth memorising: *compute $d\tilde{x}$ in the natural coordinates, express the coordinate increments as dot products with $d\tilde{X}$, read off $\tilde{F}$ by comparison with $d\tilde{x} = \tilde{F}d\tilde{X}$, then form $\underline{C}$, $\underline{B}$, $\underline{U}$, $\underline{V}$, $\underline{R}$.*

The four rules used on every page below

Everything in §3.5 is built from four facts, all from Chapter 1. They are collected here so that the computations can quote them by name, and each is then still carried out in full at its point of use.

(R1) A dyad acting on a vector. $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$. The *second* slot does the dotting; the first slot is what comes out. So on an orthonormal basis $(\underline{a} \otimes \hat{\underline{E}}_B)\hat{\underline{E}}_A = \delta_{AB}\underline{a}$: a dyad picks out one basis vector and ignores the others.

(R2) A dyad times a dyad. $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c})\underline{a} \otimes \underline{d}$. The

two *inner* slots dot; the two outer slots survive. So a product of two three-term sums has nine terms, and orthonormality kills six of them.

(R3) The box product. $[\underline{a}, \underline{b}, \underline{c}] = \underline{a} \cdot \underline{b} \times \underline{c} = \varepsilon_{ijk} a_i b_j c_k$. It is *linear in each slot separately*, so scalars come out one at a time; it is *alternating*, so it vanishes whenever two slots are equal and changes sign when two are swapped; and a cyclic permutation, being two swaps, leaves it unchanged.

(R4) The determinant. For a two-point $\tilde{F}$, equation (2.47) gives

$$\det \tilde{F} = \frac{[\tilde{F}\underline{a}, \tilde{F}\underline{b}, \tilde{F}\underline{c}]}{[\underline{a}, \underline{b}, \underline{c}]}$$

for any independent $\underline{a}, \underline{b}, \underline{c}$ in the referential space. Both box products are then formed from vectors of a single space, which is what makes the ratio meaningful.

LEMMA 3.28 (Box product of an orthonormal triad). *If $\{\underline{a}, \underline{b}, \underline{c}\}$ is orthonormal, then $[\underline{a}, \underline{b}, \underline{c}] = +1$ when the triad is right handed and -1 when it is left handed.*

Proof. Right handed means, by definition, $\underline{b} \times \underline{c} = \underline{a}$. Then by (R3)

$$[\underline{a}, \underline{b}, \underline{c}] = \underline{a} \cdot \underline{b} \times \underline{c} = \underline{a} \cdot \underline{a} = |\underline{a}|^2 = 1.$$

Left handed means $\underline{b} \times \underline{c} = -\underline{a}$, and the same line gives -1 . $\square$

In particular $[\underline{e}_1, \underline{e}_2, \underline{e}_3] = 1$ and $[\hat{\underline{E}}_1, \hat{\underline{E}}_2, \hat{\underline{E}}_3] = 1$, both bases being right handed by assumption. The cylindrical frames need one line of checking, and it is done once here.

LEMMA 3.29 (The polar frames are right-handed and orthonormal). *With $\underline{e}_r(\theta) = \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2$ and $\underline{e}_\theta(\theta) = \underline{e}'_r(\theta) = -\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2$,*

$$\underline{e}_r \times \underline{e}_\theta = \underline{e}_3, \quad [\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = 1,$$

and likewise $\hat{\underline{E}}_R \times \hat{\underline{E}}_\Theta = \hat{\underline{E}}_3$, $[\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta, \hat{\underline{E}}_3] = 1$.

Proof. Orthonormality was checked in Problem 25 of Chapter 1; $|\underline{e}_r|^2 = \cos^2 \theta + \sin^2 \theta = 1$, similarly for $\underline{e}_\theta$, and $\underline{e}_r \cdot \underline{e}_\theta = -\cos \theta \sin \theta + \sin \theta \cos \theta = 0$. For the cross product, expand bilinearly, keeping all four terms:

$$\begin{aligned} \underline{e}_r \times \underline{e}_\theta &= (\cos \theta \underline{e}_1 + \sin \theta \underline{e}_2) \times (-\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2) \\ &= -\cos \theta \sin \theta (\underline{e}_1 \times \underline{e}_1) + \cos^2 \theta (\underline{e}_1 \times \underline{e}_2) \\ &\quad - \sin^2 \theta (\underline{e}_2 \times \underline{e}_1) + \sin \theta \cos \theta (\underline{e}_2 \times \underline{e}_2). \end{aligned}$$

The first and last vanish because $\underline{v} \times \underline{v} = \underline{0}$. In the remaining two, $\underline{e}_1 \times \underline{e}_2 = \underline{e}_3$ and $\underline{e}_2 \times \underline{e}_1 = -\underline{e}_3$, so they contribute $\cos^2 \theta \underline{e}_3 + \sin^2 \theta \underline{e}_3 = \underline{e}_3$. Hence, by (R3) with a cyclic shift,

$$[\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = \underline{e}_3 \cdot \underline{e}_r \times \underline{e}_\theta = \underline{e}_3 \cdot \underline{e}_3 = 1.$$

The referential computation is the same with hats and Θ . □

3.5.1 Simple shear

What this example is for. It is the simplest deformation that is neither a pure stretch nor a pure rotation, so it is the smallest case in which the polar decomposition has anything to say. Everything in §§3.1–3.4 gets used here at least once: the characteristic equation at (3.58), the extremal property implicitly at (3.60), the square root at $\underline{U} = \sqrt{\underline{C}}$, and the uniqueness of $\underline{R}$ at (3.70). It is also homogeneous, so §3.6 applies to it, and isochoric, so the constraint (3.57) cuts the work in half. The one surprise it contains is Remark 3.33: the material rotates *backwards*, and seeing why is the point of the whole calculation.

Consider the deformation

$$\underline{x} = \underline{1}\underline{X} + \underline{u}, \tag{3.46}$$

where $\underline{1}$ is the shifter of (2.68) from $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$ and the displacement $\underline{u}$ of (2.77) is

$$\underline{u} = \gamma(\underline{E}_2 \cdot \underline{1}\underline{X})\underline{e}_1 = \gamma(\underline{1}^t \underline{E}_2 \cdot \underline{X})\underline{e}_1 = \gamma(\hat{\underline{E}}_2 \cdot \underline{X})\underline{e}_1, \tag{3.47}$$

in which γ , the *amount of shear*, is a dimensionless constant. The middle step moves $\underline{1}$ across the dot product by the definition of the transpose, and the last uses $\underline{1}^t \underline{E}_2 = \underline{1}^t \underline{1} \hat{\underline{E}}_2 = \hat{\underline{I}} \hat{\underline{E}}_2 = \hat{\underline{E}}_2$, from (2.72).

The Cartesian coordinates after deformation are $x_i = \underline{e}_i \cdot \underline{x}$, so

$$x_1 = \underline{e}_1 \cdot \underline{1}X + \gamma \underline{E}_2 \cdot \underline{1}X, \quad x_2 = \underline{e}_2 \cdot \underline{1}X, \quad x_3 = \underline{e}_3 \cdot \underline{1}X. \quad (3.48)$$

Using $\underline{1}X = X_A \underline{E}_A$, which is (2.75), and choosing $\{\underline{E}_A\} = \{\underline{e}_i\}$ so that $\underline{e}_i \cdot \underline{E}_A = \delta_{iA}$,

$$x_1 = X_1 + \gamma X_2, \quad x_2 = X_2, \quad x_3 = X_3. \quad (3.49)$$

REMARK 3.30 (reading (3.49)). Planes $X_2 = \text{const}$ slide rigidly in the $\underline{e}_1$ direction, by an amount proportional to their height X_2 . Nothing moves vertically and nothing moves out of plane. A vertical line $X_1 = \text{const}$ tilts to a line of slope $1/\gamma$; the angle it makes with the vertical has tangent γ , which is where the name “amount of shear” comes from. Note carefully that γ is a *tangent of an angle*, not the angle itself; the two agree only to first order.

Differentiating (3.46) and (3.47) at fixed t ,

$$\begin{aligned} F d\underline{X} &= d\underline{x} = \underline{1} d\underline{X} + \gamma (\hat{\underline{E}}_2 \cdot d\underline{X}) \underline{e}_1 \\ &= \underline{1} d\underline{X} + \gamma (\underline{e}_1 \otimes \hat{\underline{E}}_2) d\underline{X} = (\underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2) d\underline{X}, \end{aligned} \quad (3.50)$$

the middle step being the definition of the dyad, $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$, read backwards. Since $d\underline{X}$ is arbitrary,

$$F = \underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2. \quad (3.51)$$

The Jacobian. Rule (R4) requires the images of three independent referential vectors, so take the basis and compute all three, using (R1) on the dyad in (3.51) and $\underline{1}\hat{\underline{E}}_A = \underline{E}_A = \underline{e}_A$:

$$\begin{aligned} F \hat{\underline{E}}_1 &= \underline{1}\hat{\underline{E}}_1 + \gamma (\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_1) \underline{e}_1 = \underline{e}_1 + \gamma \cdot 0 \cdot \underline{e}_1 = \underline{e}_1, \\ F \hat{\underline{E}}_2 &= \underline{1}\hat{\underline{E}}_2 + \gamma (\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_2) \underline{e}_1 = \underline{e}_2 + \gamma \cdot 1 \cdot \underline{e}_1 = \underline{e}_2 + \gamma \underline{e}_1, \\ F \hat{\underline{E}}_3 &= \underline{1}\hat{\underline{E}}_3 + \gamma (\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_3) \underline{e}_1 = \underline{e}_3 + \gamma \cdot 0 \cdot \underline{e}_1 = \underline{e}_3. \end{aligned}$$

Only the second is moved, and it is moved by γ in the $\underline{e}_1$ direction: that is the whole of (3.49) read at the level of fibres. Now form the ratio (R4).

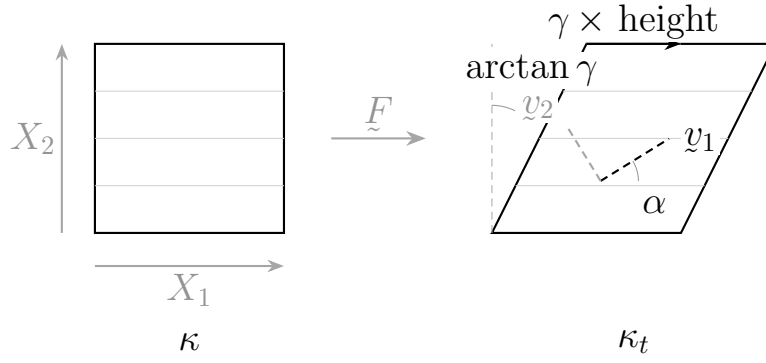


Figure 3.2: Simple shear. Horizontal material planes slide by an amount proportional to their height; the amount of shear γ is the tangent of the tilt angle of an initially vertical line. The dashed directions are the principal axes $\underline{v}_1, \underline{v}_2$ of $\underline{V}$ in κ_t , inclined at the angle α of (3.69); as $\gamma \rightarrow 0$ they approach $\pm 45^\circ$.

The denominator is $[\hat{\underline{E}}_1, \hat{\underline{E}}_2, \hat{\underline{E}}_3] = 1$ by Lemma 3.28, the referential basis being right-handed orthonormal. The numerator is expanded by linearity of the box product in its second slot, (R3):

$$\begin{aligned}
 J = \det \tilde{F} &= \frac{[\tilde{F}\hat{\underline{E}}_1, \tilde{F}\hat{\underline{E}}_2, \tilde{F}\hat{\underline{E}}_3]}{[\hat{\underline{E}}_1, \hat{\underline{E}}_2, \hat{\underline{E}}_3]} = \frac{[\underline{e}_1, \underline{e}_2 + \gamma \underline{e}_1, \underline{e}_3]}{1} \\
 &= [\underline{e}_1, \underline{e}_2, \underline{e}_3] + \gamma [\underline{e}_1, \underline{e}_1, \underline{e}_3] \\
 &= 1 + \gamma \cdot 0 = 1.
 \end{aligned} \tag{3.52}$$

The first box product is 1 by Lemma 3.28; the second vanishes because the box product is *alternating* and its first two slots are the same vector, (R3). If one prefers to see it through the cross product instead of through alternation, the same number comes out:

$$\begin{aligned}
 [\underline{e}_1, \underline{e}_2 + \gamma \underline{e}_1, \underline{e}_3] &= \underline{e}_1 \cdot (\underline{e}_2 + \gamma \underline{e}_1) \times \underline{e}_3 = \underline{e}_1 \cdot \underline{e}_2 \times \underline{e}_3 + \gamma \underline{e}_1 \cdot \underline{e}_1 \times \underline{e}_3 \\
 &= \underline{e}_1 \cdot \underline{e}_1 + \gamma \underline{e}_1 \cdot -\underline{e}_2 = 1 - \gamma \cdot 0 = 1,
 \end{aligned}$$

using $\underline{e}_2 \times \underline{e}_3 = \underline{e}_1$ and $\underline{e}_1 \times \underline{e}_3 = -(\underline{e}_3 \times \underline{e}_1) = -\underline{e}_2$.

The geometry behind the algebra is worth one sentence: the sheared block of Figure 3.2 has the same base and the same height as the reference block, and a parallelepiped's volume is base times height regardless of how far the top has slid. That is what $\gamma[\underline{e}_1, \underline{e}_1, \underline{e}_3] = 0$ is saying. By

(2.50) there is therefore no change of volume, and by (2.59) the densities agree, $\rho_\kappa = \rho$. Deformations with $J = 1$ are called *isochoric*.

The Cauchy–Green tensors. From (3.51) and Problem 17(a) of Chapter 1, $\underline{F}^t = \underline{1}^t + \gamma \hat{\underline{E}}_2 \otimes \underline{e}_1$, so

$$\begin{aligned}
\mathcal{C} &= (\underline{1}^t + \gamma \hat{\underline{E}}_2 \otimes \underline{e}_1)(\underline{1} + \gamma \underline{e}_1 \otimes \hat{\underline{E}}_2) \\
&= \underline{1}^t \underline{1} + \gamma \underline{1}^t (\underline{e}_1 \otimes \hat{\underline{E}}_2) + \gamma (\hat{\underline{E}}_2 \otimes \underline{e}_1) \underline{1} + \gamma^2 (\hat{\underline{E}}_2 \otimes \underline{e}_1)(\underline{e}_1 \otimes \hat{\underline{E}}_2) \\
&= \hat{\underline{I}} + \gamma (\underline{1}^t \underline{e}_1) \otimes \hat{\underline{E}}_2 + \gamma \hat{\underline{E}}_2 \otimes (\underline{1}^t \underline{e}_1) + \gamma^2 (\underline{e}_1 \cdot \underline{e}_1) \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2 \\
&= \hat{\underline{I}} + \gamma (\hat{\underline{E}}_1 \otimes \hat{\underline{E}}_2 + \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_1) + \gamma^2 \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2 \\
&= \hat{\underline{E}}_1 \otimes \hat{\underline{E}}_1 + \gamma (\hat{\underline{E}}_1 \otimes \hat{\underline{E}}_2 + \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_1) + (1 + \gamma^2) \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2 + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3,
\end{aligned} \tag{3.53}$$

Each line deserves its justification, since nothing here is meant to be taken on trust.

Line 2 is the distributive law: the product of two two-term sums has four terms, and they are written in the order (first)(first), (first)(second), (second)(first), (second)(second).

Line 3 treats the four terms one at a time. The first is $\underline{1}^t \underline{1} = \hat{\underline{I}}$ by (2.72). The second uses Problem 17(b) of Chapter 1, $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$, with $\underline{A} = \underline{1}^t$: only the *first* slot of the dyad is touched, giving $(\underline{1}^t \underline{e}_1) \otimes \hat{\underline{E}}_2$. The third uses Problem 17(c), $(\underline{a} \otimes \underline{b})\underline{A} = \underline{a} \otimes (\underline{A}^t \underline{b})$, with $\underline{A} = \underline{1}$ so that $\underline{A}^t = \underline{1}^t$: now only the *second* slot is touched, giving $\hat{\underline{E}}_2 \otimes (\underline{1}^t \underline{e}_1)$. The fourth is rule (R2), the dyad product: the two inner slots, $\underline{e}_1$ and $\underline{e}_1$, dot to give $\underline{e}_1 \cdot \underline{e}_1 = 1$, and the two outer slots, both $\hat{\underline{E}}_2$, survive.

Line 4 evaluates $\underline{1}^t \underline{e}_1$. From (2.68), $\underline{1} = \underline{E}_A \otimes \hat{\underline{E}}_A$, so by Problem 17(a) $\underline{1}^t = \hat{\underline{E}}_A \otimes \underline{E}_A$, and by (R1)

$$\underline{1}^t \underline{e}_1 = (\underline{E}_A \cdot \underline{e}_1) \hat{\underline{E}}_A = \delta_{A1} \hat{\underline{E}}_A = \hat{\underline{E}}_1,$$

using $\underline{E}_A = \underline{e}_A$ from (3.49). Substituting it into both the second and third terms produces the symmetric pair.

Line 5 writes out the identity, $\hat{\underline{I}} = \hat{\underline{E}}_1 \otimes \hat{\underline{E}}_1 + \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2 + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3$ by (3.11), and adds its $\hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2$ term to the $\gamma^2 \hat{\underline{E}}_2 \otimes \hat{\underline{E}}_2$ already present, giving the coefficient $1 + \gamma^2$. It is a rearrangement, not a new computation, and its only purpose is to display the matrix (3.55) entry by entry.

The book writes “similarly, the left Cauchy–Green tensor is” and gives the answer. The computation is not quite the mirror image of the one above, because the shifter now appears the other way round, so here it is in full:

$$\begin{aligned}\underline{\underline{B}} &= \underline{\underline{F}} \underline{\underline{F}}^t = (\underline{\underline{1}} + \gamma \underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2)(\underline{\underline{1}}^t + \gamma \hat{\underline{\underline{E}}}_2 \otimes \underline{\underline{e}}_1) \\ &= \underline{\underline{1}}\underline{\underline{1}}^t + \gamma \underline{\underline{1}}(\hat{\underline{\underline{E}}}_2 \otimes \underline{\underline{e}}_1) + \gamma (\underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2)\underline{\underline{1}}^t \\ &\quad + \gamma^2(\underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2)(\hat{\underline{\underline{E}}}_2 \otimes \underline{\underline{e}}_1).\end{aligned}$$

Take the four terms one at a time. The first is $\underline{\underline{I}}$, by (2.73). The second, by Problem 17(b), is $\gamma(\underline{\underline{1}}\hat{\underline{\underline{E}}}_2) \otimes \underline{\underline{e}}_1 = \gamma \hat{\underline{\underline{E}}}_2 \otimes \underline{\underline{e}}_1 = \gamma \underline{\underline{e}}_2 \otimes \underline{\underline{e}}_1$, using $\underline{\underline{1}}\hat{\underline{\underline{E}}}_2 = \hat{\underline{\underline{E}}}_2$ from (2.68) and the choice $\{\underline{\underline{E}}_A\} = \{\underline{\underline{e}}_i\}$ made at (3.49). The third, by Problem 17(c) with $\underline{\underline{A}} = \underline{\underline{1}}^t$ so that $\underline{\underline{A}}^t = \underline{\underline{1}}$, is $\gamma \underline{\underline{e}}_1 \otimes (\underline{\underline{1}}\hat{\underline{\underline{E}}}_2) = \gamma \underline{\underline{e}}_1 \otimes \underline{\underline{e}}_2$. The fourth, by the dyad product rule, is $\gamma^2(\hat{\underline{\underline{E}}}_2 \cdot \hat{\underline{\underline{E}}}_2) \underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1 = \gamma^2 \underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1$. Adding, and expanding $\underline{\underline{I}} = \underline{\underline{e}}_i \otimes \underline{\underline{e}}_i$ so as to collect the $\underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1$ terms,

$$\underline{\underline{B}} = (1 + \gamma^2)\underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1 + \underline{\underline{e}}_2 \otimes \underline{\underline{e}}_2 + \underline{\underline{e}}_3 \otimes \underline{\underline{e}}_3 + \gamma(\underline{\underline{e}}_1 \otimes \underline{\underline{e}}_2 + \underline{\underline{e}}_2 \otimes \underline{\underline{e}}_1). \quad (3.54)$$

REMARK 3.31 ($\underline{\underline{C}}$ and $\underline{\underline{B}}$ are not the same tensor with new labels). Compare (3.53) and (3.54): the coefficient $1 + \gamma^2$ sits on $\hat{\underline{\underline{E}}}_2 \otimes \hat{\underline{\underline{E}}}_2$ in $\underline{\underline{C}}$ and on $\underline{\underline{e}}_1 \otimes \underline{\underline{e}}_1$ in $\underline{\underline{B}}$. The index has moved from 2 to 1. That is not a slip; it is the rotation $\underline{\underline{R}}$ making itself felt. $\underline{\underline{C}}$ measures stretching as recorded on referential fibres, and the fibre that stretches most is $\hat{\underline{\underline{E}}}_2$ -ish; $\underline{\underline{B}}$ records it on the current fibres, and after the shear the most-stretched current direction is $\underline{\underline{e}}_1$ -ish. Anyone who writes $\underline{\underline{B}}$ by putting hats on $\underline{\underline{C}}$ will get this wrong.

In matrix form, with $C_{AB} = \hat{\underline{\underline{E}}}_A \cdot \underline{\underline{C}} \hat{\underline{\underline{E}}}_B$ and $B_{ij} = \underline{\underline{e}}_i \cdot \underline{\underline{B}} \underline{\underline{e}}_j$,

$$(C_{AB}) = \begin{pmatrix} 1 & \gamma & 0 \\ \gamma & 1 + \gamma^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (3.55)$$

$$(B_{ij}) = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (3.56)$$

Principal stretches. First, one principal vector is free. Apply $\mathcal{C}$ of (3.53) to $\hat{\underline{E}}_3$ with (R1), taking the four dyads in turn:

$$\begin{aligned}\mathcal{C}\hat{\underline{E}}_3 &= (\hat{\underline{E}}_1 \cdot \hat{\underline{E}}_3)\hat{\underline{E}}_1 + \gamma(\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_3)\hat{\underline{E}}_1 + \gamma(\hat{\underline{E}}_1 \cdot \hat{\underline{E}}_3)\hat{\underline{E}}_2 \\ &\quad + (1 + \gamma^2)(\hat{\underline{E}}_2 \cdot \hat{\underline{E}}_3)\hat{\underline{E}}_2 + (\hat{\underline{E}}_3 \cdot \hat{\underline{E}}_3)\hat{\underline{E}}_3 \\ &= 0 + 0 + 0 + 0 + \hat{\underline{E}}_3 = \hat{\underline{E}}_3,\end{aligned}$$

so $\hat{\underline{E}}_3$ is a principal vector of $\mathcal{C}$ with principal value 1, that is $\lambda_3^2 = 1$ and hence $\lambda_3 = 1$, the positive root being the one that is a stretch.

The other two are constrained before any computation, because the deformation is isochoric. Here every step is spelled out: (3.35) with $\det \underline{R} = 1$ from Step 6 turns $\det \underline{F}$ into $\det \underline{U}$; then the box product definition is applied with the principal triad of $\underline{U}$ as test triad; then $\underline{U}\underline{u}_i = \lambda_i\underline{u}_i$; then the scalars come out one slot at a time by (R3); then the common box product cancels:

$$\begin{aligned}1 = \det \underline{F} &= \det(\underline{R}\underline{U}) = (\det \underline{R})(\det \underline{U}) = 1 \cdot \det \underline{U} \\ &= \frac{[\underline{U}\underline{u}_1, \underline{U}\underline{u}_2, \underline{U}\underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} = \frac{[\lambda_1\underline{u}_1, \lambda_2\underline{u}_2, \lambda_3\underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} \\ &= \frac{\lambda_1\lambda_2\lambda_3[\underline{u}_1, \underline{u}_2, \underline{u}_3]}{[\underline{u}_1, \underline{u}_2, \underline{u}_3]} = \lambda_1\lambda_2\lambda_3.\end{aligned}\tag{3.57}$$

This is (3.17) again, applied to $\underline{U}$ instead of to a general symmetric tensor, and it says what it should: the volume ratio is the product of the three stretches, one per direction. With $\lambda_3 = 1$ it gives $\lambda_2 = \lambda_1^{-1}$: in simple shear the two in-plane stretches are reciprocal, so whatever is stretched in one principal direction is compressed by the same factor in the other.

Now solve $\det(\mathcal{C} - \lambda^2\hat{\underline{I}}) = 0$. From (3.55), expanding along the third row,

$$\begin{aligned}0 &= \begin{vmatrix} 1 - \lambda^2 & \gamma & 0 \\ \gamma & 1 + \gamma^2 - \lambda^2 & 0 \\ 0 & 0 & 1 - \lambda^2 \end{vmatrix} \\ &= (1 - \lambda^2)[(1 - \lambda^2)(1 + \gamma^2 - \lambda^2) - \gamma^2] \\ &= (1 - \lambda^2)[(1 - \lambda^2)^2 - \gamma^2\lambda^2],\end{aligned}\tag{3.58}$$

the last step because $(1 - \lambda^2)(1 + \gamma^2 - \lambda^2) - \gamma^2 = (1 - \lambda^2)(1 - \lambda^2) + \gamma^2(1 - \lambda^2) - \gamma^2 = (1 - \lambda^2)^2 - \gamma^2\lambda^2$. One root is $\lambda_3^2 = 1$ as noted. The remaining factor gives $1 - \lambda^2 = \pm\gamma\lambda$, i.e. $\lambda^2 \mp \gamma\lambda - 1 = 0$, whence $\lambda = \frac{1}{2}(\pm\gamma \pm \sqrt{\gamma^2 + 4})$; squaring the admissible (positive) roots,

$$4\lambda_{1,2}^2 = 4 + 2\gamma^2 \pm 2\gamma\sqrt{4 + \gamma^2}. \quad (3.59)$$

Identifying λ_1 with the $+$ root,

$$\lambda_1 = \frac{1}{2}(\gamma + \sqrt{4 + \gamma^2}), \quad \lambda_2 = \lambda_1^{-1}, \quad \lambda_3 = 1. \quad (3.60)$$

That $\lambda_2 = \lambda_1^{-1}$ is consistent with (3.57) can be checked directly:

$$\frac{1}{2}(\sqrt{4 + \gamma^2} - \gamma) \cdot \frac{1}{2}(\sqrt{4 + \gamma^2} + \gamma) = \frac{1}{4}(4 + \gamma^2 - \gamma^2) = 1.$$

The spectral representation of $\mathcal{C}$ is therefore

$$\mathcal{C} = \lambda_1^2 \underline{u}_1 \otimes \underline{u}_1 + \lambda_1^{-2} \underline{u}_2 \otimes \underline{u}_2 + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3, \quad (3.61)$$

with $\underline{u}_1, \underline{u}_2$ orthonormal and perpendicular to $\hat{\underline{E}}_3$:

$$\underline{u}_1 = \cos A \hat{\underline{E}}_1 + \sin A \hat{\underline{E}}_2, \quad \underline{u}_2 = -\sin A \hat{\underline{E}}_1 + \cos A \hat{\underline{E}}_2. \quad (3.62)$$

To find A , dot $\mathcal{C}\underline{u}_1 = \lambda_1^2 \underline{u}_1$ with $\hat{\underline{E}}_1$ and move $\mathcal{C}$ across using its symmetry:

$$\underline{u}_1 \cdot \mathcal{C} \hat{\underline{E}}_1 = \hat{\underline{E}}_1 \cdot \mathcal{C} \underline{u}_1 = \lambda_1^2 \hat{\underline{E}}_1 \cdot \underline{u}_1. \quad (3.63)$$

From (3.53), $\mathcal{C} \hat{\underline{E}}_1 = \hat{\underline{E}}_1 + \gamma \hat{\underline{E}}_2$, so the left side is $\cos A + \gamma \sin A$, while the right side is $\lambda_1^2 \cos A$. Hence $\gamma \sin A = (\lambda_1^2 - 1) \cos A$ and

$$\tan A = \frac{1}{\gamma}(\lambda_1^2 - 1) = \frac{1}{2}(\gamma + \sqrt{4 + \gamma^2}) = \lambda_1, \quad (3.64)$$

the middle equality following from $\lambda_1^2 = \frac{1}{4}(\gamma + \sqrt{4 + \gamma^2})^2 = \frac{1}{4}(2\gamma^2 + 4 + 2\gamma\sqrt{4 + \gamma^2}) = \frac{1}{2}(\gamma^2 + 2 + \gamma\sqrt{4 + \gamma^2})$, so that $\lambda_1^2 - 1 = \frac{1}{2}\gamma(\gamma + \sqrt{4 + \gamma^2}) = \gamma\lambda_1$.

As $\gamma \rightarrow 0$, $\tan A \rightarrow 1$ and $A \rightarrow \pi/4$: for small shears the principal axes sit at approximately $\pm 45^\circ$ to the shearing direction. This is the familiar statement that a small shear is a small extension along one diagonal and an equal contraction along the other.

REMARK 3.32 (why the second scalar equation is redundant). Dotting $\underline{C}\underline{u}_1 = \lambda_1^2 \underline{u}_1$ with $\hat{\underline{E}}_2$ instead of $\hat{\underline{E}}_1$ produces no new information, and the reason is general. Suppose $\det \underline{A} = 0$, equivalently $\underline{A}\underline{u} = \underline{0}$ for some $\underline{u} \neq \underline{0}$. That statement is equivalent to $\underline{a} \cdot \underline{A}\underline{u} = 0$ for arbitrary $\underline{a}$, since a vector orthogonal to everything is $\underline{0}$. Expanding $\underline{a}$ on an orthonormal basis,

$$a_1 \underline{e}_1 \cdot \underline{A}\underline{u} + a_2 \underline{e}_2 \cdot \underline{A}\underline{u} + a_3 \underline{e}_3 \cdot \underline{A}\underline{u} = 0 \quad (3.65)$$

for arbitrary a_1, a_2, a_3 . Here $\underline{A} = \underline{C} - \lambda_1^2 \hat{\underline{I}}$, whose determinant vanishes by construction, so the three scalar equations $\underline{e}_i \cdot \underline{A}\underline{u} = 0$ are linearly dependent; solving two of them forces the third. This is the reason an eigenvector is determined only up to scale and one uses normalisation to fix it.

The left tensor. From (3.54), $\underline{B}\underline{e}_3 = \underline{e}_3$, so

$$\underline{B} = \lambda_1^2 \underline{v}_1 \otimes \underline{v}_1 + \lambda_1^{-2} \underline{v}_2 \otimes \underline{v}_2 + \underline{e}_3 \otimes \underline{e}_3, \quad (3.66)$$

with

$$\underline{v}_1 = \cos \alpha \underline{e}_1 + \sin \alpha \underline{e}_2, \quad \underline{v}_2 = -\sin \alpha \underline{e}_1 + \cos \alpha \underline{e}_2 \quad (3.67)$$

for some angle α . Dot $\underline{B}\underline{v}_1 = \lambda_1^2 \underline{v}_1$ with $\underline{e}_2$. From (3.54), $\underline{B}\underline{v}_1 = \cos \alpha \underline{B}\underline{e}_1 + \sin \alpha \underline{B}\underline{e}_2$ with $\underline{B}\underline{e}_1 = (1 + \gamma^2)\underline{e}_1 + \gamma \underline{e}_2$ and $\underline{B}\underline{e}_2 = \gamma \underline{e}_1 + \underline{e}_2$; therefore $\underline{e}_2 \cdot \underline{B}\underline{v}_1 = \gamma \cos \alpha + \sin \alpha$, while $\lambda_1^2 \underline{e}_2 \cdot \underline{v}_1 = \lambda_1^2 \sin \alpha$. Dividing by $\sin \alpha$,

$$1 + \gamma \cot \alpha = \lambda_1^2. \quad (3.68)$$

Hence $\cot \alpha = (\lambda_1^2 - 1)/\gamma = \lambda_1$ by the same identity used in (3.64), so

$$\tan \alpha = \frac{2}{\gamma + \sqrt{4 + \gamma^2}} = \cot A; \quad \text{hence} \quad A = \frac{\pi}{2} - \alpha. \quad (3.69)$$

Accordingly $\sin A = \cos \alpha$ and $\cos A = \sin \alpha$, and $\underline{v}_1, \underline{v}_2$ are determined.

The rotation. By (3.43), $\underline{R} = \sum_i \underline{v}_i \otimes \underline{u}_i$. Substituting (3.62) and (3.67) with $\cos A = \sin \alpha$, $\sin A = \cos \alpha$ and expanding both dyads term by

term,

$$\begin{aligned}
\underline{v}_1 \otimes \underline{u}_1 &= (\cos \alpha \underline{e}_1 + \sin \alpha \underline{e}_2) \otimes (\sin \alpha \hat{\underline{E}}_1 + \cos \alpha \hat{\underline{E}}_2) \\
&= \cos \alpha \sin \alpha \underline{e}_1 \otimes \hat{\underline{E}}_1 + \cos^2 \alpha \underline{e}_1 \otimes \hat{\underline{E}}_2 \\
&\quad + \sin^2 \alpha \underline{e}_2 \otimes \hat{\underline{E}}_1 + \sin \alpha \cos \alpha \underline{e}_2 \otimes \hat{\underline{E}}_2, \\
\underline{v}_2 \otimes \underline{u}_2 &= (-\sin \alpha \underline{e}_1 + \cos \alpha \underline{e}_2) \otimes (-\cos \alpha \hat{\underline{E}}_1 + \sin \alpha \hat{\underline{E}}_2) \\
&= \sin \alpha \cos \alpha \underline{e}_1 \otimes \hat{\underline{E}}_1 - \sin^2 \alpha \underline{e}_1 \otimes \hat{\underline{E}}_2 \\
&\quad - \cos^2 \alpha \underline{e}_2 \otimes \hat{\underline{E}}_1 + \cos \alpha \sin \alpha \underline{e}_2 \otimes \hat{\underline{E}}_2.
\end{aligned}$$

Adding, and using $2 \sin \alpha \cos \alpha = \sin 2\alpha$ and $\cos^2 \alpha - \sin^2 \alpha = \cos 2\alpha$,

$$\begin{aligned}
\underline{R} &= \underline{e}_3 \otimes \hat{\underline{E}}_3 + \sin 2\alpha (\underline{e}_1 \otimes \hat{\underline{E}}_1 + \underline{e}_2 \otimes \hat{\underline{E}}_2) + \cos 2\alpha (\underline{e}_1 \otimes \hat{\underline{E}}_2 - \underline{e}_2 \otimes \hat{\underline{E}}_1). \\
&\hspace{25em} (3.70)
\end{aligned}$$

As a check, $\underline{R}$ should carry the right-handed orthonormal triad $\{\hat{\underline{E}}_A\}$ to a right-handed orthonormal triad $\{\underline{w}_A\}$. Reading the columns of (3.70), that is picking out the terms whose second slot is $\hat{\underline{E}}_A$,

$$\begin{aligned}
\underline{w}_1 &= \underline{R} \hat{\underline{E}}_1 = \sin 2\alpha \underline{e}_1 - \cos 2\alpha \underline{e}_2, \\
\underline{w}_2 &= \underline{R} \hat{\underline{E}}_2 = \cos 2\alpha \underline{e}_1 + \sin 2\alpha \underline{e}_2, \\
\underline{w}_3 &= \underline{R} \hat{\underline{E}}_3 = \underline{e}_3.
\end{aligned} \tag{3.71}$$

Each has unit length, $\sin^2 2\alpha + \cos^2 2\alpha = 1$; they are mutually orthogonal, $\underline{w}_1 \cdot \underline{w}_2 = \sin 2\alpha \cos 2\alpha - \cos 2\alpha \sin 2\alpha = 0$; and $\underline{w}_1 \times \underline{w}_2 = (\sin^2 2\alpha + \cos^2 2\alpha) \underline{e}_1 \times \underline{e}_2 = \underline{e}_3 = \underline{w}_3$, so the triad is right handed.

REMARK 3.33 (the material turns backwards). The matrix of (3.70) in the plane is $\begin{pmatrix} \sin 2\alpha & \cos 2\alpha \\ -\cos 2\alpha & \sin 2\alpha \end{pmatrix}$, which is the rotation through the angle β with $\cos \beta = \sin 2\alpha$ and $\sin \beta = -\cos 2\alpha$, i.e. $\beta = 2\alpha - \pi/2$. As $\gamma \rightarrow 0$ we found $\alpha \rightarrow \pi/4$, so $\beta \rightarrow 0$ as it must. Expanding, $\tan \alpha = 2/(\gamma + \sqrt{4 + \gamma^2})$ gives $\alpha \approx \pi/4 - \gamma/4$, hence

$$\beta \approx -\frac{\gamma}{2}.$$

The polar rotation in simple shear is *clockwise* by half the amount of shear, even though the shear pushes the top of the block to the right.

The check is independent: the skew part of $\underline{F} - \underline{I}$ is $\frac{\gamma}{2}(\underline{e}_1 \otimes \underline{e}_2 - \underline{e}_2 \otimes \underline{e}_1)$, which sends $\underline{e}_1 \mapsto -\frac{\gamma}{2}\underline{e}_2$, a clockwise turn of $\gamma/2$. Problem 9 below recovers the same β a third way, from $\tan \beta = -\gamma/2$ exactly.

3.5.2 Radial deformation of a right circular cylinder

What this example is for. Two new things. First, it is the first deformation in the chapter that is *not* homogeneous: $\underline{F}$ changes from point to point, so $\underline{C}$, $\underline{U}$, $\underline{R}$ and $\underline{J}$ are all fields and the shortcuts of §3.6 are unavailable. Second, it is written in curvilinear coordinates, in which the basis vectors themselves depend on position; handling that correctly is the entire content of (3.74) and is the skill the next two examples reuse without comment. The compensation is that the geometry does the diagonalisation for us, so no characteristic equation is ever solved: see Remark 3.35.

Let κ and κ_t be hollow right circular cylinders. Describe position in κ_t by the polar parametrisation (1.117),

$$\underline{x} = r \underline{e}_r(\theta) + z \underline{e}_3, \quad \underline{e}_r(\theta) = \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2, \quad (3.72)$$

and position in κ by

$$\underline{X} = R \hat{\underline{E}}_R(\Theta) + Z \hat{\underline{E}}_3, \quad \hat{\underline{E}}_R(\Theta) = \cos \Theta \hat{\underline{E}}_1 + \sin \Theta \hat{\underline{E}}_2. \quad (3.73)$$

Suppose circles $R = \text{const}$ map to circles $r = \text{const}$ and that the azimuthal and axial coordinates are unchanged, $\theta = \Theta$ and $z = Z$; then $r = f(R)$ for some function f , and f determines the deformation completely.

What we are about to do, and why

The goal is $\underline{F}$, because everything else in the chapter is built from it: $\underline{C} = \underline{F}^t \underline{F}$ gives the stretches, $\underline{U} = \sqrt{\underline{C}}$ and $\underline{R} = \underline{F} \underline{U}^{-1}$ split stretch from rotation, and $\underline{J} = \det \underline{F}$ gives the volume change. But $\underline{F}$ is defined by $d\underline{x} = \underline{F} d\underline{X}$, (2.34), so the only way to find it is to compute $d\underline{x}$ and

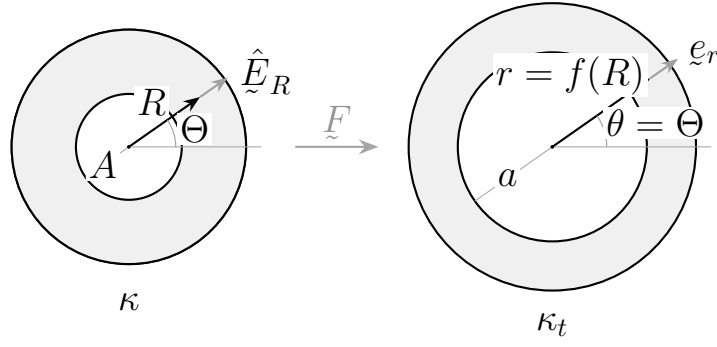


Figure 3.3: Annular cross-section of a right circular cylinder in κ and κ_t . Each material circle stays a circle and each radius stays a radius; only the radial coordinate changes, by $r = f(R)$. The inner radius A goes to $a = f(A)$.

$d\tilde{X}$ and compare. The obstacle is that the natural description here is in polar coordinates, in which the basis vectors themselves move, and the whole of (3.74) is the bookkeeping that handles that. There are four steps and it is worth naming them before doing them:

Step A. Differentiate the position $\tilde{x} = r\tilde{e}_r(\theta) + z\tilde{e}_3$, being careful that $\tilde{e}_r$ depends on θ .

Step B. Do the same for $\tilde{X}$, and invert the result to express the three referential coordinate increments $dR, d\Theta, dZ$ as dot products with $d\tilde{X}$. This is the step that puts $d\tilde{X}$ into evidence so that $\tilde{F}$ can be read off.

Step C. Use the deformation itself, $r = f(R)$, $\theta = \Theta$, $z = Z$, to replace $dr, d\theta, dz$ by $dR, d\Theta, dZ$.

Step D. Compare with $d\tilde{x} = \tilde{F}d\tilde{X}$ and strip off the arbitrary $d\tilde{X}$.

Step A. From (3.72), $\tilde{x} = r\tilde{e}_r(\theta) + z\tilde{e}_3$, in which r, θ, z all vary and $\tilde{e}_3$ is a constant vector. Differentiating with the product rule,

$$d\tilde{x} = dr\tilde{e}_r(\theta) + r d[\tilde{e}_r(\theta)] + dz\tilde{e}_3,$$

and $\tilde{e}_r$ depends on position only through θ , so the chain rule gives $d[\tilde{e}_r(\theta)] = \tilde{e}'_r(\theta) d\theta = \tilde{e}_\theta(\theta) d\theta$, by the definition $\tilde{e}_\theta = \tilde{e}'_r$ of Lemma 3.29. Hence

$$d\tilde{x} = dr\tilde{e}_r(\theta) + r d\theta\tilde{e}_\theta(\theta) + dz\tilde{e}_3.$$

Note the factor r in the middle term: it is there because moving the angle

by $d\theta$ moves the point by the *arc length* $r d\theta$, not by $d\theta$. Forgetting it is the commonest error in this computation.

Step B. Exactly the same differentiation applied to (3.73), $\tilde{X} = R \hat{\tilde{E}}_R(\Theta) + Z \hat{\tilde{E}}_3$, gives

$$d\tilde{X} = dR \hat{\tilde{E}}_R(\Theta) + R d\Theta \hat{\tilde{E}}_\Theta(\Theta) + dZ \hat{\tilde{E}}_3.$$

Now invert this. The frame $\{\hat{\tilde{E}}_R, \hat{\tilde{E}}_\Theta, \hat{\tilde{E}}_3\}$ is orthonormal by Lemma 3.29, so dotting with each member in turn picks out one coefficient and kills the other two:

$$\begin{aligned} \hat{\tilde{E}}_R \cdot d\tilde{X} &= dR (\hat{\tilde{E}}_R \cdot \hat{\tilde{E}}_R) + R d\Theta (\hat{\tilde{E}}_R \cdot \hat{\tilde{E}}_\Theta) + dZ (\hat{\tilde{E}}_R \cdot \hat{\tilde{E}}_3) \\ &= dR \cdot 1 + R d\Theta \cdot 0 + dZ \cdot 0 = dR, \\ \hat{\tilde{E}}_\Theta \cdot d\tilde{X} &= dR \cdot 0 + R d\Theta \cdot 1 + dZ \cdot 0 = R d\Theta, \\ \hat{\tilde{E}}_3 \cdot d\tilde{X} &= dR \cdot 0 + R d\Theta \cdot 0 + dZ \cdot 1 = dZ. \end{aligned}$$

These three are $(*_2)$ of Problem 25 of Chapter 1 in cylindrical form. The middle one is the one to watch: what the dot product returns is $R d\Theta$, an arc length, not $d\Theta$ itself. So whenever $d\Theta$ is needed alone one must divide by R , and that division is where the f/R in (3.75) comes from.

Step C. Differentiate the three defining relations of the deformation, $r = f(R)$, $\theta = \Theta$, $z = Z$:

$$dr = f'(R) dR, \quad d\theta = d\Theta, \quad dz = dZ.$$

Substitute these into Step A, and use $r = f(R)$ in the coefficient of the middle term:

$$d\tilde{x} = f'(R) dR \underline{e}_r(\theta) + f(R) d\Theta \underline{e}_\theta(\theta) + dZ \underline{e}_3.$$

Now replace each of dR , $d\Theta$, dZ by its expression from Step B. The first and third go straight in. The second needs the division just flagged: $d\Theta = \frac{1}{R}(R d\Theta) = \frac{1}{R} \hat{\tilde{E}}_\Theta \cdot d\tilde{X}$, so the coefficient $f(R)$ becomes f/R .

Assembling,

$$\begin{aligned}
\tilde{F} d\tilde{X} &= d\tilde{x} = dr \, \underline{e}_r(\theta) + r d\theta \, \underline{e}_\theta(\theta) + dz \, \underline{e}_3 \\
&= f'(R) dR \, \underline{e}_r(\theta) + f(R) d\Theta \, \underline{e}_\theta(\theta) + dZ \, \underline{e}_3 \\
&= f'(R) \underline{e}_r(\theta) [\hat{\underline{E}}_R(\Theta) \cdot d\tilde{X}] + (f/R) \underline{e}_\theta(\theta) [\hat{\underline{E}}_\Theta(\Theta) \cdot d\tilde{X}] \\
&\quad + \underline{e}_3 [\hat{\underline{E}}_3 \cdot d\tilde{X}].
\end{aligned} \tag{3.74}$$

Step D. Every term on the last line of (3.74) has the shape $\underline{a}(\underline{b} \cdot d\tilde{X})$, which by rule (R1) read backwards is $(\underline{a} \otimes \underline{b})d\tilde{X}$. Collecting the three,

$$\tilde{F} d\tilde{X} = \left[f' \underline{e}_r \otimes \hat{\underline{E}}_R + (f/R) \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta + \underline{e}_3 \otimes \hat{\underline{E}}_3 \right] d\tilde{X}.$$

This holds for *every* $d\tilde{X}$. Two tensors agreeing on every vector are the same tensor: if $\underline{A}d\tilde{X} = \underline{B}d\tilde{X}$ for all $d\tilde{X}$ then $(\underline{A} - \underline{B})d\tilde{X} = \underline{0}$ for all $d\tilde{X}$, so $\underline{A} - \underline{B} = \underline{0}$. Therefore the brackets may be equated, and using $\Theta = \theta$, which is the second of the defining relations and lets the two angle labels be written the same way,

$$\tilde{F} = f'(R) \underline{e}_r(\theta) \otimes \hat{\underline{E}}_R(\theta) + (f/R) \underline{e}_\theta(\theta) \otimes \hat{\underline{E}}_\Theta(\theta) + \underline{e}_3 \otimes \hat{\underline{E}}_3. \tag{3.75}$$

REMARK 3.34 (three warnings about (3.75)). *The two frames are not the same objects.* $\underline{e}_r(\theta)$ is spatial and $\hat{\underline{E}}_R(\Theta)$ referential; they carry the same angle here only because this particular deformation happens to leave the azimuth alone. In §3.5.3 they will not, and the two arguments must then be kept apart.

$\tilde{F}$ is not constant. The coefficients $f'(R)$ and f/R depend on R , and the frame vectors depend on Θ . So this deformation is *not* homogeneous in the sense of §3.6, and every quantity derived from $\tilde{F}$ below is a field.

The middle coefficient is a ratio of two arc lengths. The azimuthal stretch is f/R , the deformed arc $f d\Theta$ divided by the reference arc $R d\Theta$. It equals 1 only where $f(R) = R$; a purely radial motion that leaves radii radial still stretches the material circumferentially, and that is what a pressurised tube feels.

REMARK 3.35 ((3.75) is already a polar decomposition). Look at (3.75) against (3.43), $\tilde{F} = \sum_i \lambda_i \underline{v}_i \otimes \underline{u}_i$. The referential triad $\{\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta, \hat{\underline{E}}_3\}$ is

orthonormal and so is the spatial triad $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$, and $\tilde{F}$ is written as a sum of three dyads pairing them. So the principal stretches and both principal frames can be read off without solving any characteristic equation. This is why the cylindrical examples are the ones every textbook uses: the geometry has done the diagonalisation in advance.

The Jacobian, computed in full

Rule (R4) needs the images of three independent referential vectors. Take the frame $\{\hat{\tilde{E}}_R, \hat{\tilde{E}}_\Theta, \hat{\tilde{E}}_3\}$, which is orthonormal, and apply (3.75) to each member. Each application uses (R1) three times, once per dyad, and the inner products are Kronecker deltas because the frame is orthonormal:

$$\begin{aligned}\tilde{F}\hat{\tilde{E}}_R &= f'(\hat{\tilde{E}}_R \cdot \hat{\tilde{E}}_R) \underline{e}_r + \frac{f}{R}(\hat{\tilde{E}}_\Theta \cdot \hat{\tilde{E}}_R) \underline{e}_\theta + (\hat{\tilde{E}}_3 \cdot \hat{\tilde{E}}_R) \underline{e}_3 \\ &= f' \cdot 1 \cdot \underline{e}_r + \frac{f}{R} \cdot 0 \cdot \underline{e}_\theta + 0 \cdot \underline{e}_3 = f'(R) \underline{e}_r(\theta), \\ \tilde{F}\hat{\tilde{E}}_\Theta &= f'(\hat{\tilde{E}}_R \cdot \hat{\tilde{E}}_\Theta) \underline{e}_r + \frac{f}{R}(\hat{\tilde{E}}_\Theta \cdot \hat{\tilde{E}}_\Theta) \underline{e}_\theta + (\hat{\tilde{E}}_3 \cdot \hat{\tilde{E}}_\Theta) \underline{e}_3 \\ &= f' \cdot 0 \cdot \underline{e}_r + \frac{f}{R} \cdot 1 \cdot \underline{e}_\theta + 0 \cdot \underline{e}_3 = \frac{f}{R} \underline{e}_\theta(\theta), \\ \tilde{F}\hat{\tilde{E}}_3 &= f' \cdot 0 \cdot \underline{e}_r + \frac{f}{R} \cdot 0 \cdot \underline{e}_\theta + 1 \cdot \underline{e}_3 = \underline{e}_3.\end{aligned}$$

So the three referential frame vectors go to the three spatial frame vectors, each merely rescaled: radially by f' , azimuthally by f/R , axially by 1. Nothing is tilted, which is the content of Remark 3.35.

Both box products in (R4) are now box products of orthonormal right-handed triads, hence equal to 1 by Lemma 3.28 and Lemma 3.29:

$$[\hat{\tilde{E}}_R, \hat{\tilde{E}}_\Theta, \hat{\tilde{E}}_3] = 1, \quad [\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = 1.$$

Assuming κ occupiable, so that $J > 0$ by (2.50), and extracting the three scalars one slot at a time by the linearity of the box product (R3),

$$\begin{aligned}0 < J = \det \tilde{F} &= \frac{[\tilde{F}\hat{\tilde{E}}_R, \tilde{F}\hat{\tilde{E}}_\Theta, \tilde{F}\hat{\tilde{E}}_3]}{[\hat{\tilde{E}}_R, \hat{\tilde{E}}_\Theta, \hat{\tilde{E}}_3]} = \frac{[f'\underline{e}_r, \frac{f}{R}\underline{e}_\theta, \underline{e}_3]}{1} \\ &= f'[\underline{e}_r, \frac{f}{R}\underline{e}_\theta, \underline{e}_3] = f' \cdot \frac{f}{R} [\underline{e}_r, \underline{e}_\theta, \underline{e}_3] \\ &= f' \cdot \frac{f}{R} \cdot 1 = \frac{ff'}{R}.\end{aligned}\tag{3.76}$$

The middle two lines are the point of “linear in each slot separately”: the factor f' is pulled out of the first slot with the other two left alone, then f/R out of the second, and only then is the surviving box product of pure unit vectors evaluated.

REMARK 3.36 (reading (3.76) without any algebra). The answer can be checked against the picture. Take the small curvilinear brick between radii R and $R + dR$, azimuths Θ and $\Theta + d\Theta$, heights Z and $Z + dZ$. Its volume is

$$dV = dR \cdot R d\Theta \cdot dZ,$$

the middle factor being an arc length, radius times angle. After the deformation the radial edge has length $dr = f'dR$, the azimuthal edge has length $r d\theta = f d\Theta$, and the axial edge is unchanged, so

$$dv = f'dR \cdot f d\Theta \cdot dZ.$$

Dividing,

$$J = \frac{dv}{dV} = \frac{f'dR \cdot f d\Theta \cdot dZ}{dR \cdot R d\Theta \cdot dZ} = \frac{ff'}{R},$$

which is (3.76) again. The R in the denominator is not an accident of the algebra: it is there because the reference arc length is $R d\Theta$ while the deformed one is $f d\Theta$, so the azimuthal stretch is f/R , a ratio of two arc lengths at two different radii.

Since $f > 0$ and $R > 0$, $J > 0$ implies $f'(R) > 0$: f is strictly increasing, so $R_2 > R_1$ gives $f(R_2) > f(R_1)$. Concentric circles go to concentric circles with the ordering preserved, outer to outer and inner to inner.

If the deformation is isochoric then $J = 1$, i.e. $ff' = R$, i.e. $(\frac{1}{2}f^2)' = R$. Integrating and writing $f(R) = r$,

$$r^2 - a^2 = R^2 - A^2, \quad a = f(A), \quad (3.77)$$

A being a positive constant, since evaluating at $R = A$ gives $a^2 - a^2 = A^2 - A^2$, which fixes the constant of integration.

The book calls the area statement “an immediate consequence”. Written out: (3.77) says $r^2 - a^2 = R^2 - A^2$, and multiplying by π ,

$$\underbrace{\pi r^2 - \pi a^2}_{\text{area of the deformed annulus}} = \underbrace{\pi R^2 - \pi A^2}_{\text{area of the reference annulus}}.$$

So the annulus $[A, R]$ has exactly the area of its image $[a, r]$. That this had to happen follows from $J = 1$ and $z = Z$: a slab of the cylinder of axial thickness ΔZ over the annulus has volume (area) $\times \Delta Z$ before and (area) $\times \Delta Z$ after, since the thickness is unchanged; equality of the volumes therefore forces equality of the areas.

The transpose of (3.75), by Problem 17(a), $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$ applied to each of the three dyads, is

$$\underline{F}^t = f'(R) \hat{\underline{E}}_R(\theta) \otimes \underline{e}_r(\theta) + (f/R) \hat{\underline{E}}_\Theta(\theta) \otimes \underline{e}_\theta(\theta) + \hat{\underline{E}}_3 \otimes \underline{e}_3. \quad (3.78)$$

Now form $\underline{C} = \underline{F}^t \underline{F}$. The product of the two three-term sums has nine terms, each of the shape $(\hat{\underline{E}}_A \otimes \underline{e}_i)(\underline{e}_j \otimes \hat{\underline{E}}_B) = (\underline{e}_i \cdot \underline{e}_j) \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B$ by the dyad rule $(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c}) \underline{a} \otimes \underline{d}$. Writing $\lambda_1 = f'$, $\lambda_2 = f/R$, $\lambda_3 = 1$ and $(\hat{\underline{E}}_A) = (\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta, \hat{\underline{E}}_3)$, $(\underline{e}_i) = (\underline{e}_r, \underline{e}_\theta, \underline{e}_3)$, the nine terms are $\lambda_A \lambda_B (\underline{e}_A \cdot \underline{e}_B) \hat{\underline{E}}_A \otimes \hat{\underline{E}}_B$, and the inner product is δ_{AB} : the six off-diagonal terms vanish, for instance

$$(f' \hat{\underline{E}}_R \otimes \underline{e}_r)((f/R) \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta) = f'(f/R) (\underline{e}_r \cdot \underline{e}_\theta) \hat{\underline{E}}_R \otimes \hat{\underline{E}}_\Theta = 0,$$

and the three diagonal ones survive with their coefficients squared, for instance

$$(f' \hat{\underline{E}}_R \otimes \underline{e}_r)(f' \underline{e}_r \otimes \hat{\underline{E}}_R) = (f')^2 (\underline{e}_r \cdot \underline{e}_r) \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R = (f')^2 \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R.$$

Multiplying in the other order for $\underline{B} = \underline{F} \underline{F}^t$ produces terms $(\underline{e}_i \otimes \hat{\underline{E}}_A)(\hat{\underline{E}}_B \otimes \underline{e}_j) = (\hat{\underline{E}}_A \cdot \hat{\underline{E}}_B) \underline{e}_i \otimes \underline{e}_j$, and now it is the *referential* triad whose orthonormality kills the cross terms. What survives is

$$\begin{aligned} \underline{C} &= (f')^2 \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + (f/R)^2 \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3, \\ \underline{B} &= (f')^2 \underline{e}_r \otimes \underline{e}_r + (f/R)^2 \underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3. \end{aligned} \quad (3.79)$$

These are spectral representations, in the sense of (3.12): each is a sum of three dyads built on an orthonormal triad, so the coefficients are the

principal values and the triad members are the principal vectors. The principal stretches are therefore

$$\lambda_1 = f'(R), \quad \lambda_2 = f/R, \quad \lambda_3 = 1,$$

all three positive, since $f > 0$ and $R > 0$ give $\lambda_2 > 0$ and (3.76) gave $f' > 0$. The book says the stretch tensors “follow immediately”; the reason is Theorem 3.16. Since $\underline{C}$ is symmetric and positive definite with the spectral form above, the *unique* symmetric positive definite square root is obtained by replacing each principal value by its positive square root and leaving the axes untouched, which is (3.28):

$$\begin{aligned} \underline{U} &= f' \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + (f/R) \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3, \\ \underline{V} &= f' \underline{e}_r \otimes \underline{e}_r + (f/R) \underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3. \end{aligned} \quad (3.80)$$

Finally the rotation. The inverse of a tensor in spectral form is obtained by inverting the principal values, Remark 3.10, so

$$\underline{U}^{-1} = \frac{1}{f'} \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + \frac{R}{f} \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3,$$

which one verifies at once: $\underline{U}\underline{U}^{-1}$ has nine terms, the six off-diagonal ones vanishing on orthonormality and the three diagonal ones giving $f'(1/f')\hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + (f/R)(R/f)\hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3 = \hat{\underline{I}}$. Multiplying (3.75) on the right by it, again nine terms of which six vanish,

$$\begin{aligned} \underline{F}\underline{U}^{-1} &= f' \cdot \frac{1}{f'} (\hat{\underline{E}}_R \cdot \hat{\underline{E}}_R) \underline{e}_r \otimes \hat{\underline{E}}_R + \frac{f}{R} \cdot \frac{R}{f} (\hat{\underline{E}}_\Theta \cdot \hat{\underline{E}}_\Theta) \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta \\ &\quad + 1 \cdot 1 \cdot \underline{e}_3 \otimes \hat{\underline{E}}_3 \\ &= \underline{e}_r \otimes \hat{\underline{E}}_R + \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta + \underline{e}_3 \otimes \hat{\underline{E}}_3. \end{aligned}$$

The left factorisation gives the same thing: with $\underline{V}^{-1} = (1/f')\underline{e}_r \otimes \underline{e}_r + (R/f)\underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3$, multiplying (3.75) on the *left* produces the identical three surviving terms, now with the spatial triad supplying the Kronecker deltas. Hence

$$\underline{R} = \underline{F}\underline{U}^{-1} = \underline{V}^{-1}\underline{F} = \underline{e}_r(\theta) \otimes \hat{\underline{E}}_R(\theta) + \underline{e}_\theta(\theta) \otimes \hat{\underline{E}}_\Theta(\theta) + \underline{e}_3 \otimes \hat{\underline{E}}_3. \quad (3.81)$$

By Lemma 3.39 of §3.5.3, which applies verbatim, this $\underline{R}$ is a two-point rotation; indeed it is exactly the tensor called $\underline{Q}$ there.

Why bother interpreting $\underline{R}$ at all? Because (3.81) pairs a spatial vector with a referential one in each dyad, and such an object cannot be compared with $\underline{I}$: they live in different spaces. To say the sentence “there is no rotation here” one must first move everything into one space, and the shifter of (2.68) is the only tool for that. This is exactly the manoeuvre of §2.6, and (3.82)–(3.83) are its payoff.

So introduce the shifted vectors

$$\underline{E}_R(\Theta) = \underline{1}\hat{\underline{E}}_R(\Theta), \quad \underline{E}_\Theta(\Theta) = \underline{1}\hat{\underline{E}}_\Theta(\Theta), \quad \underline{E}_3 = \underline{1}\hat{\underline{E}}_3,$$

all three now in $\mathbb{E}^3$. Each satisfies $\underline{1}^t \underline{E}_R = \underline{1}^t \underline{1} \hat{\underline{E}}_R = \hat{\underline{I}} \hat{\underline{E}}_R = \hat{\underline{E}}_R$ by (2.72), and likewise for the other two. Problem 17(c) of Chapter 1 says $(\underline{a} \otimes \underline{b})\underline{A} = \underline{a} \otimes (\underline{A}^t \underline{b})$: post-multiplying a dyad by a tensor acts on the *second* slot, through the transpose. Apply it to each of the three dyads with $\underline{A} = \underline{1}$, so that $\underline{A}^t = \underline{1}^t$:

$$\begin{aligned} (\underline{e}_r \otimes \underline{E}_R) \underline{1} &= \underline{e}_r \otimes (\underline{1}^t \underline{E}_R) = \underline{e}_r \otimes \hat{\underline{E}}_R, \\ (\underline{e}_\theta \otimes \underline{E}_\Theta) \underline{1} &= \underline{e}_\theta \otimes (\underline{1}^t \underline{E}_\Theta) = \underline{e}_\theta \otimes \hat{\underline{E}}_\Theta, \\ (\underline{e}_3 \otimes \underline{E}_3) \underline{1} &= \underline{e}_3 \otimes (\underline{1}^t \underline{E}_3) = \underline{e}_3 \otimes \hat{\underline{E}}_3. \end{aligned}$$

Adding the three lines reproduces (3.81) exactly. Reading the addition backwards, that is factoring the common $\underline{1}$ out on the right,

$$\underline{R} = [\underline{e}_r(\theta) \otimes \underline{E}_R(\theta) + \underline{e}_\theta(\theta) \otimes \underline{E}_\Theta(\theta) + \underline{e}_3 \otimes \underline{E}_3] \underline{1}. \quad (3.82)$$

The choices $\underline{E}_R(\theta) = \underline{e}_r(\theta)$, $\underline{E}_\Theta(\theta) = \underline{e}_\theta(\theta)$, $\underline{E}_3 = \underline{e}_3$ reduce the bracketed expression to $\underline{e}_r \otimes \underline{e}_r + \underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3 = \underline{I}$, the spatial identity, by (3.11) applied to the orthonormal spatial triad. Therefore

$$\underline{R} = \underline{I} \underline{1} = \underline{1}. \quad (3.83)$$

REMARK 3.37 (what (3.83) says). There is no rotation in this deformation. Every material point moves radially outward or inward, and the local principal axes are the radial, azimuthal and axial directions both

before and after. The apparatus of shifters is needed here only to say the sentence: in the one-space reading of Chapter 2 the statement is simply $\underline{R} = \underline{I}$. Note also that when $f' = f/R$, i.e. $f = cR$, two principal stretches coincide, every direction in the $(\underline{e}_r, \underline{e}_\theta)$ plane is principal, and $\underline{U} = c(\hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta) + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3$: this is the repeated-root case anticipated in Remark 3.8, and the theory handles it without modification.

3.5.3 Azimuthal shear

What this example is for. §3.5.2 stretched the cylinder without distorting it; this one distorts it without stretching it. It is the first deformation in which a rotation is genuinely present and has to be extracted, and the first in which the amount of shear varies from point to point. The pay-off is Remark 3.45: once this is done, the next example, torsion, is the same algebra with two letters changed.

Now the radial and axial coordinates are unchanged and the azimuth is shifted by an amount depending on R :

$$r = R, \quad \theta = \Theta + \phi(R), \quad z = Z. \quad (3.84)$$

Differentiating these three relations, and remembering that ϕ depends on R alone,

$$dr = dR, \quad d\theta = d\Theta + \phi'(R) dR, \quad dz = dZ.$$

Substitute into the polar line element $d\tilde{x} = dr \underline{e}_r(\theta) + r d\theta \underline{e}_\theta(\theta) + dz \underline{e}_3$, using $r = R$ in the middle term:

$$\begin{aligned} d\tilde{x} &= dR \underline{e}_r(\theta) + R[d\Theta + \phi'(R)dR] \underline{e}_\theta(\theta) + dZ \underline{e}_3 \\ &= dR \underline{e}_r(\theta) + R\phi'(R) dR \underline{e}_\theta(\theta) + R d\Theta \underline{e}_\theta(\theta) + dZ \underline{e}_3, \end{aligned}$$

and collect the two terms carrying dR :

$$\underline{F}d\tilde{X} = [\underline{e}_r(\theta) + R\phi'(R)\underline{e}_\theta(\theta)]dR + R \underline{e}_\theta(\theta) d\Theta + \underline{e}_3 dZ. \quad (3.85)$$

REMARK 3.38 (where the two terms in the bracket come from). This bracket is the whole of the deformation, and it is worth saying in words what each term is. Move a material point outward by dR at fixed Θ and Z . Two things happen at once. Its radius grows by $dr = dR$, which is the $\underline{e}_r$ term and is no deformation at all, merely the fact that the point has moved outward. But it also lands on a cylinder that has been twisted through a *different* angle, since ϕ depends on R ; the extra azimuthal displacement is $r \Delta\theta = R\phi'(R)dR$, which is the $\underline{e}_\theta$ term. So the coefficient $\gamma = R\phi'$ is not ϕ' : the factor R is there because an angle becomes a length only after multiplication by the radius. That single factor is what makes γ vary with R even when ϕ' is constant.

Substituting $dR = \hat{E}_R \cdot d\tilde{X}$, $Rd\Theta = \hat{E}_\Theta \cdot d\tilde{X}$ and $dZ = \hat{E}_3 \cdot d\tilde{X}$, which is the referential line element of (3.74) read backwards, (3.85) becomes

$$\begin{aligned} F d\tilde{X} &= [\underline{e}_r + R\phi' \underline{e}_\theta] (\hat{E}_R \cdot d\tilde{X}) + \underline{e}_\theta (\hat{E}_\Theta \cdot d\tilde{X}) + \underline{e}_3 (\hat{E}_3 \cdot d\tilde{X}) \\ &= \left[\underline{e}_r \otimes \hat{E}_R + \underline{e}_\theta \otimes \hat{E}_\Theta + \underline{e}_3 \otimes \hat{E}_3 + R\phi' \underline{e}_\theta \otimes \hat{E}_R \right] d\tilde{X}, \end{aligned}$$

the R in $Rd\Theta$ having cancelled the R in the coefficient $R\underline{e}_\theta$ of $d\Theta$. Since $d\tilde{X}$ is arbitrary,

$$\tilde{F} = \tilde{Q} + R\phi'(R) \underline{e}_\theta(\theta) \otimes \hat{E}_R(\Theta), \quad (3.86)$$

where

$$\tilde{Q} = \underline{e}_r(\theta) \otimes \hat{E}_R(\Theta) + \underline{e}_\theta(\theta) \otimes \hat{E}_\Theta(\Theta) + \underline{e}_3 \otimes \hat{E}_3. \quad (3.87)$$

LEMMA 3.39 ($\tilde{Q}$ of (3.87) is a two-point rotation). $\tilde{Q}^t \tilde{Q} = \hat{I}$, $\tilde{Q} \tilde{Q}^t = \hat{I}$, $\det \tilde{Q} = 1$, and $\tilde{Q} \hat{E}_R = \underline{e}_r$, $\tilde{Q} \hat{E}_\Theta = \underline{e}_\theta$, $\tilde{Q} \hat{E}_3 = \underline{e}_3$.

Proof. The action on the basis is immediate from $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$ and orthonormality of $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$; for instance

$$\tilde{Q} \hat{E}_R = (\hat{E}_R \cdot \hat{E}_R) \underline{e}_r + (\hat{E}_\Theta \cdot \hat{E}_R) \underline{e}_\theta + (\hat{E}_3 \cdot \hat{E}_R) \underline{e}_3 = \underline{e}_r + 0 + 0 = \underline{e}_r.$$

By Problem 17(a), $\tilde{Q}^t = \hat{E}_R \otimes \underline{e}_r + \hat{E}_\Theta \otimes \underline{e}_\theta + \hat{E}_3 \otimes \underline{e}_3$. Multiplying out $\tilde{Q}^t \tilde{Q}$ produces nine terms, each of the form $(\hat{E}_A \otimes \underline{e}_i)(\underline{e}_j \otimes \hat{E}_B) =$

$(\underline{e}_i \cdot \underline{e}_j) \hat{E}_A \otimes \hat{E}_B$ by the dyad product rule; the six with $i \neq j$ vanish because the spatial triad is orthonormal, and the three with $i = j$ survive with coefficient 1:

$$\underset{\sim}{Q}^t \underset{\sim}{Q} = \hat{E}_R \otimes \hat{E}_R + \hat{E}_\Theta \otimes \hat{E}_\Theta + \hat{E}_3 \otimes \hat{E}_3 = \hat{I}$$

by (3.11). The same computation in the other order gives $\underset{\sim}{Q} \underset{\sim}{Q}^t = \underline{e}_r \otimes \underline{e}_r + \underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3 = \underline{I}$. Finally the determinant. Use rule (R4) with the referential frame as test triad; its images were just computed. The denominator is $[\hat{E}_R, \hat{E}_\Theta, \hat{E}_3] = 1$ and the numerator is $[\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = 1$, both by Lemma 3.28 together with Lemma 3.29, which established that each of the two frames is orthonormal and right handed. Hence

$$\det \underset{\sim}{Q} = \frac{[\underset{\sim}{Q} \hat{E}_R, \underset{\sim}{Q} \hat{E}_\Theta, \underset{\sim}{Q} \hat{E}_3]}{[\hat{E}_R, \hat{E}_\Theta, \hat{E}_3]} = \frac{[\underline{e}_r, \underline{e}_\theta, \underline{e}_3]}{1} = \frac{1}{1} = 1.$$

No scalars had to be extracted here, because $\underset{\sim}{Q}$ maps unit vectors to unit vectors: it is pure rotation with no stretching, and the determinant of a rotation is 1 precisely because a rotation moves a unit cube to a unit cube. $\square$

Since $\underline{e}_\theta(\theta) = \underset{\sim}{Q} \hat{E}_\Theta(\Theta)$, factoring $\underset{\sim}{Q}$ out on the left of (3.86) with Problem 17(b), $\underset{\sim}{A}(\underset{\sim}{a} \otimes \underset{\sim}{b}) = (\underset{\sim}{A} \underset{\sim}{a}) \otimes \underset{\sim}{b}$,

$$\underset{\sim}{F} = \underset{\sim}{Q} \hat{G}, \quad \hat{G} = \hat{I} + R\phi'(R) \hat{E}_\Theta(\Theta) \otimes \hat{E}_R(\Theta), \quad (3.88)$$

which is checked forwards: $\underset{\sim}{Q} \hat{G} = \underset{\sim}{Q} \hat{I} + R\phi' \underset{\sim}{Q} (\hat{E}_\Theta \otimes \hat{E}_R) = \underset{\sim}{Q} + R\phi' (\underset{\sim}{Q} \hat{E}_\Theta) \otimes \hat{E}_R = \underset{\sim}{Q} + R\phi' \underline{e}_\theta \otimes \hat{E}_R$. Since $\hat{E}_R(\Theta) = \underset{\sim}{Q}^t \underline{e}_r(\theta)$, factoring on the right with Problem 17(c), $(\underset{\sim}{a} \otimes \underset{\sim}{b}) \underset{\sim}{A} = \underset{\sim}{a} \otimes (\underset{\sim}{A}^t \underset{\sim}{b})$,

$$\underset{\sim}{F} = \underset{\sim}{G} \underset{\sim}{Q}, \quad \underset{\sim}{G} = \underline{I} + R\phi'(R) \underline{e}_\theta(\theta) \otimes \underline{e}_r(\theta), \quad (3.89)$$

checked the same way: $\underset{\sim}{G} \underset{\sim}{Q} = \underset{\sim}{Q} + R\phi'(\underline{e}_\theta \otimes \underline{e}_r) \underset{\sim}{Q} = \underset{\sim}{Q} + R\phi' \underline{e}_\theta \otimes (\underset{\sim}{Q}^t \underline{e}_r) = \underset{\sim}{Q} + R\phi' \underline{e}_\theta \otimes \hat{E}_R$.

The deformation is isochoric: four independent proofs

The book disposes of this in one line, “it is straightforward to show, using either representation”. Here is the whole of it, four ways, because each uses a different piece of Chapter 1 and the agreement is a check on all four. Write $\gamma = R\phi'(R)$ throughout.

(i) *From (3.88).* By the multiplicativity established at (3.34)–(3.35), valid here because the first factor is two-point and the second referential,

$$\det \underline{\underline{F}} = (\det \underline{\underline{Q}})(\det \hat{\underline{\underline{G}}}) = 1 \cdot \det \hat{\underline{\underline{G}}}.$$

By Problem 21(f) of Chapter 1, $\det(\hat{\underline{\underline{I}}} + \underline{\underline{a}} \otimes \underline{\underline{b}}) = 1 + \underline{\underline{a}} \cdot \underline{\underline{b}}$, so

$$\det \hat{\underline{\underline{G}}} = 1 + \gamma \hat{\underline{\underline{E}}}_\Theta \cdot \hat{\underline{\underline{E}}}_R = 1 + \gamma \cdot 0 = 1,$$

the inner product vanishing because $\hat{\underline{\underline{E}}}_\Theta \perp \hat{\underline{\underline{E}}}_R$. Hence $\det \underline{\underline{F}} = 1$. Note where the hypothesis enters: it is the *orthogonality of the shear pair* that kills the determinant, not the smallness of γ . A shear of any amount whatever preserves volume.

(ii) *From (3.89).* Symmetrically, $\det \underline{\underline{F}} = (\det \underline{\underline{G}})(\det \underline{\underline{Q}})$ with $\det \underline{\underline{G}} = 1 + \gamma \underline{\underline{e}}_\theta \cdot \underline{\underline{e}}_r = 1 + 0 = 1$, so again $\det \underline{\underline{F}} = 1$.

(iii) *Directly from (3.86), without factoring at all.* Problem 21(g) of Chapter 1 gives $\det(\underline{\underline{A}} + \underline{\underline{a}} \otimes \underline{\underline{b}}) = (\det \underline{\underline{A}})(1 + \underline{\underline{a}} \cdot \underline{\underline{A}}^{-t}\underline{\underline{b}})$ for invertible $\underline{\underline{A}}$. Take $\underline{\underline{A}} = \underline{\underline{Q}}$, $\underline{\underline{a}} = \gamma \underline{\underline{e}}_\theta$, $\underline{\underline{b}} = \hat{\underline{\underline{E}}}_R$. By Lemma 3.39, $\underline{\underline{Q}}^{-1} = \underline{\underline{Q}}^t$, hence $\underline{\underline{Q}}^{-t} = (\underline{\underline{Q}}^{-1})^t = (\underline{\underline{Q}}^t)^t = \underline{\underline{Q}}$, so

$$\det \underline{\underline{F}} = (\det \underline{\underline{Q}})(1 + \gamma \underline{\underline{e}}_\theta \cdot \underline{\underline{Q}} \hat{\underline{\underline{E}}}_R) = 1 \cdot (1 + \gamma \underline{\underline{e}}_\theta \cdot \underline{\underline{e}}_r) = 1.$$

(iv) *From $\underline{\underline{C}}$, with no reference to $\underline{\underline{F}}$ at all.* By Problem 21(b), $\det \underline{\underline{C}} = \det(\underline{\underline{F}}^t \underline{\underline{F}}) = (\det \underline{\underline{F}}^t)(\det \underline{\underline{F}}) = (\det \underline{\underline{F}})^2$. From (3.92) below, the matrix of $\underline{\underline{C}}$ in the ordered orthonormal basis $(\hat{\underline{\underline{E}}}_R, \hat{\underline{\underline{E}}}_\Theta, \hat{\underline{\underline{E}}}_3)$ is

$$(C_{AB}) = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

whose determinant, expanded along the third row, is $1 \cdot [(1 + \gamma^2)(1) - \gamma \cdot \gamma] = 1 + \gamma^2 - \gamma^2 = 1$. Hence $(\det \tilde{F})^2 = 1$ and $\det \tilde{F} = \pm 1$; the positive sign is selected by the occupiability of κ , (2.50).

All four give $\det \tilde{F} = 1$: by (2.50) no material volume changes, and by (2.59) the densities coincide, $\rho_\kappa = \rho$.

The image of a radial line

Take a radial material curve in κ , that is the set $\Theta = \Theta_0$, $Z = Z_0$ with R varying, whose unit tangent is $\tilde{M} = \hat{\tilde{E}}_R(\Theta_0)$. Its stretch and current tangent follow from (2.67) and (3.86), using $\hat{\tilde{E}}_R \cdot \hat{\tilde{E}}_R = 1$ and $\tilde{Q} \hat{\tilde{E}}_R = \underline{e}_r$:

$$\mu \tilde{m} = \tilde{F} \hat{\tilde{E}}_R(\Theta) = \underline{e}_r(\theta) + R\phi'(R)\underline{e}_\theta(\theta); \quad \mu = \sqrt{1 + (R\phi')^2}, \quad (3.90)$$

the length following from $|\underline{e}_r + \gamma \underline{e}_\theta|^2 = \underline{e}_r \cdot \underline{e}_r + 2\gamma \underline{e}_r \cdot \underline{e}_\theta + \gamma^2 \underline{e}_\theta \cdot \underline{e}_\theta = 1 + \gamma^2$.

REMARK 3.40 (a plane spiral, not a helix). The book calls the image a *spiral* and leaves it there. It is worth being explicit, because the next section produces a curve with a different name and the two are easy to confuse.

Parametrise the radial material line by R and push it forward through (3.84):

$$\tilde{x}(R) = R \underline{e}_r(\Theta_0 + \phi(R)) + Z_0 \underline{e}_3.$$

The axial coordinate is the constant Z_0 , so the image lies entirely in the single plane $z = Z_0$: it is a *plane* curve. Its polar equation is obtained by eliminating R through $r = R$:

$$\theta = \Theta_0 + \phi(r),$$

a curve winding around the origin with radius increasing monotonically, which is precisely what the word spiral means. Two familiar cases: $\phi(R) = k(R - A)$ gives $r = A + (\theta - \Theta_0)/k$, an Archimedean spiral; $\phi(R) = c \ln(R/A)$ gives $r = A e^{(\theta - \Theta_0)/c}$, a logarithmic spiral.

The angle ψ that the deformed curve makes with the radial direction is read off (3.90): the tangent is proportional to $\underline{e}_r + \gamma \underline{e}_\theta$, so

$$\tan \psi = \frac{\gamma |\underline{e}_\theta|}{|\underline{e}_r|} = \gamma = R\phi'(R).$$

So γ is the tangent of the angle by which the shear tilts a radius away from being radial, exactly as in (3.49) it was the tangent of the angle by which the shear tilts a vertical line. The logarithmic spiral is the case $\gamma = c$ constant, the only azimuthal shear in which the tilt is the same at every radius.

REMARK 3.41 (three curves that get confused: spiral, cylindrical helix, spherical helix). The chapter produces two curves and names them differently, and a third name is often attached to neither correctly. They are genuinely different objects.

Plane spiral (§3.5.3, above). Lies in one plane. Radius varies, axial coordinate constant. Curvature varies along it; it is not a curve of constant curvature.

Cylindrical, or circular, helix (§3.5.4, (3.98)). Lies on a cylinder $r = \text{const}$. Radius constant, axial coordinate varies linearly with the azimuth. It has *constant* curvature and *constant* torsion, and it is the only curve in three-space with both constant, which is why it is the standard example. Remark 3.44 computes both.

Spherical helix. A curve lying on a *sphere* and crossing all the meridians at one fixed angle, the loxodrome or rhumb line. Neither $\tilde{F}$ of this chapter produces one, and nothing in §3.5 is a spherical helix. The word *spherical* occurs in Chapter 3 only in §3.6.1(c), and there it qualifies a *surface*, not a curve: a spherical material *surface* is carried to an ellipsoidal surface, (3.112). A sphere going to an ellipsoid and a curve winding on a sphere are unrelated statements, and the only thing they share is the adjective. The discriminating question is always the same: *on what surface does the curve lie, and which of r, θ, z is held fixed?* Plane spiral: z fixed. Cylindrical helix: r fixed. Spherical helix: $|\underline{x}|$ fixed. Problem 5(b) below asks for the stretch of a curve with $\underline{M} = \frac{\sqrt{2}}{2}(\hat{E}_\Theta + \hat{E}_3)$, which has R fixed and is therefore a cylindrical helix, not a spiral and not a spherical anything.

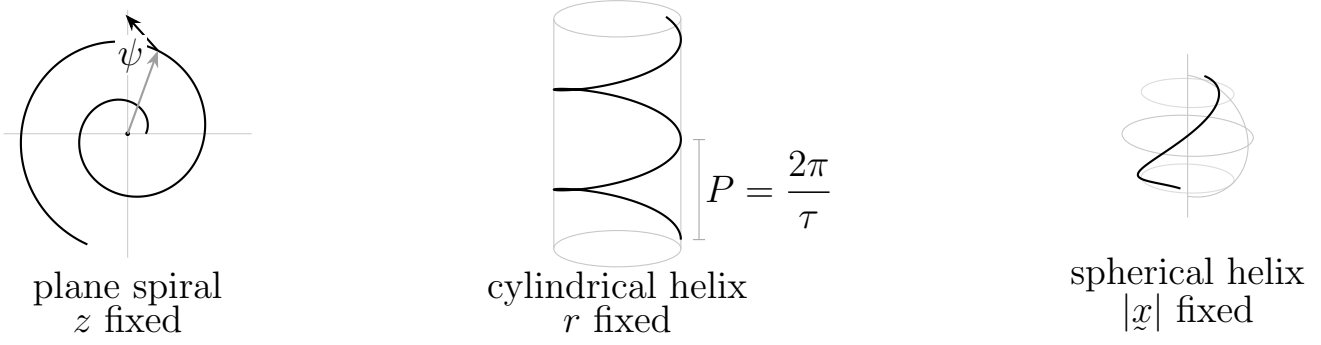


Figure C: The three curves of Remark 3.41, distinguished by which coordinate is held fixed. Left, the image of a radius under azimuthal shear: a plane curve, z constant, meeting every radius at the angle ψ with $\tan \psi = \gamma$. Centre, the image of a generator under torsion: a curve on the cylinder $r = R_0$, of constant curvature and torsion, rising by the pitch $P = 2\pi/\tau$ per turn, the same pitch on every coaxial cylinder. Right, a spherical helix, or loxodrome: a curve on a sphere crossing every meridian at one fixed angle. No deformation in this chapter produces one; it is drawn only so that the name is attached to the right object.

The Cauchy–Green tensors collapse because $\underline{Q}$ is orthogonal:

$$\underline{C} = \underline{\hat{G}}^t \underline{Q}^t \underline{Q} \underline{\hat{G}} = \underline{\hat{G}}^t \underline{\hat{I}} \underline{\hat{G}} = \underline{\hat{G}}^t \underline{\hat{G}}, \quad \underline{B} = \underline{G} \underline{Q} \underline{Q}^t \underline{G}^t = \underline{G} \underline{I} \underline{G}^t = \underline{G} \underline{G}^t. \quad (3.91)$$

Writing $\gamma(R) = R\phi'(R)$ and expanding,

$$\begin{aligned} \underline{C} &= (\underline{\hat{I}} + \gamma \underline{\hat{E}}_R \otimes \underline{\hat{E}}_\Theta)(\underline{\hat{I}} + \gamma \underline{\hat{E}}_\Theta \otimes \underline{\hat{E}}_R) \\ &= \underline{\hat{I}} + \gamma [\underline{\hat{E}}_R(\Theta) \otimes \underline{\hat{E}}_\Theta(\Theta) + \underline{\hat{E}}_\Theta(\Theta) \otimes \underline{\hat{E}}_R(\Theta)] \\ &\quad + \gamma^2(R) \underline{\hat{E}}_R(\Theta) \otimes \underline{\hat{E}}_R(\Theta), \end{aligned} \quad (3.92)$$

the last term arising from $(\underline{\hat{E}}_R \otimes \underline{\hat{E}}_\Theta)(\underline{\hat{E}}_\Theta \otimes \underline{\hat{E}}_R) = (\underline{\hat{E}}_\Theta \cdot \underline{\hat{E}}_\Theta) \underline{\hat{E}}_R \otimes \underline{\hat{E}}_R = \underline{\hat{E}}_R \otimes \underline{\hat{E}}_R$, and likewise

$$\underline{B} = \underline{I} + \gamma(R) [\underline{e}_r(\theta) \otimes \underline{e}_\theta(\theta) + \underline{e}_\theta(\theta) \otimes \underline{e}_r(\theta)] + \gamma^2(R) \underline{e}_\theta(\theta) \otimes \underline{e}_\theta(\theta). \quad (3.93)$$

Why (3.92) and (3.53) are the same formula

The book states that “on replacing $\underline{\hat{E}}_1$ by $\underline{\hat{E}}_\Theta(\Theta)$, $\underline{\hat{E}}_2$ by $\underline{\hat{E}}_R(\Theta)$ and $\underline{\hat{E}}_3$ by $-\underline{\hat{E}}_3$, $\underline{C}$ is seen to have exactly the same form as its counterpart

(3.53)". The claim is correct, but what a *replacement* of basis vectors is allowed to mean needs saying, and the substitution is short enough to do in full.

What the replacement is. Define

$$K := \hat{E}_\Theta \otimes \hat{E}_1 + \hat{E}_R \otimes \hat{E}_2 - \hat{E}_3 \otimes \hat{E}_3,$$

so that $K\hat{E}_1 = \hat{E}_\Theta$, $K\hat{E}_2 = \hat{E}_R$, $K\hat{E}_3 = -\hat{E}_3$. This is a referential tensor, and by exactly the computation of Lemma 3.39 it is orthogonal, since it pairs the orthonormal triad $\{\hat{E}_A\}$ with the triad $\{\hat{E}_\Theta, \hat{E}_R, -\hat{E}_3\}$. That second triad is orthonormal because $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$ is, and it is *right handed* because

$$\hat{E}_\Theta \times \hat{E}_R = -(\hat{E}_R \times \hat{E}_\Theta) = -\hat{E}_3,$$

which is the third member. Hence $\det K = +1$: the replacement is a *rotation of the referential frame*, not merely a relabelling.

The substitution, term by term. Apply the replacement to (3.53). The identity goes to itself:

$$\begin{aligned} \hat{I} &= \hat{E}_1 \otimes \hat{E}_1 + \hat{E}_2 \otimes \hat{E}_2 + \hat{E}_3 \otimes \hat{E}_3 \\ &\longmapsto \hat{E}_\Theta \otimes \hat{E}_\Theta + \hat{E}_R \otimes \hat{E}_R + (-\hat{E}_3) \otimes (-\hat{E}_3) \\ &= \hat{E}_R \otimes \hat{E}_R + \hat{E}_\Theta \otimes \hat{E}_\Theta + \hat{E}_3 \otimes \hat{E}_3 = \hat{I}, \end{aligned}$$

the two minus signs multiplying to +1 and (3.11) closing the sum because $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$ is orthonormal. The shear term goes to

$$\gamma(\hat{E}_1 \otimes \hat{E}_2 + \hat{E}_2 \otimes \hat{E}_1) \longmapsto \gamma(\hat{E}_\Theta \otimes \hat{E}_R + \hat{E}_R \otimes \hat{E}_\Theta),$$

and the square term to

$$\gamma^2 \hat{E}_2 \otimes \hat{E}_2 \longmapsto \gamma^2 \hat{E}_R \otimes \hat{E}_R.$$

Adding the three lines,

$$\hat{I} + \gamma(\hat{E}_R \otimes \hat{E}_\Theta + \hat{E}_\Theta \otimes \hat{E}_R) + \gamma^2 \hat{E}_R \otimes \hat{E}_R,$$

which is (3.92) exactly. Nothing is left over and nothing is missing, so the two equations are the same tensor written in two frames. In symbols the statement is $\underline{C}_{\text{azim}} = \underline{K} \underline{C}_{\text{simple}} \underline{K}^t$ with γ replaced by $\gamma(R)$.

The same works for $\underline{B}$. Replacing $\underline{e}_1$ by $\underline{e}_\theta(\theta)$, $\underline{e}_2$ by $\underline{e}_r(\theta)$ and $\underline{e}_3$ by $-\underline{e}_3$ in (3.54),

$$\begin{aligned} (1 + \gamma^2) \underline{e}_1 \otimes \underline{e}_1 &\longmapsto (1 + \gamma^2) \underline{e}_\theta \otimes \underline{e}_\theta, \\ \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3 &\longmapsto \underline{e}_r \otimes \underline{e}_r + \underline{e}_3 \otimes \underline{e}_3, \\ \gamma(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1) &\longmapsto \gamma(\underline{e}_\theta \otimes \underline{e}_r + \underline{e}_r \otimes \underline{e}_\theta), \end{aligned}$$

and adding, with $\underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_r \otimes \underline{e}_r + \underline{e}_3 \otimes \underline{e}_3 = \underline{I}$, gives $\underline{I} + \gamma(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r) + \gamma^2 \underline{e}_\theta \otimes \underline{e}_\theta$, which is (3.93). The book omits the third replacement here; it is invisible in $\underline{B}$ for the same reason it was invisible in $\underline{C}$, since $\underline{e}_3$ enters only through $\underline{e}_3 \otimes \underline{e}_3$, but it is needed for the same reason, to keep the frame right handed.

REMARK 3.42 (so is anything wrong? two caveats the book does not state). Nothing is wrong. But two things are true that the bare word “replacement” hides, and both matter later.

(i) *The sign on $\hat{\underline{E}}_3$ is invisible in $\underline{C}$ and indispensable elsewhere.* As the substitution above shows, $\hat{\underline{E}}_3$ enters (3.53) only inside $\hat{\underline{I}}$, as $\hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3$, which is insensitive to the sign. So one could drop the minus sign and still land on (3.92). One may not drop it, because without it $\det \underline{K} = -1$ and $\underline{K}$ is a reflection; conclusions about $\underline{C}$ would survive, but conclusions about the *rotation* $\underline{R}$, about the handedness of the principal triad, and about the sign of $\det \underline{F}$ would not. The minus sign is doing work that only becomes visible one equation later.

(ii) *The order of the shear pair is reversed, and this flips the sign of the rotation angle.* In (3.53) the shear direction is $\hat{\underline{E}}_1$ and the shear normal is $\hat{\underline{E}}_2$; the replacement sends these to $\hat{\underline{E}}_\theta$ and $\hat{\underline{E}}_r$ respectively. But the natural ordering of the cylindrical frame is $(\hat{\underline{E}}_r, \hat{\underline{E}}_\theta, \hat{\underline{E}}_3)$, radial first. So the pair that carries the shear appears in the *opposite* order from the one in §3.5.1, and the polar rotation comes out with the opposite sign: Remark 3.33 found $\tan \beta = -\gamma/2$ for simple shear, whereas the computation below gives $\tan \hat{\beta} = +\gamma/2$ for azimuthal shear. There is

no contradiction, and the physical check confirms it: with $\phi' > 0$ the outer cylinders lead, a radius is dragged towards increasing θ , and the material turns counterclockwise, so a positive angle is right.

(iii) *The replacement is different at every material point.* Since $\hat{\underline{E}}_R$ and $\hat{\underline{E}}_\Theta$ depend on Θ , the tensor $\underline{K}$ above is $\underline{K}(\Theta)$, and $\gamma = \gamma(R)$ depends on R . So the transfer of results from §3.5.1 is valid *pointwise* and only pointwise. The principal *stretches* depend on R alone, but the principal *directions* depend on Θ as well, and $\underline{U}$, $\underline{V}$, $\underline{R}$ are fields, not constants, as they were in the homogeneous case.

The polar factors, explicitly

The book says only that $\underline{U}$ and $\underline{V}$ “are obtained as they were in Section 3.5.1 with the indicated replacements”, and that the rotations of (3.94) “are computed in the same way”. Since Problem 9 of §3.8 gives a closed form that needs no eigenvectors, here they are.

Work in the plane spanned by $(\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta)$, the third direction being inert since $\underline{C}\hat{\underline{E}}_3 = \hat{\underline{E}}_3$. From (3.92),

$$(\underline{C}_{AB}) = \begin{pmatrix} 1 + \gamma^2 & \gamma \\ \gamma & 1 \end{pmatrix}, \quad (\hat{\underline{G}}_{AB}) = \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix},$$

in the ordered basis $(\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta)$; the matrix of $\hat{\underline{G}}$ has its γ in the (Θ, R) slot because $\hat{\underline{G}} = \hat{\underline{I}} + \gamma \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_R$. Then

$$J = \sqrt{\det \underline{C}} = \sqrt{(1 + \gamma^2) - \gamma^2} = 1, \\ I = \sqrt{\text{tr } \underline{C} + 2J} = \sqrt{2 + \gamma^2 + 2} = \sqrt{4 + \gamma^2},$$

and Problem 9 gives $\underline{U} = (\underline{C} + J\hat{\underline{I}})/I$:

$$\underline{U} = \frac{1}{\sqrt{4 + \gamma^2}} \left[2(\hat{\underline{E}}_R \otimes \hat{\underline{E}}_R + \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_\Theta) \right. \\ \left. + \gamma(\hat{\underline{E}}_R \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_R) + \gamma^2 \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R \right] + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3,$$

with matrix $\frac{1}{I} \begin{pmatrix} 2+\gamma^2 & \gamma \\ \gamma & 2 \end{pmatrix}$ in the plane. The check $\underline{U}^2 = \underline{C}$ is the computation of Remark 3.51 with the two indices interchanged. The principal

stretches are the roots of $\lambda^2 - I\lambda + J = 0$, that is

$$\lambda_{1,2} = \frac{1}{2} \left(\sqrt{4 + \gamma^2} \pm \gamma \right), \quad \lambda_3 = 1, \quad \gamma = R\phi'(R),$$

which vary from point to point, unlike §3.5.1.

For the rotation, invert $\underline{U}$ by the same identity: multiplying $\underline{U}^2 - I\underline{U} + J\hat{I} = \hat{Q}$ by $\underline{U}^{-1}$ gives $\underline{U}^{-1} = (I\hat{I} - \underline{U})/J = I\hat{I} - \underline{U}$, so in the plane

$$(U_{AB}^{-1}) = \frac{1}{I} \begin{pmatrix} (4 + \gamma^2) - (2 + \gamma^2) & -\gamma \\ -\gamma & (4 + \gamma^2) - 2 \end{pmatrix} = \frac{1}{I} \begin{pmatrix} 2 & -\gamma \\ -\gamma & 2 + \gamma^2 \end{pmatrix},$$

and therefore

$$\begin{aligned} (R_{\hat{G}})_{AB} &= (\hat{G}_{AC})(U_{CB}^{-1}) = \frac{1}{I} \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix} \begin{pmatrix} 2 & -\gamma \\ -\gamma & 2 + \gamma^2 \end{pmatrix} \\ &= \frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 2 & -\gamma \\ \gamma & 2 \end{pmatrix}, \end{aligned}$$

the (2, 1) entry being $2\gamma - \gamma = \gamma$ and the (2, 2) entry $-\gamma^2 + 2 + \gamma^2 = 2$. This is orthogonal, each column having length $\sqrt{4 + \gamma^2}/I = 1$ and the columns being orthogonal since $-2\gamma + 2\gamma = 0$, with determinant $(4 + \gamma^2)/I^2 = 1$. So $R_{\hat{G}}$ is the rotation of the $(\hat{E}_R, \hat{E}_\Theta)$ plane through the angle $\hat{\beta}$ with

$$\cos \hat{\beta} = \frac{2}{\sqrt{4 + \gamma^2}}, \quad \sin \hat{\beta} = \frac{\gamma}{\sqrt{4 + \gamma^2}}, \quad \tan \hat{\beta} = \frac{\gamma}{2} = \frac{R\phi'(R)}{2},$$

positive, that is counterclockwise, as Remark 3.42(ii) predicted.

The spatial side is identical with hats removed. From (3.93), $(B_{ij}) = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix}$ in the ordered basis $(\underline{e}_r, \underline{e}_\theta)$, with the same I and J since $\underline{B} = \underline{R}\underline{C}\underline{R}^t$ has the same invariants, so

$$\underline{V} = \frac{2(\underline{e}_r \otimes \underline{e}_r + \underline{e}_\theta \otimes \underline{e}_\theta) + \gamma(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r) + \gamma^2 \underline{e}_\theta \otimes \underline{e}_\theta}{\sqrt{4 + \gamma^2}} + \underline{e}_3 \otimes \underline{e}_3,$$

and $R_G = \underline{V}^{-1}\underline{G}$ has the same matrix $\frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 2 & -\gamma \\ \gamma & 2 \end{pmatrix}$ in $(\underline{e}_r, \underline{e}_\theta)$. Notice which dyad carries the γ^2 : it is $\hat{E}_R \otimes \hat{E}_R$ in $\underline{U}$ and $\underline{e}_\theta \otimes \underline{e}_\theta$ in $\underline{V}$, the index

having moved from the shear normal to the shear direction, exactly as Remark 3.31 says it must.

Finally, from (3.91) the right polar factorisation of $\hat{G}$ and the left polar factorisation of G are

$$\hat{G} = \hat{R}_{\hat{G}} \hat{U}, \quad G = \hat{V} \hat{R}_G, \quad (3.94)$$

with $\hat{R}_{\hat{G}}$ referential and $\hat{R}_G$ spatial. Substituting into (3.88) and (3.89),

$$\hat{F} = \hat{Q} \hat{G} = \hat{Q} \hat{R}_{\hat{G}} \hat{U}, \quad \hat{F} = \hat{G} \hat{Q} = \hat{V} \hat{R}_G \hat{Q}.$$

In the first, $\hat{U}$ is symmetric and positive definite and $\hat{Q} \hat{R}_{\hat{G}}$ is a rotation, being a product of rotations by Problem 2 of §3.8; in the second, $\hat{V}$ is symmetric positive definite and $\hat{R}_G \hat{Q}$ is a rotation for the same reason. Both are therefore polar decompositions of $\hat{F}$, and by the uniqueness proved in Step 7 of §3.4 their rotation factors must be the one and only $\hat{R}$:

$$\hat{R} = \hat{Q} \hat{R}_{\hat{G}} = \hat{R}_G \hat{Q}. \quad (3.95)$$

The two expressions for $\hat{R}$ are equal as a consequence, not by inspection; equating them gives the useful by-product $\hat{R}_G = \hat{Q} \hat{R}_{\hat{G}} \hat{Q}^t$, which says that the spatial and referential shear rotations are the same rotation seen in the two frames, and explains why their matrices came out identical above.

3.5.4 Torsion

What this example is for. It is the last of the four and it is included mainly to be recognised: nothing new happens. Torsion differs from azimuthal shear only in *which* coordinate the azimuth is made to depend on, Z instead of R , and every formula transfers with $\hat{E}_R \mapsto \hat{E}_3$. What is genuinely new is the geometric reading, the generator becoming a helix, which Remarks 3.43 and 3.44 develop well past the book.

Consider

$$r = R, \quad \theta = \Theta + \tau Z, \quad z = Z, \quad (3.96)$$

where the constant τ is the *twist*, the rate of change of azimuth with axial coordinate. A cross-section $Z = \text{const}$ is rotated rigidly through the angle τZ , different sections by different amounts, which is what twisting a bar does.

The book compresses this into “the usual procedure leads again to (3.88) and (3.89)”. Written out, the usual procedure is this. Differentiating (3.96),

$$dr = dR, \quad d\theta = d\Theta + \tau dZ, \quad dz = dZ,$$

the middle relation because τ is a constant and θ depends on Z as well as on Θ . Substituting into the polar line element with $r = R$,

$$\begin{aligned} d\tilde{x} &= dR \underline{e}_r(\theta) + R[d\Theta + \tau dZ] \underline{e}_\theta(\theta) + dZ \underline{e}_3 \\ &= dR \underline{e}_r(\theta) + R d\Theta \underline{e}_\theta(\theta) + [R\tau \underline{e}_\theta(\theta) + \underline{e}_3] dZ, \end{aligned}$$

the terms in dZ having been collected. Now insert $dR = \hat{E}_R \cdot d\tilde{X}$, $R d\Theta = \hat{E}_\Theta \cdot d\tilde{X}$, $dZ = \hat{E}_3 \cdot d\tilde{X}$:

$$\begin{aligned} d\tilde{x} &= \underline{e}_r(\hat{E}_R \cdot d\tilde{X}) + \underline{e}_\theta(\hat{E}_\Theta \cdot d\tilde{X}) + (R\tau \underline{e}_\theta + \underline{e}_3)(\hat{E}_3 \cdot d\tilde{X}) \\ &= \left[\underline{e}_r \otimes \hat{E}_R + \underline{e}_\theta \otimes \hat{E}_\Theta + \underline{e}_3 \otimes \hat{E}_3 + R\tau \underline{e}_\theta \otimes \hat{E}_3 \right] d\tilde{X}, \end{aligned}$$

so that $\tilde{F} = \tilde{Q} + R\tau \underline{e}_\theta \otimes \hat{E}_3$ with the same $\tilde{Q}$ of (3.87). Compare this with (3.86): the extra dyad now has $\hat{E}_3$ in its second slot instead of $\hat{E}_R$, and that is the *only* difference between torsion and azimuthal shear. Factoring $\tilde{Q}$ out on either side exactly as before, using $\underline{e}_\theta = \tilde{Q} \hat{E}_\Theta$ on the left and $\hat{E}_3 = \tilde{Q}^t \underline{e}_3$ on the right, gives (3.88) and (3.89) again, now with

$$\hat{G} = \hat{I} + R\tau \hat{E}_\Theta(\Theta) \otimes \hat{E}_3, \quad \mathcal{G} = \mathcal{I} + R\tau \underline{e}_\theta(\theta) \otimes \underline{e}_3. \quad (3.97)$$

This is again a shearing deformation, of variable amount $\gamma(R) = R\tau$, now on the orthonormal pair $(\hat{E}_\Theta, \hat{E}_3)$, and it is isochoric by any of the four arguments given for azimuthal shear; the first of them reads $\det \hat{G} = 1 + R\tau \hat{E}_\Theta \cdot \hat{E}_3 = 1 + 0 = 1$.

REMARK 3.43 (what a *generator* of a cylinder is). A right circular cylinder is a *ruled* surface: it is swept out by translating a straight line along a circle, and the straight lines so produced, the ones lying in the surface and parallel to the axis, are its *generators*, or generatrices. In the referential coordinates of (3.73) a generator is the material line

$$R = R_0, \quad \Theta = \Theta_0, \quad Z \text{ varying}, \quad \underline{\tilde{X}}(Z) = R_0 \hat{\underline{\tilde{E}}}_R(\Theta_0) + Z \hat{\underline{\tilde{E}}}_3,$$

and its unit tangent is $\underline{\tilde{M}} = d\underline{\tilde{X}}/dZ = \hat{\underline{\tilde{E}}}_3$, since R_0 and Θ_0 are fixed and $\hat{\underline{\tilde{E}}}_R(\Theta_0)$ is therefore a constant vector. That is why the book takes $\underline{\tilde{M}} = \hat{\underline{\tilde{E}}}_3$: it is not an arbitrary choice of test direction but the tangent to the one family of *straight* material lines that the undeformed cylinder contains. A cylinder has two such rulings only if it is degenerate; the circular cylinder has exactly one, the axial family, and the circles $Z = \text{const}$ are of course not straight.

A generator therefore maps to

$$\mu \underline{\tilde{m}} = \underline{\tilde{F}} \hat{\underline{\tilde{E}}}_3 = \underline{\tilde{Q}} \hat{\underline{\tilde{E}}}_3 + R\tau \underline{\tilde{e}}_\theta = \underline{\tilde{e}}_3 + R\tau \underline{\tilde{e}}_\theta(\theta); \quad \mu = \sqrt{1 + (R\tau)^2}, \quad (3.98)$$

using $\underline{\tilde{Q}} \hat{\underline{\tilde{E}}}_3 = \underline{\tilde{e}}_3$ from Lemma 3.39 and $\hat{\underline{\tilde{E}}}_3 \cdot \hat{\underline{\tilde{E}}}_3 = 1$, with $|\underline{\tilde{e}}_3 + \gamma \underline{\tilde{e}}_\theta|^2 = 1 + \gamma^2$ by orthonormality.

REMARK 3.44 (the image is a circular helix, and everything about it). The book says the generator goes to “a helix” and stops. Here is the curve and its geometry, none of which is in the text.

Push the generator forward through (3.96). A point at axial station Z has $r = R_0$, $\theta = \Theta_0 + \tau Z$, $z = Z$, so

$$\underline{\tilde{x}}(Z) = R_0 \underline{\tilde{e}}_r(\Theta_0 + \tau Z) + Z \underline{\tilde{e}}_3.$$

The radius is constant, so the curve lies on the cylinder $r = R_0$, and the azimuth is an affine function of the axial coordinate. That is the definition of a *circular*, or cylindrical, helix. Differentiating, and using $d\underline{\tilde{e}}_r/d\theta = \underline{\tilde{e}}_\theta$ together with $d\theta/dZ = \tau$,

$$\frac{d\underline{\tilde{x}}}{dZ} = R_0\tau \underline{\tilde{e}}_\theta + \underline{\tilde{e}}_3, \quad \left| \frac{d\underline{\tilde{x}}}{dZ} \right| = \sqrt{1 + R_0^2\tau^2} = \mu,$$

which is (3.98) obtained a second way, by differentiating the curve rather than by applying $\tilde{F}$ to its tangent. The agreement is the content of (2.67).

Pitch. One full turn needs $\Delta\theta = 2\pi$, hence $\Delta Z = 2\pi/\tau$. So the pitch is

$$P = \frac{2\pi}{\tau},$$

independent of R_0 : every generator of every coaxial cylinder becomes a helix of the same pitch, which is exactly what it means for the cross-sections to rotate rigidly. The helices differ only in radius.

Helix angle. The angle α_h between the deformed generator and the axis satisfies

$$\tan \alpha_h = \frac{R_0\tau |\underline{e}_\theta|}{|\underline{e}_3|} = R_0\tau = \gamma.$$

So γ is again the tangent of a tilt angle, as it was in (3.49) for the vertical line and in Remark 3.40 for the radius. The outer fibres are tilted most, the axis not at all, which is why a twisted bar fails at its surface.

Curvature and torsion. Writing $a = R_0$ and $b = 1/\tau$, the curve is the standard helix $(a \cos t, a \sin t, bt)$ with $t = \tau Z$, whose curvature and geometric torsion are

$$\kappa = \frac{a}{a^2 + b^2} = \frac{R_0\tau^2}{1 + R_0^2\tau^2}, \quad \varsigma = \frac{b}{a^2 + b^2} = \frac{\tau}{1 + R_0^2\tau^2}.$$

Both are constants along the curve, and the circular helix is the only curve in three-space for which both are constant and non-zero. Two checks: as $\tau \rightarrow 0$ both tend to 0 and the helix straightens back into the generator; and $\kappa/\varsigma = R_0\tau = \gamma$, so the ratio of curvature to torsion is the amount of shear.

REMARK 3.45 (three shears, one algebra). Sections 3.5.1, 3.5.3 and 3.5.4 are the same deformation three times. Each has $\hat{G} = \hat{I} + \gamma \underline{a} \otimes \underline{b}$ with $\underline{a}, \underline{b}$ orthonormal: $(\hat{E}_1, \hat{E}_2)$ for simple shear, $(\hat{E}_\Theta, \hat{E}_R)$ for azimuthal shear, $(\hat{E}_\Theta, \hat{E}_3)$ for torsion. Every consequence therefore follows once and for all from that single form. In particular $\det \hat{G} = 1$ by Problem 21(f), the principal stretches are $\frac{1}{2}(\gamma \pm \sqrt{4 + \gamma^2})$ and 1 by (3.60), and

the stretch tensors are given by Problem 9 of §3.8 in closed form. The only difference between the three is which pair of orthonormal directions carries the shear, and whether γ is constant. Recognising this saves solving three characteristic equations.

3.6 Homogeneous deformations

Where we are and why this section exists

Sections 3.1 to 3.4 built the machinery, $\underline{F} = \underline{R}\underline{U}$, and §3.5 exercised it on four deformations chosen because they can be written down. The obvious complaint is that real deformations cannot be written down. This section answers the complaint. It isolates the deformations for which $\underline{F}$ does not vary from point to point, proves three things about them, and then observes, at (3.102), that *every* deformation looks like one of these under a microscope. So the three properties proved here are the local structure of every deformation there is, and that is why the section is worth the trouble.

The definition and its integrated form

A deformation is *homogeneous* if $\underline{F}$ is the same at every material point, i.e. independent of $\underline{X}$. What does that buy? Start from the definition of $\underline{F}$, which is $d\underline{x} = \underline{F} d\underline{X}$, and ask whether the right-hand side is itself the differential of something. In general it is not, because $\underline{F}$ varies. If $\underline{F}$ is uniform it is, because a constant tensor passes through a differential untouched: $d(\underline{F}\underline{X}) = \underline{F} d\underline{X}$ exactly, with no term $(d\underline{F})\underline{X}$ to worry about. Hence

$$d\underline{x} = \underline{F} d\underline{X} = d(\underline{F}\underline{X}), \quad (3.99)$$

so $d(\underline{x} - \underline{F}\underline{X}) = \underline{0}$, and a field whose differential vanishes identically is constant in $\underline{X}$, though it may still depend on t , which the differential at fixed t cannot see. Calling that constant $\underline{c}(t)$,

$$\underline{x} = \underline{\chi}_\kappa(\underline{X}, t) = \underline{F}(t)\underline{X} + \underline{c}(t). \quad (3.100)$$

So homogeneous means *affine*: a fixed linear map plus a translation. The translation $\underline{c}$ is a rigid displacement of the whole body and carries no deformation, which is why it never appeared in §3.5, where only $\underline{F}$ was ever computed. Simple shear, §3.5.1, is homogeneous, as (3.49) shows directly; the three cylindrical deformations of §§3.5.2–3.5.4 are not, since in each the coefficients of $\underline{F}$ depend on R .

For homogeneous deformations, write (3.100) at two material points and subtract:

$$\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t) = [\underline{F}\underline{Y} + \underline{c}] - [\underline{F}\underline{X} + \underline{c}] = \underline{F}(\underline{Y} - \underline{X}),$$

the translations cancelling, which gives

$$\chi_\kappa(\underline{Y}, t) = \chi_\kappa(\underline{X}, t) + \underline{F}(t)(\underline{Y} - \underline{X}), \quad (3.101)$$

exactly, for *any* two points however far apart. For a general differentiable deformation (1.127) and (1.128) give only

$$\chi_\kappa(\underline{Y}, t) = \chi_\kappa(\underline{X}, t) + \underline{F}(\underline{X}, t)(\underline{Y} - \underline{X}) + o(\underline{Y} - \underline{X}). \quad (3.102)$$

So every smooth deformation is approximately homogeneous in a small enough neighbourhood of a material point. That is the reason for studying the class: it describes the local structure of everything else.

3.6.1 Some properties of homogeneous deformations

(a) Material planes deform to planes; parallel planes map to parallel planes

Consider a fixed plane in κ with uniform unit normal $\underline{N}$. Its differential equation is

$$0 = d\underline{X} \cdot \underline{N} = d(\underline{X} \cdot \underline{N}), \quad (3.103)$$

the second form because $\underline{N}$ is constant, and this integrates to

$$\underline{X} \cdot \underline{N} = D, \quad (3.104)$$

D being the perpendicular distance from the origin to the plane, and parallel planes being distinguished by the value of D alone.

Invert (3.100) to get $\tilde{X} = \tilde{F}^{-1}(\tilde{x} - \tilde{c})$ and substitute into (3.104). Moving $\tilde{F}^{-1}$ across the dot product by the definition of the transpose, $\tilde{A}\tilde{u} \cdot \tilde{v} = \tilde{u} \cdot \tilde{A}^t\tilde{v}$ applied with $\tilde{A} = \tilde{F}^{-1}$ and $\tilde{A}^t = \tilde{F}^{-t}$ by Problem 18 of Chapter 1, and then distributing,

$$D = \tilde{F}^{-1}(\tilde{x} - \tilde{c}) \cdot \tilde{N} = (\tilde{x} - \tilde{c}) \cdot \tilde{F}^{-t}\tilde{N} = \tilde{x} \cdot \tilde{F}^{-t}\tilde{N} - \tilde{c} \cdot \tilde{F}^{-t}\tilde{N},$$

so, moving the last term to the other side,

$$\tilde{x} \cdot \tilde{F}^{-t}\tilde{N} = d, \quad d = D + \tilde{c} \cdot \tilde{F}^{-t}\tilde{N}. \quad (3.105)$$

The left-hand side is linear in $\tilde{x}$ with a constant coefficient vector, so this is the equation of a plane in κ_t whose normal is parallel to $\tilde{F}^{-t}\tilde{N}$; the Piola–Nanson formulas (2.54) and (2.55) say the same thing, since $\tilde{n} da = \tilde{F}^*\tilde{N} dA = J\tilde{F}^{-t}\tilde{N} dA$. Different values of D give different values of d with the same coefficient vector, so parallel planes map to parallel planes.

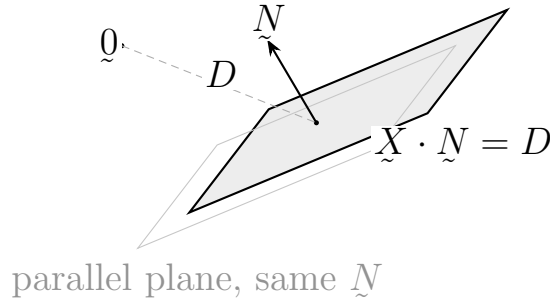


Figure 3.4: A material plane in κ . It is fixed by its unit normal $\tilde{N}$ and the perpendicular distance D from the origin; parallel planes share $\tilde{N}$ and differ only in D . Under a homogeneous deformation the whole family goes to a parallel family with common normal along $\tilde{F}^{-t}\tilde{N}$.

(b) Straight lines deform to straight lines; parallel lines map to parallel lines

The method is the same as in (a), one dimension lower: characterise the object by a differential equation, integrate it, push it through (3.100), and recognise what comes out.

A straight line is the curve whose unit tangent never changes. Writing S for arclength, that is $d\tilde{X}/dS = \tilde{M}$ with $\tilde{M}$ a fixed vector, or

$$d\tilde{X} = \tilde{M} dS = d(\tilde{M}S), \quad (3.106)$$

the second form because $\tilde{M}$ is constant and so passes through the differential. Hence $d(\tilde{X} - \tilde{M}S) = 0$, the bracket is constant, and naming that constant $\tilde{X}_0 = \tilde{X}(0)$,

$$\tilde{X}(S) = \tilde{M}S + \tilde{X}_0, \quad \tilde{X}_0 = \tilde{X}(0). \quad (3.107)$$

Now push it through (3.100), term by term, using linearity of $\tilde{F}$:

$$\tilde{x}(S) = \tilde{F}\tilde{X}(S) + \tilde{c} = \tilde{F}(\tilde{M}S + \tilde{X}_0) + \tilde{c} = S \tilde{F}\tilde{M} + \underbrace{\tilde{F}\tilde{X}_0 + \tilde{c}}_{=: \tilde{x}_0}.$$

The image is therefore of the form (constant vector) times (scalar parameter) plus (fixed point), which is again a straight line. To display it in the same normalised form as (3.107), write $\tilde{F}\tilde{M}$ as its length times its direction, $\tilde{F}\tilde{M} = \mu \tilde{m}$ with

$$\mu = |\tilde{F}\tilde{M}|, \quad \tilde{m} = \mu^{-1}\tilde{F}\tilde{M}, \quad |\tilde{m}| = 1,$$

μ being the stretch of the line by (2.67), and put $s = \mu S$. Then $S \tilde{F}\tilde{M} = S\mu \tilde{m} = s \tilde{m}$ and

$$\tilde{x}(s) = s \tilde{m} + \tilde{x}_0, \quad (3.108)$$

in which s is arclength on the deformed line, since $\tilde{m}$ is a unit vector and $|d\tilde{x}| = |\tilde{m}| ds = ds$.

Two consequences read straight off. The deformed tangent $\tilde{m}$ depends only on $\tilde{M}$, not on $\tilde{X}_0$; so *parallel* lines in κ , which by definition share $\tilde{M}$ and differ only in $\tilde{X}_0$, go to lines sharing $\tilde{m}$ and differing only in $\tilde{x}_0$: they stay parallel. And the stretch $\mu = |\tilde{F}\tilde{M}|$ likewise depends only on $\tilde{M}$: every line of a parallel family is stretched by the same factor.

REMARK 3.46 (the converse, which the book does not state). Property (b) is proved in one direction: homogeneous implies straight lines to straight lines. The converse is also true, and it is the more useful half,

because it turns a *geometric* observation into a statement about $\underline{F}$. Suppose only that the deformation is invertible and twice differentiable and that it carries every straight material line onto a straight line. Then it is homogeneous.

Proof. Let $\underline{X}(S) = \underline{X}_0 + S\underline{M}$ be any line. Its image is by hypothesis a straight line, so $\underline{x}(S) = \underline{x}_0 + g(S)\underline{m}$ for some scalar function g and some fixed unit vector $\underline{m}$ depending on $\underline{M}$ but not on S . Differentiating and using the chain rule,

$$\underline{F}(\underline{X}_0 + S\underline{M})\underline{M} = \underline{x}'(S) = g'(S)\underline{m}.$$

So for every $\underline{X}$ and every $\underline{M}$, the vector $\underline{F}(\underline{X})\underline{M}$ points along a direction $\underline{m}(\underline{M})$ that does not depend on $\underline{X}$: only its length does. Write $\underline{F}(\underline{X})\underline{M} = \mu(\underline{X}, \underline{M})\underline{m}(\underline{M})$. Fix two points $\underline{X}_1$ and $\underline{X}_2$ and put $\underline{A} = \underline{F}(\underline{X}_1)^{-1}\underline{F}(\underline{X}_2)$. Then, for every $\underline{M}$,

$$\underline{A}\underline{M} = \underline{F}(\underline{X}_1)^{-1}[\mu(\underline{X}_2, \underline{M})\underline{m}(\underline{M})] = \frac{\mu(\underline{X}_2, \underline{M})}{\mu(\underline{X}_1, \underline{M})}\underline{M},$$

so every vector is a principal vector of $\underline{A}$. A tensor with that property is a multiple of the identity: for independent $\underline{u}, \underline{v}$ with $\underline{A}\underline{u} = \alpha\underline{u}$, $\underline{A}\underline{v} = \beta\underline{v}$ and $\underline{A}(\underline{u} + \underline{v}) = \sigma(\underline{u} + \underline{v})$, subtracting gives $(\alpha - \sigma)\underline{u} + (\beta - \sigma)\underline{v} = \underline{0}$, hence $\alpha = \beta = \sigma$ by independence. Therefore

$$\underline{F}(\underline{X}) = c(\underline{X})\underline{F}_0, \quad \underline{F}_0 := \underline{F}(\underline{X}_1),$$

for a positive scalar field c with $c(\underline{X}_1) = 1$.

It remains to show c is constant, and here the fact that $\underline{F}$ is a *gradient* does the work. From $F_{iA} = \chi_{i,A}$ and the equality of mixed partial derivatives, $F_{iA,B} = \chi_{i,AB} = \chi_{i,BA} = F_{iB,A}$. Applying this to $F_{iA} = c(\underline{F}_0)_{iA}$, in which $\underline{F}_0$ is constant,

$$c_{,B}(\underline{F}_0)_{iA} = c_{,A}(\underline{F}_0)_{iB} \quad \text{for all } i, A, B.$$

Read with i fixed, this says $\underline{g} \otimes \underline{f}_i = \underline{f}_i \otimes \underline{g}$, where $\underline{g} = \text{Grad } c$ and $\underline{f}_i$ is the i th row of $\underline{F}_0$; by Problem 17(a) that forces $\underline{g}$ parallel to $\underline{f}_i$ or $\underline{g} = \underline{0}$. The three rows of the invertible $\underline{F}_0$ are linearly independent, so no single

non-zero $\underline{g}$ can be parallel to all three. Hence $\text{Grad } c = \underline{0}$, c is constant, $c \equiv 1$, and $\underline{F} \equiv \underline{F}_0$. $\square$

The moral is that “straight lines stay straight” is not one property of homogeneous deformations among many: it *characterises* them. The same is true of (a): a deformation carrying every material plane to a plane is homogeneous, by the same argument applied to $\underline{F}^{-t}$ and normals. Property (c) is not characterising, since every deformation carries small spheres to approximate ellipsoids by (3.102); what is special in the homogeneous case is that the statement is exact and the ellipsoid is the same at every point.

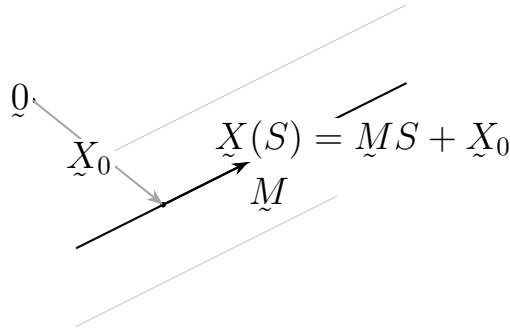


Figure 3.5: A material line in κ , fixed by a point $\underline{X}_0$ on it and its uniform unit tangent $\underline{M}$. Parallel lines share $\underline{M}$. A homogeneous deformation sends the family to a parallel family with common tangent $\underline{m} = \underline{F}\underline{M}/|\underline{F}\underline{M}|$ and common stretch $\mu = |\underline{F}\underline{M}|$.

(c) A spherical material surface is mapped to the surface of an ellipsoid

Consider a unit sphere in κ centred at $\underline{X}$, and let $\underline{Y}$ be a point on it, so that the radius vector is $\hat{\underline{r}} = \underline{Y} - \underline{X}$ with $|\hat{\underline{r}}| = 1$. After deformation the centre and the point go to $\underline{x} = \underline{\chi}_\kappa(\underline{X}, t)$ and $\underline{y} = \underline{\chi}_\kappa(\underline{Y}, t)$, and (3.101) gives

$$\underline{r} = \underline{F}\hat{\underline{r}}, \quad \underline{r} = \underline{y} - \underline{x}. \quad (3.109)$$

Hence $\hat{\underline{r}} = \underline{F}^{-1}\underline{r}$ and

$$1 = \hat{\underline{r}} \cdot \hat{\underline{r}} = \underline{F}^{-1}\underline{r} \cdot \underline{F}^{-1}\underline{r} = \underline{r} \cdot \underline{F}^{-t}\underline{F}^{-1}\underline{r} = \underline{r} \cdot \underline{B}^{-1}\underline{r}, \quad (3.110)$$

using $\tilde{B}^{-1} = (\tilde{F}\tilde{F}^t)^{-1} = \tilde{F}^{-t}\tilde{F}^{-1}$, which is Problem 19 of Chapter 1 combined with Problem 18. Substituting the spectral representation

$$\tilde{B}^{-1} = \sum_i \lambda_i^{-2} \underline{v}_i \otimes \underline{v}_i \quad (3.111)$$

from (3.42) and Remark 3.10, the substitution runs as follows. Apply $\tilde{B}^{-1}$ to $\underline{r}$ using $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$:

$$\tilde{B}^{-1}\underline{r} = \sum_i \lambda_i^{-2} (\underline{v}_i \cdot \underline{r}) \underline{v}_i,$$

then dot with $\underline{r}$ and move the scalars out:

$$\underline{r} \cdot \tilde{B}^{-1}\underline{r} = \sum_i \lambda_i^{-2} (\underline{v}_i \cdot \underline{r}) (\underline{r} \cdot \underline{v}_i) = \sum_i \lambda_i^{-2} (\underline{v}_i \cdot \underline{r})^2.$$

Writing $r_i = \underline{v}_i \cdot \underline{r}$ for the components of $\underline{r}$ on the spatial principal frame, and setting the whole thing equal to 1 by (3.110),

$$1 = (r_1/\lambda_1)^2 + (r_2/\lambda_2)^2 + (r_3/\lambda_3)^2, \quad (3.112)$$

which is the equation of an ellipsoid with axes along $\underline{v}_i$ and semi-axis lengths λ_i , centred at $\underline{x}$.

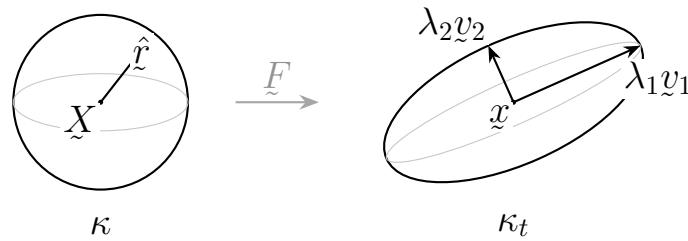


Figure 3.6: A unit sphere of material in κ is carried by a homogeneous deformation to an ellipsoid in κ_t whose axes lie along the spatial principal directions $\underline{v}_i$ and whose semi-axes have lengths equal to the principal stretches λ_i . Equation (3.112) is the algebraic statement, and $\tilde{B}^{-1}$ is the tensor that carries it.

REMARK 3.47 (what the numerical test in Figure D actually checks). Panels (c) and (d) are the substance; (a) and (b) only make the result

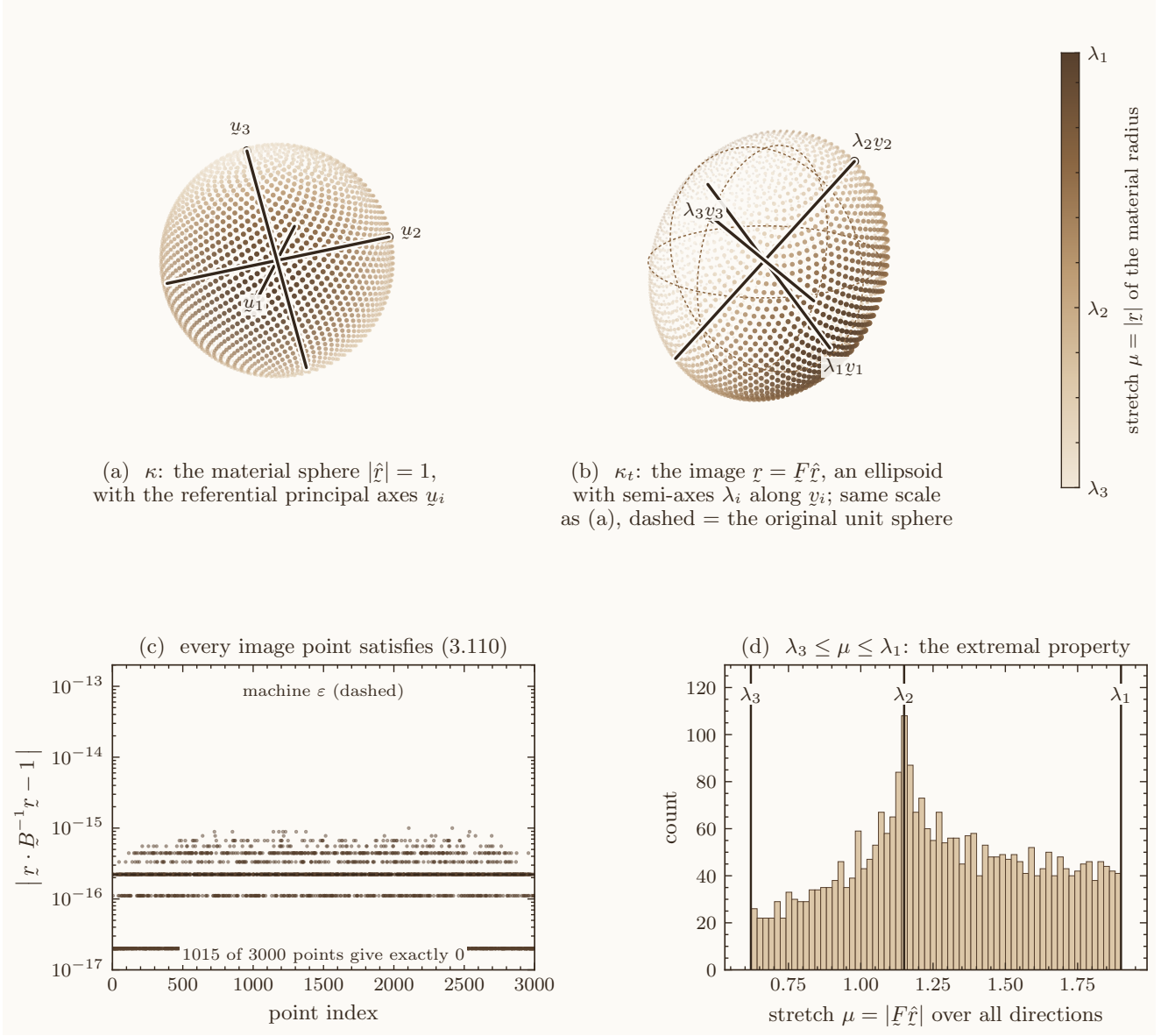


Figure D: Property (c) tested numerically, by the script `sphere_to_ellipsoid.py`. Three thousand points are spread over the unit material sphere by a golden-angle spiral and pushed forward by one fixed $\underline{\tilde{F}} = \underline{\tilde{R}}\underline{\tilde{U}}$ with principal stretches $\lambda = (1.90, 1.15, 0.62)$ and $J = \det \underline{\tilde{F}} = 1.3547$. Every marker in the figure is an individual vector point, not a raster image. (a) the sphere in κ , each point shaded by the stretch that material radius will undergo, with the referential principal axes u_i . (b) the image in κ_t , at the same scale, with the original sphere retained as dashed great circles for comparison; the point cloud lies on an ellipsoid with semi-axes λ_i along v_i , exactly as (3.112) asserts. (c) the residual $|\underline{\tilde{r}} \cdot \underline{\tilde{B}}^{-1} \underline{\tilde{r}} - 1|$ of (3.110) for every one of the three thousand image points, at or below machine precision throughout, with 1015 of them exactly zero. (d) the distribution of the stretch $\mu = |\underline{\tilde{F}} \hat{r}|$ over all directions, confined between λ_3 and λ_1 as the extremal property (3.26) requires, with λ_2 falling inside.

visible. Four separate predictions of the text are tested, and all four hold to about 10^{-15} :

- (3.110), $\underline{r} \cdot \underline{B}^{-1} \underline{r} = 1$, panel (c);
- (3.112), $\sum_i (r_i/\lambda_i)^2 = 1$ with $r_i = \underline{v}_i \cdot \underline{r}$, the same statement resolved on the principal frame;
- (3.17), $\lambda_1 \lambda_2 \lambda_3 = J$, so that the ellipsoid volume $\frac{4}{3}\pi \lambda_1 \lambda_2 \lambda_3$ is exactly J times the sphere volume;
- (3.44), $\underline{F} \underline{u}_i = \lambda_i \underline{v}_i$, so the axes really are the images of the referential principal directions.

One check is deliberately *not* at machine precision, and it should not be. The semi-axis lengths measured from the cloud, $\max_{\text{cloud}} |\underline{v}_i \cdot \underline{r}|$, come out about 4×10^{-4} below λ_i . That is not an error in (3.112) but a consequence of sampling: with finitely many points, none lands exactly on an axis tip, so the measured semi-axis always undershoots. The script confirms the diagnosis by refining the cloud, and the discrepancy falls from 3.3×10^{-4} at $N = 3000$ to 1.8×10^{-5} at $N = 30\,000$ and 1.5×10^{-6} at $N = 300\,000$, that is like $1/N$. Distinguishing a discretisation error from a genuine one is the whole skill of testing a formula numerically.

The script also runs the test backwards. It builds $\underline{F}$ from a chosen $\underline{R}$ and $\underline{U}$, then throws them away and recovers them from $\underline{F}$ alone by Theorem 3.23, that is by the closed form $\underline{U} = (-\underline{C}^2 + (i_1^2 - i_2)\underline{C} + i_1 i_3 \hat{\underline{I}})/(\underline{i}_1 \underline{i}_2 - \underline{i}_3)$ with $\underline{i}_1$ from the quartic $(*_2)$. The recovered $\underline{U}$ and $\underline{R}$ agree with the originals to 5×10^{-15} , $\underline{R}^t \underline{R} = \hat{\underline{I}}$ and $\det \underline{R} = 1$ hold to the same order, and both $\underline{F} = \underline{R} \underline{U}$ and $\underline{F} = \underline{V} \underline{R}$ are reproduced. So the closed form of §3.4 is not merely derived, it is executed.

REMARK 3.48 (the ellipsoid is the picture of the polar decomposition). Property (c) is $\underline{F} = \underline{V} \underline{R}$ drawn. The rotation $\underline{R}$ takes the unit sphere to the unit sphere, since it preserves lengths, and therefore contributes nothing to the shape; all of the shape comes from $\underline{V}$, which stretches the sphere by λ_i along $\underline{v}_i$. Conversely, if one measures the ellipsoid in κ_t one recovers $\underline{V}$ completely and $\underline{R}$ not at all. This is exactly the sense in which the ellipsoid, and not $\underline{F}$, is what the material experiences.

3.7 Rigid-body motions

Where we are and why this section closes the chapter

The chapter opened with a promise: separate the part of $\underline{F}$ that a material feels from the part it does not. §3.4 delivered the separation, $\underline{F} = \underline{R}\underline{U}$, and asserted that $\underline{R}$ is the part that costs nothing. That assertion has not yet been proved. This section proves it, in the sharpest possible form: the motions in which *nothing at all* happens to the material are exactly the motions with $\underline{U} = \hat{\underline{I}}$, that is exactly the motions in which $\underline{F}$ is a rotation. Once that is established, the case for writing constitutive laws in terms of $\underline{C}$ rather than $\underline{F}$ is complete, and the chapter is finished.

The logic runs the opposite way from §3.6. There we assumed something about $\underline{F}$ and deduced geometry; here we assume something purely geometric, that distances are preserved, and deduce what $\underline{F}$ must be.

A motion is *rigid* if it preserves the distance between every pair of material points, i.e. if $|\underline{\chi}_\kappa(\underline{Y}, t) - \underline{\chi}_\kappa(\underline{X}, t)| = |\underline{Y} - \underline{X}|$, or, squaring to remove the square root,

$$[\underline{\chi}_\kappa(\underline{Y}, t) - \underline{\chi}_\kappa(\underline{X}, t)] \cdot [\underline{\chi}_\kappa(\underline{Y}, t) - \underline{\chi}_\kappa(\underline{X}, t)] = (\underline{Y} - \underline{X}) \cdot (\underline{Y} - \underline{X}) \quad (3.113)$$

for all $\underline{X}, \underline{Y}$. Squaring is not cosmetic: the square root in the definition is awkward to differentiate at $\underline{Y} = \underline{X}$, whereas the quadratic form on each side of (3.113) is a polynomial and differentiates cleanly.

The plan. We have a functional equation, (3.113), holding for all pairs of points, and we want a statement about $\underline{F}$. The route is to differentiate twice, once with respect to $\underline{Y}$ and once with respect to $\underline{X}$, because each differentiation converts a difference of positions into an $\underline{F}$, by the definition $d\underline{\chi}_\kappa = \underline{F} d\underline{X}$. Two differentiations therefore produce a relation between two $\underline{F}$'s, and that relation will turn out to say both that $\underline{F}$ is orthogonal and that it does not vary.

First differentiation. Differentiate (3.113). Both sides have the form $\underline{a} \cdot \underline{a}$, whose differential is $d(\underline{a} \cdot \underline{a}) = d\underline{a} \cdot \underline{a} + \underline{a} \cdot d\underline{a} = 2\underline{a} \cdot d\underline{a}$, the two

terms being equal because the dot product is symmetric. The factor 2 appears on both sides and cancels, leaving

$$[\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] \cdot d[\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] = (\underline{Y} - \underline{X}) \cdot d(\underline{Y} - \underline{X}), \quad (3.114)$$

in which the derivative may be taken with respect to $\underline{X}$ or to $\underline{Y}$; we choose $\underline{Y}$ first.

So fix $\underline{X}$, which makes $\chi_\kappa(\underline{X}, t)$ a constant vector, and let $\underline{Y}$ vary. Then

$$d[\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] = d\chi_\kappa(\underline{Y}, t) - 0 = \underline{F}(\underline{Y}, t) d\underline{Y}, \quad d(\underline{Y} - \underline{X}) = d\underline{Y},$$

the first by the definition (2.34) of the deformation gradient at the point $\underline{Y}$. Substituting into (3.114),

$$[\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] \cdot \underline{F}(\underline{Y}, t) d\underline{Y} = (\underline{Y} - \underline{X}) \cdot d\underline{Y} \quad (3.115)$$

for all $d\underline{Y}$. Move $\underline{F}$ to the other factor by the definition of the transpose, $\underline{a} \cdot \underline{F}\underline{b} = \underline{F}^t \underline{a} \cdot \underline{b}$, and collect:

$$\left(\underline{F}^t(\underline{Y}, t) [\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] - (\underline{Y} - \underline{X}) \right) \cdot d\underline{Y} = 0 \quad \text{for all } d\underline{Y}.$$

A vector orthogonal to every vector is 0 , so the bracket vanishes:

$$\underline{F}^t(\underline{Y}, t) [\chi_\kappa(\underline{Y}, t) - \chi_\kappa(\underline{X}, t)] = \underline{Y} - \underline{X}. \quad (3.116)$$

Second differentiation. Now reverse the roles: fix $\underline{Y}$, so that $\underline{F}^t(\underline{Y}, t)$ and $\chi_\kappa(\underline{Y}, t)$ are both constant, and differentiate (3.116) with respect to $\underline{X}$. On the left, the only $\underline{X}$ -dependent piece is $-\chi_\kappa(\underline{X}, t)$, whose differential is $-\underline{F}(\underline{X}, t) d\underline{X}$, and the constant tensor $\underline{F}^t(\underline{Y}, t)$ multiplies it. On the right, the only $\underline{X}$ -dependent piece is $-\underline{X}$. Hence

$$-\underline{F}^t(\underline{Y}, t) \underline{F}(\underline{X}, t) d\underline{X} = -d\underline{X} \quad \text{for all } d\underline{X},$$

and cancelling the minus signs and using once more that two tensors agreeing on every vector are equal,

$$\underline{F}^t(\underline{Y}, t) \underline{F}(\underline{X}, t) = \hat{I} \quad \text{for all } \underline{X}, \underline{Y} \in \kappa. \quad (3.117)$$

Putting $\underline{\underline{Y}} = \underline{\underline{X}}$,

$$\underline{\underline{F}}^t(\underline{\underline{X}}, t) \underline{\underline{F}}(\underline{\underline{X}}, t) = \underline{\underline{\hat{I}}} \quad \text{for all } \underline{\underline{X}} \in \kappa, \quad (3.118)$$

so $\underline{\underline{F}}(\underline{\underline{X}}, t)$ is orthogonal at every $\underline{\underline{X}}$ and $\underline{\underline{F}}^t = \underline{\underline{F}}^{-1}$. Substituting this back into (3.117) gives $\underline{\underline{F}}^{-1}(\underline{\underline{Y}}, t) \underline{\underline{F}}(\underline{\underline{X}}, t) = \underline{\underline{\hat{I}}}$, i.e.

$$\underline{\underline{F}}(\underline{\underline{Y}}, t) = \underline{\underline{F}}(\underline{\underline{X}}, t) \quad \text{for all } \underline{\underline{X}}, \underline{\underline{Y}} \in \kappa, \quad (3.119)$$

so $\underline{\underline{F}}$ does not depend on $\underline{\underline{X}}$ at all:

$$\underline{\underline{F}}(\underline{\underline{X}}, t) = \underline{\underline{Q}}(t), \quad (3.120)$$

a uniform orthogonal tensor, and a rotation if κ is occupiable, by (2.50). Rigid motions are therefore homogeneous, and (3.100) gives

$$\underline{\underline{x}} = \underline{\underline{\chi}}_\kappa(\underline{\underline{X}}, t) = \underline{\underline{Q}}(t) \underline{\underline{X}} + \underline{\underline{c}}(t). \quad (3.121)$$

Conversely, substituting (3.121) into the left side of (3.113) gives $\underline{\underline{Q}}(\underline{\underline{Y}} - \underline{\underline{X}}) \cdot \underline{\underline{Q}}(\underline{\underline{Y}} - \underline{\underline{X}}) = (\underline{\underline{Y}} - \underline{\underline{X}}) \cdot \underline{\underline{Q}}^t \underline{\underline{Q}}(\underline{\underline{Y}} - \underline{\underline{X}}) = (\underline{\underline{Y}} - \underline{\underline{X}}) \cdot (\underline{\underline{Y}} - \underline{\underline{X}})$, so (3.121) is necessary *and* sufficient for rigidity.

The polar decomposition is trivial in this case:

$$\underline{\underline{R}} = \underline{\underline{Q}}, \quad \underline{\underline{U}} = \underline{\underline{\hat{I}}}, \quad \underline{\underline{V}} = \underline{\underline{I}}, \quad (3.122)$$

and of course $\underline{\underline{C}} = \underline{\underline{\hat{I}}}$ and $\underline{\underline{B}} = \underline{\underline{I}}$. Uniqueness makes this immediate: $\underline{\underline{Q}} = \underline{\underline{Q}} \underline{\underline{\hat{I}}}$ is a factorisation of the required type, hence *the* one.

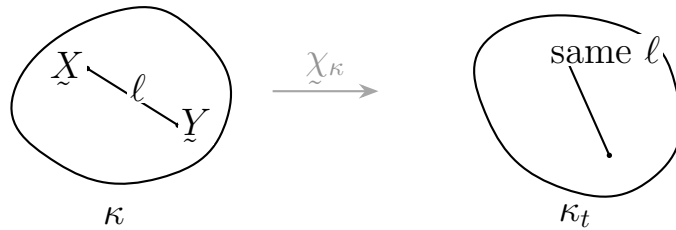


Figure 3.7: Rigid-body motion. Every material distance is preserved, so the body may translate and turn but cannot change shape. The computation (3.113) to (3.120) shows that this forces $\underline{\underline{F}}$ to be a uniform rotation, hence $\underline{\underline{U}} = \underline{\underline{\hat{I}}}$ and $\underline{\underline{C}} = \underline{\underline{\hat{I}}}$: no stretching anywhere.

REMARK 3.49 (why this is the theorem the whole chapter was for). (3.122) says: rigid motion $\iff \underline{U} = \underline{\hat{I}} \iff \underline{C} = \underline{\hat{I}} \iff \underline{E} = \underline{\hat{O}}$. The strain tensors vanish exactly on motions in which nothing happens to the material. That is the property one demands of a strain measure, and it is why $\underline{E} = \frac{1}{2}(\underline{C} - \underline{\hat{I}})$ was defined with the $\underline{\hat{I}}$ subtracted. Note what is *not* true: $\underline{F} = \underline{I}$ is not the condition for rigidity, since $\underline{F} = \underline{Q}(t)$ with $\underline{Q} \neq \underline{I}$ is rigid too. A constitutive law written in terms of $\underline{F}$ would therefore assign energy to a body that is merely being carried around; a law written in terms of $\underline{C}$ cannot. This is the whole reason for §3.4.

3.8 Problems

All nine problems of the chapter are solved below in the book's order and notation. They fall into three groups, and it helps to know which is which before starting.

Problems 1, 4, 5, 6 are further worked deformations, in the style of §3.5: write down $\underline{x}$, differentiate, read off $\underline{F}$, then extract what is asked. The recipe of §3.5 applies verbatim, and rules (R1)–(R4) are used throughout. Problem 1 adds area change through Piola–Nanson, Problem 4 is spherically symmetric rather than cylindrical, Problem 5 is a fourth shear, and Problem 6 is bending.

Problems 2, 7, 8 are algebra, filling gaps that §§3.1–3.4 left open: that products of rotations are rotations, which was used at (3.95); that the invariants are the elementary symmetric functions of the principal values, which was used at (3.17) and in Remark 3.11; and the Cayley–Hamilton theorem, which is what makes Theorem 3.23 and Problem 9 possible.

Problems 3 and 9 are the pay-off. Problem 3 completes torsion, and Problem 9 produces the closed form for $\underline{U}$ that removes eigenvectors from the subject entirely, in two dimensions; Theorem 3.23 did the same in three.

PROBLEM 1 (Triaxial stretch: area change and normals). Consider the deformation $x_1 = \lambda_1 X_1$, $x_2 = \lambda_2 X_2$, $x_3 = \lambda_3 X_3$ with constant λ_i

and $\{\underline{e}_i\} = \{\underline{E}_A\}$. (a) Obtain $\underline{F}$ and state a restriction on the λ_i for the deformation to be physically possible. (b) With $\lambda_3 = \lambda_2$ and the deformation isochoric, consider a plane material surface in κ whose unit normal is inclined at angle ϕ to the 1-axis. Find, in terms of λ_1 and ϕ , the ratio of its deformed area to its undeformed area. (c) What is the unit normal to that surface after deformation?

Solution. **(a)** Differentiating the three relations at fixed t , $dx_i = \lambda_i dX_i$ (no sum), and $dX_A = \hat{E}_A \cdot d\underline{X}$, so

$$d\underline{x} = dx_i \underline{e}_i = \sum_{A=1}^3 \lambda_A \underline{e}_A (\hat{E}_A \cdot d\underline{X}) = \left(\sum_{A=1}^3 \lambda_A \underline{e}_A \otimes \hat{E}_A \right) d\underline{X},$$

whence

$$\underline{F} = \lambda_1 \underline{e}_1 \otimes \hat{E}_1 + \lambda_2 \underline{e}_2 \otimes \hat{E}_2 + \lambda_3 \underline{e}_3 \otimes \hat{E}_3.$$

This is already in the form (3.43) with orthonormal frames on both sides, so if the λ_A are positive they *are* the principal stretches, $\underline{U} = \sum \lambda_A \hat{E}_A \otimes \hat{E}_A$, $\underline{V} = \sum \lambda_A \underline{e}_A \otimes \underline{e}_A$ and $\underline{R} = \underline{1}$. The Jacobian is obtained as follows. First the images of the referential frame, each by (R1) applied to the three dyads:

$$\underline{F} \hat{E}_1 = \lambda_1 \underline{e}_1, \quad \underline{F} \hat{E}_2 = \lambda_2 \underline{e}_2, \quad \underline{F} \hat{E}_3 = \lambda_3 \underline{e}_3,$$

since for instance $\underline{F} \hat{E}_1 = \lambda_1 (\hat{E}_1 \cdot \hat{E}_1) \underline{e}_1 + \lambda_2 (\hat{E}_2 \cdot \hat{E}_1) \underline{e}_2 + \lambda_3 (\hat{E}_3 \cdot \hat{E}_1) \underline{e}_3 = \lambda_1 \underline{e}_1$. Then rule (R4), with the denominator $[\hat{E}_1, \hat{E}_2, \hat{E}_3] = 1$ by Lemma 3.28 and the scalars taken out one slot at a time by (R3):

$$\begin{aligned} J = \det \underline{F} &= \frac{[\underline{F} \hat{E}_1, \underline{F} \hat{E}_2, \underline{F} \hat{E}_3]}{[\hat{E}_1, \hat{E}_2, \hat{E}_3]} = \frac{[\lambda_1 \underline{e}_1, \lambda_2 \underline{e}_2, \lambda_3 \underline{e}_3]}{1} \\ &= \lambda_1 \lambda_2 \lambda_3 [\underline{e}_1, \underline{e}_2, \underline{e}_3] = \lambda_1 \lambda_2 \lambda_3 \cdot 1 = \lambda_1 \lambda_2 \lambda_3. \end{aligned}$$

The restriction on the λ_A

Two requirements, excluding different things.

Invertibility. If $\lambda_1 = 0$ then $\hat{E}_1$ and $2\hat{E}_1$ both map to $\underline{0}$, contradicting (2.1). So $\lambda_A \neq 0$ and $J \neq 0$.

Sign. Along a motion, $\underline{F}(t_0) = \underline{1}$ by (2.79), so $J(t_0) = 1 > 0$. Now $J(t)$ is continuous and never zero, so by the intermediate value theorem it cannot change sign. Hence

$$\boxed{J = \lambda_1 \lambda_2 \lambda_3 > 0}$$

This is weaker than $\lambda_A > 0$. Assuming no signs,

$$\underline{C} = \sum_A \lambda_A^2 \hat{\underline{E}}_A \otimes \hat{\underline{E}}_A, \quad \underline{U} = \sqrt{\underline{C}} = \sum_A |\lambda_A| \hat{\underline{E}}_A \otimes \hat{\underline{E}}_A,$$

the positive root by Theorem 3.16, and with $s_A = \text{sgn } \lambda_A$,

$$\underline{R} = \underline{F} \underline{U}^{-1} = \sum_A \frac{\lambda_A}{|\lambda_A|} \underline{e}_A \otimes \hat{\underline{E}}_A = \sum_A s_A \underline{e}_A \otimes \hat{\underline{E}}_A,$$

$$\det \underline{R} = s_1 s_2 s_3 = \text{sgn } J.$$

So $\underline{R}$ is a rotation exactly when $J > 0$. The four sign patterns:

(s_A)	J	$\underline{R}$	
$(+, +, +)$	> 0	$\underline{1}$	pure stretch
$(-, +, +)$	< 0	$\text{diag}(-1, 1, 1)$	mirror, excluded
$(-, -, +)$	> 0	$\text{diag}(-1, -1, 1)$	half turn about $\underline{e}_3$, admissible
$(-, -, -)$	< 0	$-\underline{1}$	inversion, excluded

The stretches are $|\lambda_A|$, not λ_A . Parts (b) and (c) speak of stretches, so take $\lambda_A > 0$ below.

(b) Isochoric with $\lambda_3 = \lambda_2$ means $\lambda_1 \lambda_2^2 = 1$, so

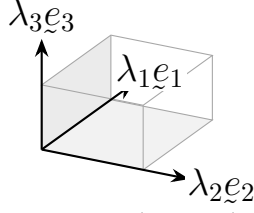
$$\lambda_2 = \lambda_3 = \lambda_1^{-1/2}.$$

Take the normal in the $(1, 2)$ plane without loss of generality, since the 2 and 3 directions are now equivalent:

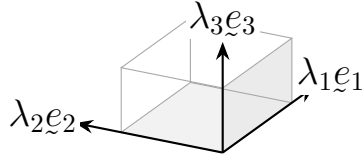
$$\underline{N} = \cos \phi \hat{\underline{E}}_1 + \sin \phi \hat{\underline{E}}_2.$$

The Piola–Nanson formula (2.54) with (2.55) gives $\underline{n} \, da = \underline{F}^* \underline{N} \, dA$ and $\underline{F}^* = J \underline{F}^{-t} = \underline{F}^{-t}$ since $J = 1$. From the dyadic form of $\underline{F}$,

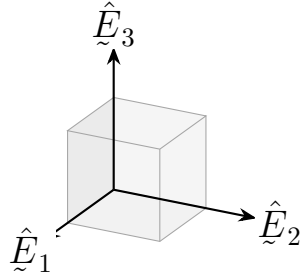
$$\underline{F}^{-1} = \sum_A \lambda_A^{-1} \hat{\underline{E}}_A \otimes \underline{e}_A, \quad \underline{F}^{-t} = \sum_A \lambda_A^{-1} \underline{e}_A \otimes \hat{\underline{E}}_A,$$



$\lambda = (-1.5, 1.1, 0.7)$: $[\tilde{F}\hat{E}_1, \tilde{F}\hat{E}_2, \tilde{F}\hat{E}_3] = -1.155$
 now left handed; $\tilde{R}$ is a mirror $\times$



$\lambda = (-1.5, -1.1, 0.7)$: $[\tilde{F}\hat{E}_1, \tilde{F}\hat{E}_2, \tilde{F}\hat{E}_3] = +1.155$
 still right handed; $\tilde{R}$ is the half turn about e_3 $\checkmark$



κ : the unit cube,
 $[\hat{E}_1, \hat{E}_2, \hat{E}_3] = +1$

Figure E: Why the restriction is $\lambda_1\lambda_2\lambda_3 > 0$ and not $\lambda_A > 0$. Bottom, the reference cube with its right-handed frame. Middle, two of the λ 's negative: the frame is turned through π about e_3 but is still right handed, the box product is still positive, and the deformation is a genuine one, a pure stretch followed by a half turn. Top, one λ negative: the frame has become left handed, the box product is negative, and $\tilde{R}$ is a mirror. No continuous motion starting from $\tilde{F} = \underline{1}$ can reach it, because J would have to pass through zero and $\tilde{F}$ would cease to be invertible at that instant.

the first because

$$\left(\sum_A \lambda_A^{-1} \hat{E}_A \otimes \underline{e}_A \right) \left(\sum_B \lambda_B \underline{e}_B \otimes \hat{E}_B \right) = \sum_{A,B} \lambda_A^{-1} \lambda_B \delta_{AB} \hat{E}_A \otimes \hat{E}_B = \hat{I}.$$

Therefore

$$\tilde{F}^* \tilde{N} = \lambda_1^{-1} \cos \phi \underline{e}_1 + \lambda_2^{-1} \sin \phi \underline{e}_2,$$

and taking the length of both sides of $\tilde{n} da = \tilde{F}^* \tilde{N} dA$, with $|\tilde{n}| = 1$,

$$\frac{da}{dA} = |\tilde{F}^* \tilde{N}| = \sqrt{\lambda_1^{-2} \cos^2 \phi + \lambda_2^{-2} \sin^2 \phi} = \boxed{\sqrt{\lambda_1^{-2} \cos^2 \phi + \lambda_1 \sin^2 \phi}}$$

where $\lambda_2^{-2} = (\lambda_1^{-1/2})^{-2} = \lambda_1$ was used.

Checks. At $\lambda_1 = 1$ the deformation is the identity and the formula gives $\sqrt{\cos^2 \phi + \sin^2 \phi} = 1$. At $\phi = 0$ the surface is normal to the stretching axis and the ratio is λ_1^{-1} , which is right: that face has area $\lambda_2 \lambda_3 = \lambda_1^{-1}$ times its original. At $\phi = \pi/2$ the ratio is $\lambda_1^{1/2} = \lambda_1 \lambda_3 = \lambda_1 \lambda_1^{-1/2}$, also right.

(c) Dividing $\tilde{F}^* \tilde{N}$ by its length,

$$\tilde{n} = \frac{\lambda_1^{-1} \cos \phi \underline{e}_1 + \lambda_1^{1/2} \sin \phi \underline{e}_2}{\sqrt{\lambda_1^{-2} \cos^2 \phi + \lambda_1 \sin^2 \phi}},$$

using $\lambda_2^{-1} = \lambda_1^{1/2}$. Equivalently, the deformed normal makes an angle ψ with the 1-axis given by

$$\tan \psi = \lambda_1^{3/2} \tan \phi.$$

Note that the normal does *not* rotate the way the material does: for $\lambda_1 > 1$ it turns *away* from the stretching axis, because $\tilde{F}^{-t}$, not $\tilde{F}$, transports normals. This is the standard trap that Piola–Nanson exists to avoid. $\square$

PROBLEM 2 (Products of orthogonal tensors). Prove that the product of two orthogonal tensors is orthogonal, and that the product of two rotations is a rotation.

Solution. Let $\underline{Q}_1, \underline{Q}_2$ be orthogonal, meaning $\underline{Q}_k^t \underline{Q}_k = \underline{I}$. Using $(\underline{AB})^t = \underline{B}^t \underline{A}^t$ from Problem 20 of Chapter 1 and associativity,

$$(\underline{Q}_1 \underline{Q}_2)^t (\underline{Q}_1 \underline{Q}_2) = \underline{Q}_2^t \underline{Q}_1^t \underline{Q}_1 \underline{Q}_2 = \underline{Q}_2^t \underline{I} \underline{Q}_2 = \underline{Q}_2^t \underline{Q}_2 = \underline{I},$$

so $\underline{Q}_1 \underline{Q}_2$ is orthogonal.

Two gaps to close. First, $\underline{Q}^t \underline{Q} = \underline{I}$ implies $\underline{Q} \underline{Q}^t = \underline{I}$: if $\underline{Q} \underline{a} = \underline{0}$ then $\underline{a} = \underline{Q}^t \underline{Q} \underline{a} = \underline{0}$, so $\underline{Q}$ is injective, hence bijective in finite dimensions, hence $\underline{Q}^{-1} = \underline{Q}^t$ and $\underline{Q} \underline{Q}^t = \underline{I}$. Second, $\det \underline{Q}^t \det \underline{Q} = \det \underline{I} = 1$ with $\det \underline{Q}^t = \det \underline{Q}$ gives $(\det \underline{Q})^2 = 1$, so $\det \underline{Q} = \pm 1$.

What orthogonality means. The three statements

$$\underline{Q}^t \underline{Q} = \underline{I}, \quad \underline{Q} \underline{a} \cdot \underline{Q} \underline{b} = \underline{a} \cdot \underline{b} \quad \forall \underline{a}, \underline{b}, \quad |\underline{Q} \underline{a}| = |\underline{a}| \quad \forall \underline{a}$$

are equivalent. The first gives the second by $\underline{Q} \underline{a} \cdot \underline{Q} \underline{b} = \underline{a} \cdot \underline{Q}^t \underline{Q} \underline{b} = \underline{a} \cdot \underline{b}$; the second gives the third at $\underline{b} = \underline{a}$; and the third gives the second by polarisation,

$$\underline{a} \cdot \underline{b} = \frac{1}{2}(|\underline{a} + \underline{b}|^2 - |\underline{a}|^2 - |\underline{b}|^2),$$

each term of which is preserved. The second then gives the first, since $\underline{a} \cdot (\underline{Q}^t \underline{Q} - \underline{I}) \underline{b} = 0$ for all $\underline{a}, \underline{b}$. So a product of orthogonal tensors is orthogonal for a reason needing no algebra: doing two length-preserving maps in succession preserves lengths.

A *rotation* is an orthogonal tensor with $\det = +1$. By Problem 21(b) of Chapter 1,

$$\det(\underline{Q}_1 \underline{Q}_2) = (\det \underline{Q}_1)(\det \underline{Q}_2) = (+1)(+1) = +1,$$

so a product of rotations is a rotation.

The version actually used at (3.95) is the two-point one: if $\underline{Q}_1 \in \text{Orth}(\hat{\mathbb{E}}^3, \mathbb{E}^3)$ and $\underline{Q}_2 \in \text{Orth}(\hat{\mathbb{E}}^3, \hat{\mathbb{E}}^3)$ then $\underline{Q}_1 \underline{Q}_2 \in \text{Orth}(\hat{\mathbb{E}}^3, \mathbb{E}^3)$. The computation is identical, the only change being that $\underline{Q}_2^t \underline{Q}_2 = \hat{\underline{I}}$ and $\underline{Q}_1^t \underline{Q}_1 = \underline{I}$, so the chain reads $(\underline{Q}_1 \underline{Q}_2)^t (\underline{Q}_1 \underline{Q}_2) = \underline{Q}_2^t \hat{\underline{I}} \underline{Q}_2 = \underline{I}$; and the determinant statement uses the extension recorded in Remark 3.20. $\square$

PROBLEM 3 (Torsion: Cauchy–Green tensors and principal stretches). Evaluate the left and right Cauchy–Green tensors for the torsion problem. Find the principal stretches in terms of the twist τ . Verify that azimuthal shear and torsion are isochoric.

Solution. From (3.88) and (3.97), $\hat{F} = \hat{Q}\hat{G} = \hat{G}\hat{Q}$ with

$$\hat{G} = \hat{I} + \gamma \hat{E}_\Theta \otimes \hat{E}_3, \quad \hat{G} = \hat{I} + \gamma \underline{e}_\theta \otimes \underline{e}_3, \quad \gamma = \gamma(R) = R\tau.$$

Since $\hat{Q}^t \hat{Q} = \hat{I}$ and $\hat{Q}\hat{Q}^t = \hat{I}$, exactly as in (3.91),

$$\hat{C} = \hat{G}^t \hat{Q}^t \hat{Q} \hat{G} = \hat{G}^t \hat{G}, \quad \hat{B} = \hat{G} \hat{Q} \hat{Q}^t \hat{G}^t = \hat{G} \hat{G}^t.$$

With $\hat{G}^t = \hat{I} + \gamma \hat{E}_3 \otimes \hat{E}_\Theta$,

$$\begin{aligned} \hat{C} &= (\hat{I} + \gamma \hat{E}_3 \otimes \hat{E}_\Theta)(\hat{I} + \gamma \hat{E}_\Theta \otimes \hat{E}_3) \\ &= \hat{I} + \gamma \hat{E}_\Theta \otimes \hat{E}_3 + \gamma \hat{E}_3 \otimes \hat{E}_\Theta + \gamma^2 (\hat{E}_\Theta \cdot \hat{E}_\Theta) \hat{E}_3 \otimes \hat{E}_3 \\ &= \hat{I} + \gamma [\hat{E}_\Theta(\Theta) \otimes \hat{E}_3 + \hat{E}_3 \otimes \hat{E}_\Theta(\Theta)] + \gamma^2 \hat{E}_3 \otimes \hat{E}_3, \end{aligned}$$

and in the same way

$$\hat{B} = \hat{I} + \gamma [\underline{e}_\theta(\theta) \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_\theta(\theta)] + \gamma^2 \underline{e}_\theta(\theta) \otimes \underline{e}_\theta(\theta).$$

Note again the shift of the squared term from $\hat{E}_3$ in $\hat{C}$ to $\underline{e}_\theta$ in $\hat{B}$, exactly as in Remark 3.31.

Principal stretches. In the orthonormal basis $\{\hat{E}_R, \hat{E}_\Theta, \hat{E}_3\}$ the matrix of $\hat{C}$ is

$$(C_{AB}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \gamma \\ 0 & \gamma & 1 + \gamma^2 \end{pmatrix},$$

which is (3.55) with the roles of the axes permuted: the shear acts on the pair $(\hat{E}_\Theta, \hat{E}_3)$ and $\hat{E}_R$ is inert. Hence one principal stretch is 1, with principal direction $\hat{E}_R$, and the other two solve the same quadratic as before. Reading (3.60) with $\gamma = R\tau$,

$$\boxed{\lambda_1 = \frac{1}{2}(R\tau + \sqrt{4 + R^2\tau^2}), \quad \lambda_2 = \lambda_1^{-1}, \quad \lambda_3 = 1.}$$

The stretches depend on R : fibres near the axis are barely deformed, fibres near the outer surface most. At $R = 0$ all three are 1, which is right, since the axis of a twisted bar is not deformed at all.

Isochoric. For torsion, by Problem 21(f) of Chapter 1,

$$\det \hat{G} = \det(\hat{I} + \gamma \hat{E}_\Theta \otimes \hat{E}_3) = 1 + \gamma \hat{E}_\Theta \cdot \hat{E}_3 = 1 + 0 = 1,$$

and $\det \underline{\tilde{F}} = (\det \underline{\tilde{Q}})(\det \underline{\hat{G}}) = 1 \cdot 1 = 1$. For azimuthal shear the same computation with (3.88) gives $\det \underline{\hat{G}} = 1 + \gamma \underline{\hat{E}}_\Theta \cdot \underline{\hat{E}}_R = 1$. Both deformations are isochoric.

An independent check: $\det \underline{\tilde{C}} = (\det \underline{\tilde{F}})^2$ should be 1. Expanding the matrix above along the first row, $\det(C_{AB}) = 1 \cdot [(1)(1 + \gamma^2) - \gamma^2] = 1$. Consistent.

Geometrically the reason is the one visible in Remark 3.45: every shear $\underline{\hat{I}} + \gamma \underline{\hat{a}} \otimes \underline{\hat{b}}$ with $\underline{\hat{a}} \perp \underline{\hat{b}}$ slides planes over one another without moving any of them closer, so no volume can change. $\square$

PROBLEM 4 (Spherically symmetric deformation). Consider the static deformation $\underline{x} = \underline{\chi}_\kappa(\underline{X}) = \lambda(R)\underline{1}\underline{X}$, where $R = |\underline{X}|$ and $\lambda(R)$ is given. (a) Describe what it does to a spherical annulus $A \leq R \leq B$. (b) Derive $\underline{\tilde{F}}$ and state the restrictions on λ . (c) Find $\lambda(R)$ if the deformation is isochoric with $\lambda(A) = a/A$.

Solution. **(a)** Since $|\underline{x}| = \lambda(R)|\underline{1}\underline{X}| = \lambda(R)R$ and the direction of $\underline{x}$ is that of $\underline{1}\underline{X}$, every material point moves along its own radius; the sphere $R = \text{const}$ goes to the sphere $r = \lambda(R)R$. The annulus $A \leq R \leq B$ therefore becomes the annulus $a \leq r \leq b$ with $a = \lambda(A)A$ and $b = \lambda(B)B$, and each concentric material sphere remains a concentric sphere. The deformation is a purely radial inflation or contraction, with no rotation and no distortion of the angular structure. It is the natural model for a pressurised spherical shell.

Not a similarity: λ depends on R , so $\Delta r/\Delta R$ varies. Every direction perpendicular to $\underline{N}$ is equivalent, so there are only two distinct stretches, the second repeated: the repeated-root case of Remark 3.8.

(b) Write $\underline{N} = \underline{X}/R$, the referential radial unit vector, so $\underline{1}\underline{X} = R\underline{1}\underline{N}$. Differentiating $\underline{x} = \lambda(R)\underline{1}\underline{X}$,

$$d\underline{x} = \lambda'(R) dR \underline{1}\underline{X} + \lambda(R) \underline{1} d\underline{X}.$$

The increment dR follows from $R^2 = \underline{X} \cdot \underline{X}$: differentiating, $2R dR = 2\underline{X} \cdot d\underline{X}$, so $dR = (\underline{X}/R) \cdot d\underline{X} = \underline{N} \cdot d\underline{X}$. Hence

$$d\underline{x} = \lambda'R(\underline{1}\underline{N})(\underline{N} \cdot d\underline{X}) + \lambda\underline{1} d\underline{X} = [\lambda\underline{1} + R\lambda'(\underline{1}\underline{N}) \otimes \underline{N}] d\underline{X},$$

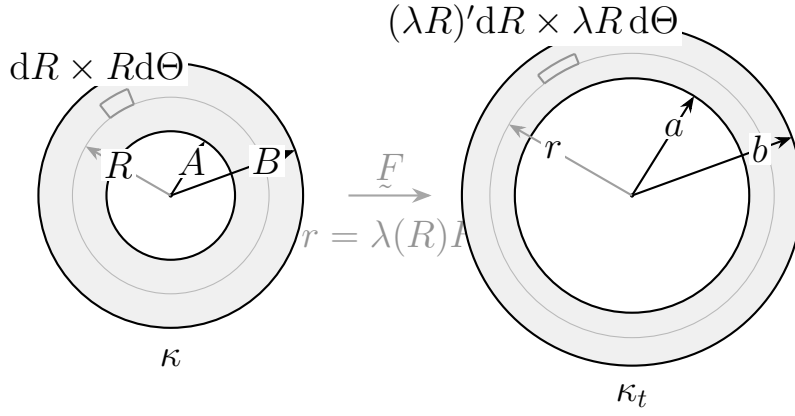


Figure F: Problem 4, on a diametral section. Every point moves along its own radius, so $A \leq R \leq B$ goes to $a \leq r \leq b$ with $a = \lambda(A)A$, $b = \lambda(B)B$. The element shows the two stretches, $dR \mapsto (\lambda R)'dR$ and $R d\Theta \mapsto \lambda R d\Theta$; with the second, equal, tangential stretch, $J = \lambda^2(\lambda R)'$.

so, writing $\underline{n}_0 = \underline{1}N$ for the shifted radial direction,

$$\underline{F} = \lambda(R) \underline{1} + R\lambda'(R) \underline{n}_0 \otimes \underline{N}.$$

Its action splits cleanly. On the radial direction,

$$\underline{F}\underline{N} = \lambda\underline{n}_0 + R\lambda'(\underline{N} \cdot \underline{N})\underline{n}_0 = (\lambda + R\lambda')\underline{n}_0 = (\lambda R)'\underline{n}_0,$$

so the radial stretch is $(\lambda R)' = dr/dR$ with $r = \lambda R$; on any $\underline{M} \perp \underline{N}$,

$$\underline{F}\underline{M} = \lambda\underline{1}\underline{M} + R\lambda'(\underline{N} \cdot \underline{M})\underline{n}_0 = \lambda\underline{1}\underline{M},$$

so the hoop stretch is λ , twice. The deformation gradient is therefore already in the form (3.43), with

$$\underline{U} = (\lambda R)'\underline{N} \otimes \underline{N} + \lambda(\hat{\underline{I}} - \underline{N} \otimes \underline{N}), \quad \underline{R} = \underline{1},$$

using $\hat{\underline{I}} - \underline{N} \otimes \underline{N} = \mathbb{P}_{(\underline{N})}$, the projection of Problem 10 of Chapter 1 onto the tangent plane of the sphere. For the Jacobian, take as test triad $\{\underline{N}, \underline{M}_1, \underline{M}_2\}$, any right-handed orthonormal frame whose first member is the radial direction, so that $\underline{M}_1, \underline{M}_2$ are tangent to the sphere. The images were just computed: $\underline{F}\underline{N} = (\lambda R)'\underline{n}_0$, $\underline{F}\underline{M}_1 = \lambda\underline{1}\underline{M}_1$, $\underline{F}\underline{M}_2 = \lambda\underline{1}\underline{M}_2$. Since $\underline{1}$ is an isometry with $\det \underline{1} = 1$, the triad $\{\underline{n}_0, \underline{1}\underline{M}_1, \underline{1}\underline{M}_2\}$

is again right-handed orthonormal, so both box products in (R4) equal 1 by Lemma 3.28, and pulling out one scalar per slot,

$$\begin{aligned} J = \det \underline{\underline{F}} &= \frac{[(\lambda R)' \underline{\underline{n}}_0, \lambda \underline{\underline{1}} \underline{\underline{M}}_1, \lambda \underline{\underline{1}} \underline{\underline{M}}_2]}{[\underline{\underline{N}}, \underline{\underline{M}}_1, \underline{\underline{M}}_2]} = \frac{(\lambda R)' \lambda \lambda \cdot 1}{1} \\ &= \lambda^2 (\lambda R)' = \frac{r^2 r'}{R^2}, \quad r = \lambda R, \end{aligned}$$

the last form because $r^2/R^2 = \lambda^2$ and $r' = (\lambda R)'$. Physical possibility requires $J > 0$, hence

$$\lambda(R) > 0 \quad \text{and} \quad (\lambda R)' > 0,$$

i.e. $r(R)$ must be positive and strictly increasing: distinct material spheres must remain distinct and must not turn inside out.

(c) Isochoric means $J = 1$, i.e. $r^2 r' = R^2$, i.e. $\frac{1}{3}(r^3)' = \frac{1}{3}(R^3)'$. Integrating,

$$r^3 - R^3 = \text{const.}$$

The condition $\lambda(A) = a/A$ says $r(A) = \lambda(A)A = a$, so the constant is $a^3 - A^3$ and

$$r^3 = R^3 + a^3 - A^3, \quad \text{that is} \quad \boxed{\lambda(R) = \frac{(R^3 + a^3 - A^3)^{1/3}}{R}}.$$

Checks. $\lambda(A) = (A^3 + a^3 - A^3)^{1/3}/A = a/A$ as required. If $a = A$ then $\lambda \equiv 1$ and the deformation is the identity, as it must be, since an isochoric radial deformation that fixes the inner surface fixes everything. If $a > A$ the shell is inflated, and $\lambda(R) \rightarrow 1$ as $R \rightarrow \infty$: the disturbance dies away, which is the familiar statement that an incompressible medium transmits a cavity expansion as a $1/R^3$ perturbation. $\square$

PROBLEM 5 (Telescopic, or anti-plane, shear). Consider $\underline{\underline{x}} = \underline{\underline{\chi}}_\kappa(\underline{\underline{X}}, t) = \underline{\underline{1}} \underline{\underline{X}} + w(R, t) \underline{\underline{e}}_3$ with $\underline{\underline{X}} = R \hat{\underline{\underline{E}}}_R(\Theta) + Z \hat{\underline{\underline{E}}}_3$. (a) Show that $\underline{\underline{F}} = \underline{\underline{1}} + \gamma(R, t) \underline{\underline{e}}_3 \otimes \hat{\underline{\underline{E}}}_R(\Theta)$ with $\gamma = \partial w / \partial R$, derive $\underline{\underline{C}}$ and $\underline{\underline{B}}$, and justify the terminology. (b) Find the stretch μ and current tangent $\underline{\underline{m}}$ of the material curve with $\underline{\underline{M}} = \frac{\sqrt{2}}{2} [\hat{\underline{\underline{E}}}_\Theta + \hat{\underline{\underline{E}}}_3]$, and the same for $\underline{\underline{M}} = \frac{\sqrt{2}}{2} [\hat{\underline{\underline{E}}}_R + \hat{\underline{\underline{E}}}_\Theta]$.

Solution. **(a)** Differentiating at fixed t and using $dR = \hat{E}_R \cdot dX$,

$$d\tilde{x} = \underline{1} dX + \frac{\partial w}{\partial R} dR \underline{e}_3 = [\underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R(\Theta)] dX, \quad \gamma = \frac{\partial w}{\partial R},$$

so $\tilde{F} = \underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R(\Theta)$ as claimed. Choose the shifter so that $\underline{1}\hat{E}_3 = \underline{e}_3$ and $\underline{1}\hat{E}_R = \underline{e}_r$, which is the alignment of (2.79).

For $\tilde{C}$, transpose and multiply, using Problems 17(b),(c) and $\underline{1}^t \underline{e}_3 = \hat{E}_3$:

$$\begin{aligned} \tilde{C} = \tilde{F}^t \tilde{F} &= (\underline{1}^t + \gamma \hat{E}_R \otimes \underline{e}_3)(\underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R) \\ &= \hat{I} + \gamma (\underline{1}^t \underline{e}_3) \otimes \hat{E}_R + \gamma \hat{E}_R \otimes (\underline{1}^t \underline{e}_3) + \gamma^2 (\underline{e}_3 \cdot \underline{e}_3) \hat{E}_R \otimes \hat{E}_R \\ &= \hat{I} + \gamma [\hat{E}_3 \otimes \hat{E}_R(\Theta) + \hat{E}_R(\Theta) \otimes \hat{E}_3] + \gamma^2 \hat{E}_R(\Theta) \otimes \hat{E}_R(\Theta), \end{aligned}$$

and similarly, with $\underline{1}\hat{E}_R = \underline{e}_r$,

$$\begin{aligned} \tilde{B} = \tilde{F} \tilde{F}^t &= (\underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R)(\underline{1}^t + \gamma \hat{E}_R \otimes \underline{e}_3) \\ &= \hat{I} + \gamma (\underline{1}\hat{E}_R) \otimes \underline{e}_3 + \gamma \underline{e}_3 \otimes (\underline{1}\hat{E}_R) + \gamma^2 \underline{e}_3 \otimes \underline{e}_3 \\ &= \hat{I} + \gamma [\underline{e}_r(\theta) \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_r(\theta)] + \gamma^2 \underline{e}_3 \otimes \underline{e}_3. \end{aligned}$$

This is again a simple shear of amount $\gamma(R, t)$, now on the orthonormal pair $(\hat{E}_3, \hat{E}_R)$, so everything in Remark 3.45 applies; in particular $\det \tilde{F} = 1 + \gamma \underline{e}_3 \cdot \underline{1}\hat{E}_R = 1 + \gamma \underline{e}_3 \cdot \underline{e}_r = 1$ by Problem 21(g) of Chapter 1, so the deformation is isochoric.

Terminology. A material point at (R, Θ, Z) moves to $(R, \Theta, Z + w(R))$. Its radial and azimuthal coordinates do not change at all, and the displacement is purely axial, i.e. purely *out of* the (R, Θ) plane: hence *anti-plane* shear. Since w depends on R alone, each concentric material cylinder $R = \text{const}$ slides rigidly along the axis by a different amount, exactly as the sections of a telescope slide over one another: hence *telescopic* shear.

(b) Both computations use (2.67), $\mu \tilde{m} = \tilde{F} \tilde{M}$, with $\tilde{F} \hat{E}_R = \underline{e}_r + \gamma \underline{e}_3$, $\tilde{F} \hat{E}_\Theta = \underline{e}_\theta$, $\tilde{F} \hat{E}_3 = \underline{e}_3$, which follow from $\tilde{F} = \underline{1} + \gamma \underline{e}_3 \otimes \hat{E}_R$ and $\hat{E}_R \cdot \hat{E}_\Theta = \hat{E}_R \cdot \hat{E}_3 = 0$, $\hat{E}_R \cdot \hat{E}_R = 1$.

First curve, $\tilde{M} = \frac{\sqrt{2}}{2}(\hat{E}_\Theta + \hat{E}_3)$:

$$\tilde{F} \tilde{M} = \frac{\sqrt{2}}{2}(\tilde{F} \hat{E}_\Theta + \tilde{F} \hat{E}_3) = \frac{\sqrt{2}}{2}(\underline{e}_\theta + \underline{e}_3), \quad \mu = |\tilde{F} \tilde{M}| = \frac{\sqrt{2}}{2} \sqrt{1+1} = 1,$$

so $\mu = 1$ and $\underline{m} = \frac{\sqrt{2}}{2}(\underline{e}_\theta + \underline{e}_3)$: *this helix does not stretch at all*, whatever γ . The reason is visible in $\underline{C}$: the shear couples $\hat{E}_R$ to $\hat{E}_3$, and this $\underline{M}$ has no $\hat{E}_R$ component. Confirming with (2.83),

$$\underline{C}\underline{M} = \frac{\sqrt{2}}{2}(\hat{E}_\Theta + \hat{E}_3 + \gamma\hat{E}_R), \quad \mu^2 = \underline{M} \cdot \underline{C}\underline{M} = \frac{1}{2}(1 + 1 + 0) = 1. \checkmark$$

Second curve, $\underline{M} = \frac{\sqrt{2}}{2}(\hat{E}_R + \hat{E}_\Theta)$:

$$\underline{F}\underline{M} = \frac{\sqrt{2}}{2}(\underline{e}_r + \gamma\underline{e}_3 + \underline{e}_\theta), \quad \mu = \frac{\sqrt{2}}{2}\sqrt{1 + \gamma^2 + 1} = \sqrt{1 + \frac{\gamma^2}{2}},$$

and

$$\underline{m} = \frac{\underline{e}_r(\theta) + \underline{e}_\theta(\theta) + \gamma\underline{e}_3}{\sqrt{2 + \gamma^2}}.$$

Confirming with $\underline{C}$: $\underline{C}\hat{E}_R = (1 + \gamma^2)\hat{E}_R + \gamma\hat{E}_3$ and $\underline{C}\hat{E}_\Theta = \hat{E}_\Theta$, so

$$\begin{aligned} \mu^2 &= \underline{M} \cdot \underline{C}\underline{M} = \frac{1}{2}(\hat{E}_R + \hat{E}_\Theta) \cdot [(1 + \gamma^2)\hat{E}_R + \gamma\hat{E}_3 + \hat{E}_\Theta] \\ &= \frac{1}{2}[(1 + \gamma^2) + 1] = 1 + \frac{\gamma^2}{2} \quad \checkmark \end{aligned}$$

So the curve inclined at 45° between the radial and azimuthal directions does stretch, and by a factor that grows with the local shear. $\square$

PROBLEM 6 (Flexure of a block). Consider the static deformation $\underline{X} \rightarrow \underline{x}$ with $\underline{X} = X_A \hat{E}_A$ and $\underline{x} = r \underline{e}_r(\theta) + z \underline{e}_3$, where $r = f(X_1) > 0$, $\theta = g(X_2)$, $z = Z = X_3$. (a) Describe the images of the lines $X_1 = \text{const}$ and $X_2 = \text{const}$ and justify calling this a flexure. (b) Derive $\underline{F}$. (c) Show $\det \underline{F} > 0 \iff ff'g' > 0$. (d) Assuming $f > 0$, $g' > 0$, obtain $\underline{R}, \underline{U}, \underline{V}$ by inspection, and find f, g if the deformation is isochoric.

Solution. **(a)** A line $X_1 = c_1$, along which X_2 varies, maps to the set $r = f(c_1) = \text{const}$ with $\theta = g(X_2)$ sweeping an interval: a *circular arc* of radius $f(c_1)$ centred on the axis. A line $X_2 = c_2$, along which X_1 varies, maps to $\theta = g(c_2) = \text{const}$ with $r = f(X_1)$ varying: a *radial straight segment*. So the two families of coordinate lines of the block go to concentric arcs and radial rays respectively, and a rectangular block is carried into an annular wedge. That is precisely bending: parallel

planes become coaxial cylinders and normal cross-sections remain plane and normal. Hence *flexure*.

Explicitly $(X_1, X_2) \mapsto (r, \theta) = (f(X_1), g(X_2))$, so

lines $X_1 = \text{const} \mapsto \text{arcs } r = \text{const},$

lines $X_2 = \text{const} \mapsto \text{rays } \theta = \text{const}.$

A block $0 \leq X_1 \leq T, 0 \leq X_2 \leq L$ becomes the wedge $f(0) \leq r \leq f(T), g(0) \leq \theta \leq g(L)$. The bend angle is $g(L) - g(0)$; the ring closes when it equals 2π .

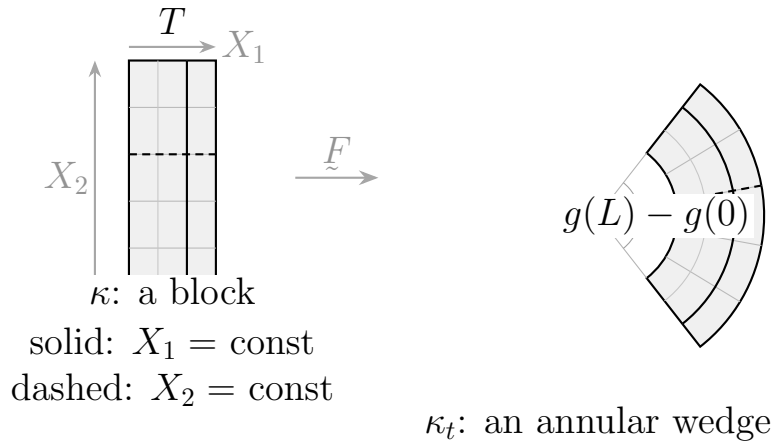


Figure G: Problem 6: flexure. The through-thickness coordinate X_1 becomes the radius, $r = f(X_1)$, so lines $X_1 = \text{const}$ (solid) become concentric arcs; the along-length coordinate X_2 becomes the azimuth, $\theta = g(X_2)$, so lines $X_2 = \text{const}$ (dashed) become radial rays. A block of thickness T and length L is wrapped into a wedge subtending $g(L) - g(0)$. Cross-sections stay plane and normal to the arcs, which is why $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$ and $\{\hat{\underline{E}}_A\}$ pair off in $\underline{F}$ with no shear between them.

(b) In polar coordinates $d\tilde{x} = dr \underline{e}_r + r d\theta \underline{e}_\theta + dz \underline{e}_3$, with $dr = f'(X_1)dX_1$, $d\theta = g'(X_2)dX_2$, $dz = dX_3$, and $dX_A = \hat{\underline{E}}_A \cdot d\tilde{X}$. Therefore

$$d\tilde{x} = f' \underline{e}_r (\hat{\underline{E}}_1 \cdot d\tilde{X}) + f g' \underline{e}_\theta (\hat{\underline{E}}_2 \cdot d\tilde{X}) + \underline{e}_3 (\hat{\underline{E}}_3 \cdot d\tilde{X}),$$

and since $d\tilde{X}$ is arbitrary,

$$\underline{F} = f'(X_1) \underline{e}_r \otimes \hat{\underline{E}}_1 + f(X_1) g'(X_2) \underline{e}_\theta \otimes \hat{\underline{E}}_2 + \underline{e}_3 \otimes \hat{\underline{E}}_3$$

with the primes denoting derivatives with respect to the indicated argument.

(c) Apply rule (R4) with the referential frame $\{\hat{E}_1, \hat{E}_2, \hat{E}_3\}$. First its images, each by (R1) on the three dyads of $\tilde{F}$:

$$\begin{aligned}\tilde{F}\hat{E}_1 &= f'(\hat{E}_1 \cdot \hat{E}_1)\underline{e}_r + fg'(\hat{E}_2 \cdot \hat{E}_1)\underline{e}_\theta + (\hat{E}_3 \cdot \hat{E}_1)\underline{e}_3 = f'\underline{e}_r, \\ \tilde{F}\hat{E}_2 &= f' \cdot 0 \cdot \underline{e}_r + fg' \cdot 1 \cdot \underline{e}_\theta + 0 \cdot \underline{e}_3 = fg'\underline{e}_\theta, \\ \tilde{F}\hat{E}_3 &= f' \cdot 0 \cdot \underline{e}_r + fg' \cdot 0 \cdot \underline{e}_\theta + 1 \cdot \underline{e}_3 = \underline{e}_3.\end{aligned}$$

The denominator is $[\hat{E}_1, \hat{E}_2, \hat{E}_3] = 1$ by Lemma 3.28. In the numerator, pull out f' from the first slot, then fg' from the second, then evaluate the surviving box product of the right-handed orthonormal polar frame by Lemma 3.29:

$$\begin{aligned}\det \tilde{F} &= \frac{[f'\underline{e}_r, fg'\underline{e}_\theta, \underline{e}_3]}{[\hat{E}_1, \hat{E}_2, \hat{E}_3]} = \frac{f'[\underline{e}_r, fg'\underline{e}_\theta, \underline{e}_3]}{1} \\ &= f'(fg')[\underline{e}_r, \underline{e}_\theta, \underline{e}_3] = f'(fg') \cdot 1 = ff'g' .\end{aligned}$$

Hence $\det \tilde{F} > 0 \iff ff'g' > 0$. The three factors are the three stretches: f' radially, fg' azimuthally, 1 axially, and their product is the volume ratio exactly as in (3.17).

(d) With $f > 0$ and $g' > 0$, part (c) forces $f' > 0$, so all three coefficients in $\tilde{F}$ are positive. But then $\tilde{F}$ is already displayed in the form (3.43), $\tilde{F} = \sum_i \lambda_i \underline{v}_i \otimes \underline{u}_i$, with orthonormal referential triad $\{\hat{E}_1, \hat{E}_2, \hat{E}_3\}$ and orthonormal spatial triad $\{\underline{e}_r, \underline{e}_\theta, \underline{e}_3\}$. By the uniqueness of the polar decomposition, the factors can be read off:

$$\begin{aligned}\lambda_1 &= f', & \lambda_2 &= fg', & \lambda_3 &= 1, \\ \underline{U} &= f'\hat{E}_1 \otimes \hat{E}_1 + fg'\hat{E}_2 \otimes \hat{E}_2 + \hat{E}_3 \otimes \hat{E}_3, \\ \underline{V} &= f'\underline{e}_r \otimes \underline{e}_r + fg'\underline{e}_\theta \otimes \underline{e}_\theta + \underline{e}_3 \otimes \underline{e}_3, \\ \underline{R} &= \underline{e}_r \otimes \hat{E}_1 + \underline{e}_\theta \otimes \hat{E}_2 + \underline{e}_3 \otimes \hat{E}_3.\end{aligned}$$

That these do the job is checked at once: $\underline{R}\underline{U} = \sum \lambda_i \underline{v}_i \otimes \underline{u}_i = \tilde{F}$ by Problem 17(b), $\underline{U}$ is symmetric with positive principal values, and $\underline{R}$ pairs two right-handed orthonormal triads so $\underline{R}^t \underline{R} = \hat{I}$ and $\det \underline{R} = 1$. Uniqueness (Step 7 of §3.4) says there is nothing else to find.

Isochoric case. $\det \underline{F} = f f' g' = 1$ gives

$$f(X_1) f'(X_1) = \frac{1}{g'(X_2)}.$$

The left side depends only on X_1 , the right only on X_2 , so both equal a constant c , and $c > 0$ since $f, f', g' > 0$. Then

$$g' = \frac{1}{c} \implies g(X_2) = \frac{X_2}{c} + g_0,$$

a linear map of the height coordinate to the azimuth, and

$$f f' = c \implies \frac{1}{2}(f^2)' = c \implies f(X_1) = \sqrt{2cX_1 + d}$$

for a constant d making $f > 0$ on the block. So an isochoric flexure bends the block through an angle proportional to its height and maps the thickness coordinate by a square root, exactly the profile (3.77) found for isochoric radial deformation, as it must be, since both express conservation of area in the cross-section. $\square$

PROBLEM 7 (Invariants in terms of principal values; Cayley–Hamilton). Let $\underline{A}$ be symmetric. (a) Show that $I_1 = \lambda_1 + \lambda_2 + \lambda_3$, $I_2 = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3$, $I_3 = \lambda_1\lambda_2\lambda_3$. (b) Show that $\underline{A}^3 - I_1\underline{A}^2 + I_2\underline{A} - I_3\underline{I} = \underline{O}$.

Solution. **(a)** Use the spectral representation (3.12) throughout, together with $\text{tr}(\underline{u} \otimes \underline{v}) = \underline{u} \cdot \underline{v}$ from Theorem 1.51 of Chapter 1.

For I_1 ,

$$I_1 = \text{tr } \underline{A} = \text{tr} \left(\sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i \right) = \sum_i \lambda_i \underline{u}_i \cdot \underline{u}_i = \sum_i \lambda_i = \lambda_1 + \lambda_2 + \lambda_3,$$

the trace being linear and each $\underline{u}_i$ a unit vector.

For I_2 we need $\text{tr}(\underline{A}^2)$. By Remark 3.10, $\underline{A}^2 = \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i$, so $\text{tr}(\underline{A}^2) = \sum_i \lambda_i^2$. Then, by (3.3)₂,

$$\begin{aligned} I_2 &= \frac{1}{2}[(\text{tr } \underline{A})^2 - \text{tr}(\underline{A}^2)] = \frac{1}{2} \left[\left(\sum_i \lambda_i \right)^2 - \sum_i \lambda_i^2 \right] \\ &= \frac{1}{2} \left[\sum_i \lambda_i^2 + 2(\lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3) - \sum_i \lambda_i^2 \right] \\ &= \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3, \end{aligned}$$

the square having been expanded in full and the pure squares cancelling.

For I_3 , equation (3.17) already gives $I_3 = \det \underline{A} = \lambda_1 \lambda_2 \lambda_3$.

These three are the elementary symmetric functions of the principal values, which is what one expects: writing (3.2) as $-(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) = 0$ and expanding gives $-\lambda^3 + (\sum \lambda_i)\lambda^2 - (\sum_{i < j} \lambda_i \lambda_j)\lambda + \lambda_1 \lambda_2 \lambda_3$, and comparing coefficients with (3.2) reproduces all three at once.

(b) Each principal value satisfies the characteristic equation (3.2), so

$$-\lambda_i^3 + I_1 \lambda_i^2 - I_2 \lambda_i + I_3 = 0, \quad \text{i.e.} \quad \lambda_i^3 - I_1 \lambda_i^2 + I_2 \lambda_i - I_3 = 0$$

for $i = 1, 2, 3$. Now evaluate the tensor expression using Remark 3.10 for the powers and (3.11) for the identity:

$$\begin{aligned} \underline{A}^3 - I_1 \underline{A}^2 + I_2 \underline{A} - I_3 \underline{I} &= \sum_i \lambda_i^3 \underline{u}_i \otimes \underline{u}_i - I_1 \sum_i \lambda_i^2 \underline{u}_i \otimes \underline{u}_i \\ &\quad + I_2 \sum_i \lambda_i \underline{u}_i \otimes \underline{u}_i - I_3 \sum_i \underline{u}_i \otimes \underline{u}_i \\ &= \sum_i (\lambda_i^3 - I_1 \lambda_i^2 + I_2 \lambda_i - I_3) \underline{u}_i \otimes \underline{u}_i \\ &= \sum_i 0 \cdot \underline{u}_i \otimes \underline{u}_i = \underline{O}. \end{aligned}$$

This is the *Cayley–Hamilton* formula. It holds for non-symmetric tensors too, though the proof above does not cover that case, since it used a spectral representation. It is meaningful only for tensors mapping a space to itself: for a two-point $\underline{F}$ the term $\underline{F}^2$ does not exist, as $\underline{F}$ maps $\hat{\mathbb{E}}^3$ to $\mathbb{E}^3$ and cannot be applied twice, so the formula cannot even be written. $\square$

PROBLEM 8 (Two-dimensional Cayley–Hamilton). Let $\underline{A}$ be symmetric on a two-dimensional space, with spectral decomposition $\underline{A} = \sum_{i=1}^2 \lambda_i \underline{u}_i \otimes \underline{u}_i$. Prove $\underline{A}^2 - (\text{tr } \underline{A})\underline{A} + (\det \underline{A})\underline{I} = \underline{O}$.

Solution. In two dimensions $\underline{I} = \underline{u}_1 \otimes \underline{u}_1 + \underline{u}_2 \otimes \underline{u}_2$, by the same verification as (3.11), and by the argument of Remark 3.10, $\underline{A}^2 = \sum_{i=1}^2 \lambda_i^2 \underline{u}_i \otimes \underline{u}_i$. The two invariants are

$$\text{tr } \underline{A} = \lambda_1 + \lambda_2, \quad \det \underline{A} = \frac{[\underline{A}\underline{u}_1, \underline{A}\underline{u}_2]}{[\underline{u}_1, \underline{u}_2]} = \frac{[\lambda_1 \underline{u}_1, \lambda_2 \underline{u}_2]}{[\underline{u}_1, \underline{u}_2]} = \lambda_1 \lambda_2,$$

where $[\cdot, \cdot]$ is the two-dimensional area form, linear in each slot and alternating, so the scalars come out and the common factor cancels. Substituting,

$$\begin{aligned} \underline{\underline{A}}^2 - (\text{tr } \underline{\underline{A}})\underline{\underline{A}} + (\det \underline{\underline{A}})\underline{\underline{I}} &= \sum_{i=1}^2 \left[\lambda_i^2 - (\lambda_1 + \lambda_2)\lambda_i + \lambda_1\lambda_2 \right] \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i \\ &= \sum_{i=1}^2 (\lambda_i - \lambda_1)(\lambda_i - \lambda_2) \underline{\underline{u}}_i \otimes \underline{\underline{u}}_i. \end{aligned}$$

For $i = 1$ the first factor vanishes and for $i = 2$ the second does, so every term is zero and the sum is $\underline{\underline{O}}$. Explicitly, for $i = 1$: $\lambda_1^2 - \lambda_1^2 - \lambda_2\lambda_1 + \lambda_1\lambda_2 = 0$; for $i = 2$: $\lambda_2^2 - \lambda_1\lambda_2 - \lambda_2^2 + \lambda_1\lambda_2 = 0$. $\square$

REMARK 3.50 (three consequences, used at once). *(i) The characteristic equation.* Feeding $\underline{\underline{u}}_i$ to the identity gives $\lambda_i^2 - (\text{tr } \underline{\underline{A}})\lambda_i + \det \underline{\underline{A}} = 0$, so the principal values are

$$\lambda_{1,2} = \frac{1}{2} \left(\text{tr } \underline{\underline{A}} \pm \sqrt{(\text{tr } \underline{\underline{A}})^2 - 4 \det \underline{\underline{A}}} \right),$$

the discriminant being $(\lambda_1 - \lambda_2)^2 \geq 0$, hence real roots, as Theorem 3.4 promises.

(ii) The inverse without cofactors. If $\det \underline{\underline{A}} \neq 0$, multiply the identity by $\underline{\underline{A}}^{-1}$:

$$\underline{\underline{A}} - (\text{tr } \underline{\underline{A}})\underline{\underline{I}} + (\det \underline{\underline{A}})\underline{\underline{A}}^{-1} = \underline{\underline{O}} \implies \underline{\underline{A}}^{-1} = \frac{(\text{tr } \underline{\underline{A}})\underline{\underline{I}} - \underline{\underline{A}}}{\det \underline{\underline{A}}}.$$

This is what Problem 9 uses to invert $\underline{\underline{U}}$.

(iii) Every power reduces. $\underline{\underline{A}}^2 = (\text{tr } \underline{\underline{A}})\underline{\underline{A}} - (\det \underline{\underline{A}})\underline{\underline{I}}$, so by induction $\underline{\underline{A}}^n = p_n \underline{\underline{A}} + q_n \underline{\underline{I}}$ for scalars p_n, q_n : in two dimensions no tensor function ever needs more than $\underline{\underline{I}}$ and $\underline{\underline{A}}$. The three-dimensional analogue, Problem 7(b), reduces everything to $\underline{\underline{I}}$, $\underline{\underline{A}}$, $\underline{\underline{A}}^2$, and that is what Theorem 3.23 exploits.

Without the spectral form. The identity also follows from components alone, which shows it does not need symmetry. For $(A_{ij}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$\text{tr } \underline{\underline{A}} = a + d$ and $\det \underline{\underline{A}} = ad - bc$, so

$$\begin{aligned} \underline{\underline{A}}^2 &= \begin{pmatrix} a^2 + bc & b(a + d) \\ c(a + d) & d^2 + bc \end{pmatrix}, & (\text{tr } \underline{\underline{A}})\underline{\underline{A}} &= \begin{pmatrix} a(a + d) & b(a + d) \\ c(a + d) & d(a + d) \end{pmatrix}, \\ (\det \underline{\underline{A}})\underline{\underline{I}} &= \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix}. \end{aligned}$$

The off-diagonal entries cancel between the first two. The diagonal ones give

$$\begin{aligned} (1, 1) : & \quad a^2 + bc - a^2 - ad + ad - bc = 0, \\ (2, 2) : & \quad d^2 + bc - ad - d^2 + ad - bc = 0, \end{aligned}$$

so the sum is $\underline{\underline{O}}$.

PROBLEM 9 (The stretch tensor without eigenvectors). Show that the two-dimensional right stretch tensor $\underline{\underline{U}}$ and right Cauchy–Green tensor $\underline{\underline{C}}$, both mapping $\hat{\mathbb{E}}^2$ to itself, satisfy $\underline{\underline{I}}\underline{\underline{U}} = \underline{\underline{C}} + J\hat{\underline{\underline{I}}}$ with $I = \text{tr } \underline{\underline{U}}$, $J = \det \underline{\underline{U}}$; show $J^2 = \det \underline{\underline{C}}$ and $I^2 = \text{tr } \underline{\underline{C}} + 2J$. Apply this to the simple shear $\underline{\underline{F}} = \underline{\underline{1}} + \gamma \underline{\underline{e}}_1 \otimes \hat{\underline{\underline{E}}}_2$, obtain $\underline{\underline{U}}$, build the three-dimensional stretch and rotation, recover the principal stretches, and derive $\underline{\underline{V}}$ directly. Do the same for azimuthal shear and torsion.

Solution. The identity. Apply Problem 8 to $\underline{\underline{U}}$, which is symmetric and positive definite:

$$\underline{\underline{U}}^2 - (\text{tr } \underline{\underline{U}})\underline{\underline{U}} + (\det \underline{\underline{U}})\hat{\underline{\underline{I}}} = \hat{\underline{\underline{O}}}.$$

Since $\underline{\underline{U}}^2 = \underline{\underline{C}}$ by definition of the right stretch tensor, and writing $I = \text{tr } \underline{\underline{U}}$, $J = \det \underline{\underline{U}}$,

$$\boxed{\underline{\underline{I}}\underline{\underline{U}} = \underline{\underline{C}} + J\hat{\underline{\underline{I}}}}$$

which determines $\underline{\underline{U}}$ as soon as the two scalars I and J are known, and $I > 0$ because $\underline{\underline{U}}$ is positive definite so $\text{tr } \underline{\underline{U}} = \lambda_1 + \lambda_2 > 0$.

The two scalars are obtained from $\underline{\underline{C}}$ alone. By Problem 21(b) of Chapter 1,

$$\det \underline{\underline{C}} = \det(\underline{\underline{U}}\underline{\underline{U}}) = (\det \underline{\underline{U}})^2 = J^2, \quad \text{so} \quad J = +\sqrt{\det \underline{\underline{C}}}$$

with the positive root because $J = \lambda_1 \lambda_2 > 0$. And

$$\text{tr } \underline{\underline{C}} = \text{tr}(U^2) = \lambda_1^2 + \lambda_2^2 = (\lambda_1 + \lambda_2)^2 - 2\lambda_1 \lambda_2 = I^2 - 2J,$$

so

$$I = +\sqrt{\text{tr } \underline{\underline{C}} + 2J}.$$

Hence $\underline{\underline{U}} = (\underline{\underline{C}} + J\underline{\underline{\hat{I}}})/I$ is computed from $\underline{\underline{C}}$ by two square roots of scalars, with no characteristic equation and no eigenvectors. That is the point of the problem.

Verification, in the abstract. That this $\underline{\underline{U}}$ really is $\sqrt{\underline{\underline{C}}}$ can be checked without ever mentioning λ_i . Square it and use Cayley–Hamilton for $\underline{\underline{C}}$, which by Problem 8 reads $\underline{\underline{C}}^2 = (\text{tr } \underline{\underline{C}})\underline{\underline{C}} - (\det \underline{\underline{C}})\underline{\underline{\hat{I}}} = (I^2 - 2J)\underline{\underline{C}} - J^2\underline{\underline{\hat{I}}}$:

$$\begin{aligned} I^2 \underline{\underline{U}}^2 &= (\underline{\underline{C}} + J\underline{\underline{\hat{I}}})^2 = \underline{\underline{C}}^2 + 2J\underline{\underline{C}} + J^2\underline{\underline{\hat{I}}} \\ &= [(I^2 - 2J)\underline{\underline{C}} - J^2\underline{\underline{\hat{I}}}] + 2J\underline{\underline{C}} + J^2\underline{\underline{\hat{I}}} = I^2 \underline{\underline{C}}, \end{aligned}$$

so $\underline{\underline{U}}^2 = \underline{\underline{C}}$. It is symmetric, being a linear combination of the symmetric $\underline{\underline{C}}$ and $\underline{\underline{\hat{I}}}$. Its principal values are $(\lambda_i^2 + J)/I$, and with $J = \lambda_1 \lambda_2$, $I = \lambda_1 + \lambda_2$,

$$\frac{\lambda_1^2 + \lambda_1 \lambda_2}{\lambda_1 + \lambda_2} = \frac{\lambda_1(\lambda_1 + \lambda_2)}{\lambda_1 + \lambda_2} = \lambda_1 > 0,$$

and likewise λ_2 ; so $\underline{\underline{U}}$ is positive definite, and by Theorem 3.16 it is *the* square root. Nothing was assumed about $\underline{\underline{C}}$ beyond symmetry and positive definiteness.

Simple shear. With $\underline{\underline{F}} = \underline{\underline{1}} + \gamma \underline{\underline{e}}_1 \otimes \underline{\underline{\hat{E}}}_2$ the two-dimensional part of (3.53) is

$$\underline{\underline{C}} = \underline{\underline{\hat{I}}} + \gamma(\underline{\underline{\hat{E}}}_1 \otimes \underline{\underline{\hat{E}}}_2 + \underline{\underline{\hat{E}}}_2 \otimes \underline{\underline{\hat{E}}}_1) + \gamma^2 \underline{\underline{\hat{E}}}_2 \otimes \underline{\underline{\hat{E}}}_2, \quad (C_{AB}) = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix}.$$

Then

$$\begin{aligned} \det \underline{\underline{C}} &= (1)(1 + \gamma^2) - \gamma^2 = 1 \Rightarrow J = 1, \\ \text{tr } \underline{\underline{C}} &= 2 + \gamma^2 \Rightarrow I = \sqrt{2 + \gamma^2 + 2} = \sqrt{4 + \gamma^2}, \end{aligned}$$

and therefore

$$\boxed{\underline{\underline{U}} = \frac{1}{\sqrt{4 + \gamma^2}} \left[2\underline{\underline{\hat{I}}} + \gamma(\underline{\underline{\hat{E}}}_1 \otimes \underline{\underline{\hat{E}}}_2 + \underline{\underline{\hat{E}}}_2 \otimes \underline{\underline{\hat{E}}}_1) + \gamma^2 \underline{\underline{\hat{E}}}_2 \otimes \underline{\underline{\hat{E}}}_2 \right].}$$

REMARK 3.51 (a misprint in the problem statement). The book prints the last term as $\gamma^2 \hat{E}_1 \otimes \hat{E}_1$. That cannot be right, and the check is immediate. In matrix form the printed version is $\underline{U} = \frac{1}{I} \begin{pmatrix} 2 + \gamma^2 & \gamma \\ \gamma & 2 \end{pmatrix}$, whose square is

$$\frac{1}{4 + \gamma^2} \begin{pmatrix} (2 + \gamma^2)^2 + \gamma^2 & * \\ * & \gamma^2 + 4 \end{pmatrix} = \begin{pmatrix} 1 + \gamma^2 & \gamma \\ \gamma & 1 \end{pmatrix} = \underline{B}^{(2)} \neq \underline{C}.$$

With $\gamma^2 \hat{E}_2 \otimes \hat{E}_2$ instead, $\underline{U} = \frac{1}{I} \begin{pmatrix} 2 & \gamma \\ \gamma & 2 + \gamma^2 \end{pmatrix}$ and

$$\begin{aligned} \underline{U}^2 &= \frac{1}{4 + \gamma^2} \begin{pmatrix} 4 + \gamma^2 & 2\gamma + \gamma(2 + \gamma^2) \\ \gamma(2 + \gamma^2) + 2\gamma & \gamma^2 + (2 + \gamma^2)^2 \end{pmatrix} \\ &= \frac{1}{4 + \gamma^2} \begin{pmatrix} 4 + \gamma^2 & \gamma(4 + \gamma^2) \\ \gamma(4 + \gamma^2) & (4 + \gamma^2)(1 + \gamma^2) \end{pmatrix}, \end{aligned}$$

the last entry because $\gamma^2 + 4 + 4\gamma^2 + \gamma^4 = 4 + 5\gamma^2 + \gamma^4 = (4 + \gamma^2)(1 + \gamma^2)$. Cancelling, $\underline{U}^2 = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix} = \underline{C}$. So the printed expression is in fact $\underline{V}$ with hats attached, and the subscript on the squared term should be 2, not 1.

Principal stretches recovered. The principal values of $\underline{U}$ solve $\lambda^2 - I\lambda + J = 0$ by Problem 8, that is $\lambda^2 - \sqrt{4 + \gamma^2} \lambda + 1 = 0$, so

$$\lambda = \frac{\sqrt{4 + \gamma^2} \pm \sqrt{4 + \gamma^2 - 4}}{2} = \frac{1}{2}(\sqrt{4 + \gamma^2} \pm \gamma),$$

which is (3.60).

Three-dimensional stretch and rotation. The third direction is inert, so

$$\underline{U}^{(3)} = \frac{2\hat{I}^{(2)} + \gamma(\hat{E}_1 \otimes \hat{E}_2 + \hat{E}_2 \otimes \hat{E}_1) + \gamma^2 \hat{E}_2 \otimes \hat{E}_2}{\sqrt{4 + \gamma^2}} + \hat{E}_3 \otimes \hat{E}_3.$$

For the rotation, invert $\underline{U}$ using the same identity: from $\underline{U}^2 - I\underline{U} + J\hat{I} = \hat{Q}$, multiply by $\underline{U}^{-1}$ to get $\underline{U} - I\hat{I} + J\underline{U}^{-1} = \hat{Q}$, i.e.

$$\underline{U}^{-1} = \frac{I\hat{I} - \underline{U}}{J} = I\hat{I} - \underline{U} = \frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 2 + \gamma^2 & -\gamma \\ -\gamma & 2 \end{pmatrix},$$

since $J = 1$ and $I^2 = 4 + \gamma^2$. Then, with $(F_{iA}) = \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix}$,

$$\begin{aligned} (R_{iA}) &= (F_{iB})(U_{BA}^{-1}) = \frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 + \gamma^2 & -\gamma \\ -\gamma & 2 \end{pmatrix} \\ &= \frac{1}{\sqrt{4 + \gamma^2}} \begin{pmatrix} 2 & \gamma \\ -\gamma & 2 \end{pmatrix}, \end{aligned}$$

the $(1, 1)$ entry being $2 + \gamma^2 - \gamma^2 = 2$ and the $(1, 2)$ entry $-\gamma + 2\gamma = \gamma$. This is orthogonal: each column has length $\sqrt{4 + \gamma^2}/\sqrt{4 + \gamma^2} = 1$, the columns are orthogonal since $2\gamma - 2\gamma = 0$, and $\det = (4 + \gamma^2)/(4 + \gamma^2) = 1$. So $\underline{R}$ is the plane rotation through β with

$$\cos \beta = \frac{2}{\sqrt{4 + \gamma^2}}, \quad \sin \beta = \frac{-\gamma}{\sqrt{4 + \gamma^2}}, \quad \tan \beta = -\frac{\gamma}{2},$$

confirming exactly and to all orders the estimate $\beta \approx -\gamma/2$ of Remark 3.33. Comparing with (3.70) gives $\sin 2\alpha = 2/\sqrt{4 + \gamma^2}$ and $\cos 2\alpha = \gamma/\sqrt{4 + \gamma^2}$, which is consistent with (3.69): at $\gamma = 1$, $\tan \alpha = 2/(1 + \sqrt{5}) = 0.618$, so $\sin 2\alpha = 2(0.618)/(1 + 0.618^2) = 0.894 = 2/\sqrt{5}$ and $\cos 2\alpha = (1 - 0.382)/1.382 = 0.447 = 1/\sqrt{5}$.

The left stretch, directly. The same identity applies to $\underline{V}$, since $\underline{V}^2 = \underline{B}$ and $\underline{B}$ has the same invariants as $\underline{C}$ (indeed $\underline{B} = \underline{R}\underline{C}\underline{R}^t$). Hence $\underline{IV} = \underline{B} + \underline{JI}$ with the same I, J , and from (3.56),

$$\underline{V} = \frac{1}{\sqrt{4 + \gamma^2}} \left[2\underline{I} + \gamma(\underline{e}_1 \otimes \underline{e}_2 + \underline{e}_2 \otimes \underline{e}_1) + \gamma^2 \underline{e}_1 \otimes \underline{e}_1 \right],$$

the squared term now sitting on $\underline{e}_1$, as Remark 3.31 predicts and Remark 3.51 confirms.

Azimuthal shear. By (3.92), in the plane spanned by $\{\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta\}$ with $\gamma = R\phi'$,

$$(C_{AB}) = \begin{pmatrix} 1 + \gamma^2 & \gamma \\ \gamma & 1 \end{pmatrix} \quad \text{in the ordered basis } (\hat{\underline{E}}_R, \hat{\underline{E}}_\Theta),$$

so again $\det \underline{C} = 1$, $J = 1$, $\text{tr } \underline{C} = 2 + \gamma^2$, $I = \sqrt{4 + \gamma^2}$, and

$$\underline{U} = \frac{2\hat{\underline{I}} + \gamma(\hat{\underline{E}}_R \otimes \hat{\underline{E}}_\Theta + \hat{\underline{E}}_\Theta \otimes \hat{\underline{E}}_R) + \gamma^2 \hat{\underline{E}}_R \otimes \hat{\underline{E}}_R}{\sqrt{4 + \gamma^2}} + \hat{\underline{E}}_3 \otimes \hat{\underline{E}}_3,$$

the $\hat{E}_3$ term restoring the inert third direction. The rotation follows from $\underline{R}_{\hat{G}} = \hat{G}\underline{U}^{-1}$ with $\underline{U}^{-1} = I\hat{I} - \underline{U}$ in the plane, and then $\underline{R} = \underline{Q}\underline{R}_{\hat{G}}$ by (3.95). Note that here γ depends on R , so $\underline{U}$ and $\underline{R}$ do too: the deformation is not homogeneous, and the principal axes turn as one moves outward through the cylinder.

Torsion. By Problem 3, in the plane spanned by $\{\hat{E}_\Theta, \hat{E}_3\}$ with $\gamma = R\tau$,

$$(C_{AB}) = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{pmatrix} \quad \text{in the ordered basis } (\hat{E}_\Theta, \hat{E}_3),$$

so $J = 1$, $I = \sqrt{4 + R^2\tau^2}$ and

$$\underline{U} = \frac{2\hat{I} + R\tau(\hat{E}_\Theta \otimes \hat{E}_3 + \hat{E}_3 \otimes \hat{E}_\Theta) + R^2\tau^2 \hat{E}_3 \otimes \hat{E}_3}{\sqrt{4 + R^2\tau^2}} + \hat{E}_R \otimes \hat{E}_R,$$

with $\underline{V}$ obtained by the same formula from $\underline{B}$, replacing $\hat{E}_\Theta \rightarrow \underline{e}_\theta$, $\hat{E}_3 \rightarrow \underline{e}_3$ and moving the squared term onto $\underline{e}_\theta$:

$$\underline{V} = \frac{2\underline{I} + R\tau(\underline{e}_\theta \otimes \underline{e}_3 + \underline{e}_3 \otimes \underline{e}_\theta) + R^2\tau^2 \underline{e}_\theta \otimes \underline{e}_\theta}{\sqrt{4 + R^2\tau^2}} + \underline{e}_r \otimes \underline{e}_r.$$

The rotation is $\underline{R} = \underline{Q}\underline{R}_{\hat{G}}$ with $\underline{R}_{\hat{G}} = \hat{G}\underline{U}^{-1}$, computed exactly as above; in the $(\hat{E}_\Theta, \hat{E}_3)$ plane its matrix is $\frac{1}{\sqrt{4+\gamma^2}} \begin{pmatrix} 2 & \gamma \\ -\gamma & 2 \end{pmatrix}$ with $\gamma = R\tau$, so the local shear-induced rotation is through $\arctan(-R\tau/2)$, growing linearly with radius near the axis. $\square$

Closing summary

The logical spine of the chapter, in one paragraph. A symmetric tensor is three numbers on three orthogonal axes, (3.12); it is positive definite exactly when the three numbers are positive, Theorem 3.13; a positive definite symmetric tensor has exactly one positive definite symmetric square root, Theorem 3.16. Applying these to $\underline{C} = \underline{F}^t \underline{F}$, which Chapter

2 proved symmetric and positive definite, produces $\underline{U} = \sqrt{\underline{C}}$, and then $\underline{R} := \underline{F}\underline{U}^{-1}$ is forced to be a rotation, (3.37), and the factorisation is unique, Step 7. The numbers λ_i are the stretches of material fibres momentarily aligned with the principal axes, (3.44); the rotation is the part of $\underline{F}$ that does nothing to the material, and $\underline{U} = \hat{\underline{I}}$ characterises the motions in which nothing at all happens, (3.122).

Two conceptual points that the algebra does not announce. First, a *material* vector is one carrying no time dependence when written as a function of $\underline{X}$: it is glued to matter, and only its image $\underline{F}\underline{M}$ moves, Remark 3.26. Second, the principal directions $\underline{u}_i$ of $\underline{U}$ are *not* material, even though they live in the referential space; they sweep across the material as the deformation proceeds, at a rate computed explicitly in Remark 3.27. Confusing the two is the commonest error in reading (3.44).

Three shears, one algebra: simple shear, azimuthal shear and torsion are the same tensor $\hat{\underline{I}} + \gamma \underline{a} \otimes \underline{b}$ with $\underline{a} \perp \underline{b}$, on three different pairs of orthonormal directions, Remark 3.45; and Problem 9 gives all of them a closed form, $\underline{U} = (\underline{C} + J\hat{\underline{I}})/I$, with no eigenvector computation anywhere.